

# Supersymmetry\*

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# Contents

|          |                                                                               |           |
|----------|-------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction: Why Supersymmetry?</b>                                       | <b>7</b>  |
| <b>2</b> | <b>The Lorentz group.</b>                                                     |           |
|          | <b>The supersymmetry algebra.</b>                                             |           |
|          | <b>Objects and Conventions</b>                                                | <b>15</b> |
| 2.1      | Lorentz Group Breviarium . . . . .                                            | 15        |
| 2.2      | The little group . . . . .                                                    | 17        |
| 2.3      | More on the Lorentz group . . . . .                                           | 19        |
| 2.4      | Weyl spinors in the dotted (Van der Waerden) notation . . . . .               | 23        |
| 2.5      | The Coleman-Mandula No-Go theorem in brief . . . . .                          | 25        |
| 2.6      | Graded algebras . . . . .                                                     | 26        |
| 2.7      | The $N = 1$ supersymmetry algebra . . . . .                                   | 27        |
| 2.8      | Representations of the supersymmetry algebra . . . . .                        | 27        |
| 2.9      | Irreducible representations of massive and massless particle states . . . . . | 29        |
| 2.9.1    | The Lorentz group, helicity and massless particles . . . . .                  | 29        |
| 2.9.2    | Space inversion and time reversal . . . . .                                   | 30        |
| 2.9.3    | Representation of massive particles . . . . .                                 | 31        |
| 2.9.4    | Representation of massless particles . . . . .                                | 33        |
| <b>3</b> | <b>Superfields</b>                                                            | <b>37</b> |
| 3.1      | Grassmann variables and superspace . . . . .                                  | 37        |
| 3.2      | The graded Lie group . . . . .                                                | 39        |
| 3.3      | Chiral fields . . . . .                                                       | 42        |
| <b>4</b> | <b>The Wess-Zumino model</b>                                                  | <b>51</b> |
| 4.1      | The Wess-Zumino Lagrangian . . . . .                                          | 51        |
| 4.2      | The on-shell Lagrangian and supersymmetry . . . . .                           | 52        |
| 4.3      | The Noether supercurrent . . . . .                                            | 53        |

|           |                                                                                               |            |
|-----------|-----------------------------------------------------------------------------------------------|------------|
| 4.4       | Spontaneous supersymmetry breaking . . . . .                                                  | 54         |
| <b>5</b>  | <b>Supersymmetry breaking in detail</b>                                                       | <b>59</b>  |
| 5.1       | Superpotentials and Kähler potentials . . . . .                                               | 59         |
| 5.2       | O’Raifeartaigh models:<br>F-term spontaneous supersymmetry breaking . . . . .                 | 62         |
| 5.3       | Introducing R-symmetry . . . . .                                                              | 67         |
| 5.4       | The Nelson-Seiberg “Theorem” . . . . .                                                        | 70         |
| 5.5       | A toy model with a metastable vacuum . . . . .                                                | 72         |
| 5.6       | Appendix: Kähler geometry . . . . .                                                           | 73         |
| <b>6</b>  | <b>Supersymmetric gauge theories</b>                                                          | <b>79</b>  |
| 6.1       | Vector superfields: the Abelian case . . . . .                                                | 79         |
| 6.2       | Coupling to a chiral superfield . . . . .                                                     | 83         |
| 6.3       | Supersymmetric Electrodynamics [3] . . . . .                                                  | 89         |
| 6.4       | Non-Abelian gauge groups . . . . .                                                            | 90         |
| 6.5       | Mass formulæ . . . . .                                                                        | 93         |
| 6.6       | Basics of supersymmetry breaking in generalized Wess-Zumino models . . . .                    | 96         |
| <b>7</b>  | <b>Path-integral quantization and effective field theories I</b>                              | <b>101</b> |
| 7.1       | Path-integral formulation of Quantum Field Theories . . . . .                                 | 101        |
| 7.2       | Effective field theories: the 1PI effective action . . . . .                                  | 106        |
| <b>8</b>  | <b>Path-integral quantization and effective field theories II</b>                             | <b>125</b> |
| 8.1       | The Coleman-Weinberg mechanism [1] . . . . .                                                  | 125        |
| 8.2       | Effective field theories: the Wilsonian effective action . . . . .                            | 131        |
| <b>9</b>  | <b>Quantum aspects of supersymmetric models</b>                                               | <b>141</b> |
| 9.1       | Brief description of the supergraph approach to non-renormalization theorems                  | 141        |
| 9.2       | Another way to derive non renormalization theorems for supersymmetric the-<br>ories . . . . . | 143        |
| 9.3       | The supersymmetric effective potential . . . . .                                              | 146        |
| 9.4       | Lifting the pseudo-moduli . . . . .                                                           | 148        |
| <b>10</b> | <b>Soft supersymmetry breaking and gauge mediation</b>                                        | <b>153</b> |
| 10.1      | The need of soft supersymmetrye breaking . . . . .                                            | 153        |
| 10.2      | Supersymmetry breaking in supergravity models . . . . .                                       | 155        |
| 10.3      | The bulding blocks in gauge mediation . . . . .                                               | 157        |

|                                                          |            |
|----------------------------------------------------------|------------|
| <b>11 Extended (<math>N &gt; 1</math>) supersymmetry</b> | <b>165</b> |
| 11.1 Problems . . . . .                                  | 171        |
| <b>12 Solitons in condensed matter</b>                   | <b>175</b> |



# Chapter 1

## Introduction: Why Supersymmetry?

28/3/10

Both in periods in which scientists work in a framework of “normal science” or during “revolutions” (to appeal for a moment to Kuhn’s simplification) it is supposed that they try to explain some established phenomena or to solve some contradictory data. A counterexample of such a simplistic view, among many others, is Einstein’s General relativity, an intellectual achievement driven by the logic of theoretical development and not by the pressure of existing data to be explained. Now, General relativity was proposed by Einstein in 1915 and the physics community accepted that it was confirmed just four years later<sup>1</sup>.

The discovery in the early 1970’s of supersymmetry, an invariance of the theory under interchange of fermions and bosons, took also place without the aim of explaining any experimental data. Now, 40 years elapsed, an enormous theoretical effort was invested, more than 40,000 papers were published but none of them reports the experimental discovery of supersymmetry. In spite of this, many of us agree that supersymmetry is playing in the beginning of the 21<sup>st</sup> century the dominant role that relativity and quantum physics played in the 20<sup>th</sup> (at least in the domain of high energy physics).

The following is a too brief time line of supersymmetry from its origin to the proposal of the Minimal Supersymmetric Standard model whose predictions started to be tested this year in the Large Hadron Collider at CERN:

- 1970- Observation by Pierre Ramond of supersymmetry as a world-sheet 2 dimensional symmetry [1]. (There were contemporary developments by Jean-Loup Gervais, André Neveu, Benji Sakita, John Schwarz along the same lines [2]-[3].)

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<sup>1</sup>In fact, Eddington’s calculations on the deflection of light by the Sun during the 1919 solar eclipse were, according to many specialists, plagued by systematic error and possibly confirmation bias. In fact, modern reanalysis of the dataset suggests that Eddington’s analysis was accurate.

- 1971- Yuri Golfand and Evgeny Likhtman [4] found the superextension of Poincaré algebra and constructed the first 4-dimensional field theory with supersymmetry (quantum electrodynamics of spinors and scalars).
- 1971- Dmitry Volkov and Vladimir Akulov [5] proposed to explain the masslessness of the neutrino by interpreting it as a Goldstone fermion related to supersymmetry.
- 1973- Julius Wess and Bruno Zumino [6] defined supergauge transformations and described different supersymmetric multiplets [1]. They also presented explicit supersymmetric models, in particular the well-honored Wess-Zumino model.
- 1974
  - Abdus Salam and John Strathdee proposed the superspace formulation of supersymmetric field theories.
  - A proof of non-renormalization theorems in supersymmetric field theories is given by Sergio Ferrara, Jean Iliopoulos, Wess and Zumino.
  - The construction of supersymmetric gauge theories is achieved by Ferrara, Salam, Strathdee and Zumino
  - Pierre Fayet and Iliopoulos constructed the first renormalizable supersymmetric theory with spontaneously broken supersymmetry.
- 1975- Rudolf Haag, Jan Lopuszanski, and Martin Sohnius published their proof that weakening the assumptions of the Coleman-Mandula theorem by allowing both commuting and anticommuting symmetry generators, there is a nontrivial extension of the Poincaré algebra, namely the supersymmetry algebra [7].
- 1975-1976- Proposal of local supersymmetry and construction of supergravity theories by Daniel Freedman, Peter van Nieuwenhuizen, Ferrara, Stanley Deser and Zumino. [8]-[9].
- 1976- Ferdinando Gliozzi, Joel Scherk, and David Olive realized that the original Ramond and Neveu Schwarz-strings were separately inconsistent and needed to be combined in a theory with no tachyon and having space-time supersymmetry [10].
- 1977-1979- The first attempts by Fayet to extend the Standard Model to the supersymmetric case were published [11].
- 1978-1979 Discoveries of strong/weak dualities among supersymmetric theories by Claus Montonen, David Olive and Edward Witten [12]-[13].



- 1981- Savas Dimopoulos and Howard Georgi put forward the first version of what today is called the Minimal Supersymmetric Standard Model (MSSM) [14].
- 1983- Dimopoulos, Stuart Raby and Frank Wilczek show the unification of coupling constants in supersymmetric unified theories [15].
- 1994- Nathan Seiberg and Witten obtained exact results in supersymmetric QCD: the Seiberg- Witten theory [16] and Seiberg dualities [17]-[18].
- 1997- Juan Martín Maldacena discovers the gauge/supergravity-superstring correspondence and proposes the Maldacena conjecture.

We stop here without recording more recent achievements and instead enumerate the reason why supersymmetry is taken seriously today:

- The Hierarchy problem
- TeV supersymmetry yields unification of gauge couplings
- TeV supersymmetry can account for the observed dark matter density
- Supersymmetry emerges frequently from string theory and is perhaps relevant for a fundamental theory of quantum gravity.

## • The hierarchy problem

In a theory aiming to unify all known interactions the Planck mass  $M_P$  sets the “natural” scale<sup>2</sup>,

$$M_P = \sqrt{\hbar c/G_N} \approx 1.2209 \times 10^{19} \text{ GeV}/c^2 = 2.17644(11) \times 10^{-8} \text{ kg} \quad (1.0.1)$$

This is enormous in comparison with the 300 GeV scale of electroweak symmetry breaking.

The question is why the Higgs boson is so much lighter than the Planck mass (or the grand unification energy): Starting from a Higgs Lagrangian with a given bare mass, one would expect that renormalization would produce large quantum contributions to the square of the Higgs boson mass  $m_H^2$  making the renormalized (measurable) mass huge, comparable to the scale  $M_P$  at which new physics appears, unless there is an incredible fine-tuning cancellation between the renormalization corrections and the bare mass.

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<sup>2</sup>For example the mass of a flea is about  $4000M_P$ .

A way to avoid such a suspicious fine tuning is to appeal to a well-known mechanism, namely the fact that some new symmetry protects a given mass from quantum corrections. For ordinary symmetries (not relating bosons with fermions) this is very common. For example one can see that a global symmetry, chiral symmetry, when exact, prevents the electron to be massive. But if it is broken at a scale  $\sim \mu$  then the electron mass cannot become huge but it is fixed by such scale,  $m_e = \mu$ . Local (gauge) symmetry prevents the mass of the proton to grow with quantum corrections. In general such kind of mechanisms usually work with fermions and not with bosons, like the Higgs field. The fact that supersymmetry relates fermions and bosons makes possible a similar mechanism to work for the Higgs field.

## • TeV supersymmetry yields unification of gauge couplings

At the quantum level coupling constants are not fixed but they “run” with the physical scale at which measures are done. Their evolution is ruled by the “renormalization group”. When we say that in the Standard model electromagnetic, weak and strong interactions are unified at a certain scale  $\Lambda$  we mean that the corresponding running coupling constants become identical at a scale of the order of  $\Lambda$ . Now, this unification of the three gauge coupling constants of the (ordinary) Standard Model is sensitive to the present particle content of the theory and in fact the 3 constants do not quite meet together at a common energy scale. In contrast, within the Minimal Supersymmetric Standard model coupling constants do unify reasonably (at a scale  $\Lambda \sim 2 \times 10^{16}$  GeV).

## • Account for dark matter

According to present observations of structures larger than galaxies, as well as Big Bang cosmology, the vast majority of the mass in the observable universe does not behave as ordinary matter. In particular, this matter is undetectable by its emitted electromagnetic radiation, but its presence can be inferred from gravitational effects on visible matter and background radiation. This is the reason why it is called dark matter.

Dark matter abundance is at least five times larger than that of the baryonic matter and corresponds to 83% – 90% of the total density of the observable Universe. Many lines of reasoning suggest that the dark matter consists of some new, as yet undiscovered, weakly-interacting massive particle (WIMP). Of the many WIMP candidates, perhaps the best motivated and certainly the most theoretically developed is the neutralino, the lightest superpartner in many supersymmetric theories. The neutralino is a combination of three superpartners: the photino (supersymmetric partner of the photon), the zino ( $Z_0$  boson partner) and the higgsino (Higgs boson partner). Recent results from CERN indicate that the neutralino mass is larger than 46 GeV/c<sup>2</sup>.

## • Supersymmetry and string theory

We already signaled that it was in a string context that supersymmetry was first introduced by 1970, when Ramond added fermions to a bosonic string model. This led him to formulate a two-dimensional supersymmetry that cancel certain wrong-sign states. A short time later Schwarz and Neveu added another sector to the fermi theory. Gliozzi, Scherk, and Olive realized in 1976 that the original Ramond and Neveu Schwarz-strings were separately inconsistent and needed to be combined in a supersymmetric theory. In fact, superstring theory is "super" in the same sense as it is in supersymmetry. It is a theory that includes fermionic string vibration in addition to the bosonic ones.

In the history of string theory there was a *first revolution*, a period of important discoveries in string theory between 1984 and 1986. During this period it was realized that string theory was capable of describing all elementary particles as well as the interactions between them. The revolution was started by a discovery of "anomaly cancelation" in type I string theory via the Green-Schwarz mechanism in 1984 [20]. Several other ground-breaking discoveries, such as the heterotic string, were made in 1985.

The second revolution in string theory, called *superstring revolution*, took place between 1994 and 1997. The different versions of superstring theory were unified, as long hoped, by new equivalences. The different theories of strings were also connected to a new 11-dimensional theory called M-theory. New objects called branes were discovered as inevitable ingredients of string theory. Their analysis - especially the analysis of a special type of branes called D-branes - led to the AdS/CFT correspondence, the microscopic understanding of the thermodynamic properties of black holes, and many other developments.

After this long introduction, I think one can agree with Steven Weinberg when he writes in the Preface of the third volume of "The Quantum Theory of Fields", after signaling the absence of experimental confirmation of the role of supersymmetry in Nature,

*Because of the intrinsic attractiveness of supersymmetry and the possibility it offer of resolving the hierarchy problem, I and many other physicists are reasonably confident that supersymmetry will be found to be relevant to the real world and perhaps soon.[21]*

Postscript: Supersymmetry is commonly referred to as *relating fermions with bosons* although its algebra does not require (as we shall see) the existence of fundamental bosons... Bosons can be taken as composite objects constructed from fermions at the prize of accepting nonlinear realizations of supersymmetry. Just as the mathematical existence of the group  $SU(3)$  does not imply that the strange quark, exists the existence of supersymmetry does not necessitate of bosons.



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# Chapter 2

## The Lorentz group.

## The supersymmetry algebra.

## Objects and Conventions

28/3/10

### 2.1 Lorentz Group Breviarium

A Lorentz coordinate transformation is a linear transformation of the form

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu} + a^{\mu}, \quad \mu, \nu = 0, 1, 2, 3 \quad (2.1.1)$$

with  $a_{\mu}$  an arbitrary constant 4-vector and  $\Lambda^{\mu}_{\nu}$  a constant matrix satisfying

$$g_{\mu\nu} \Lambda^{\mu}_{\rho} \Lambda^{\nu}_{\sigma} = g_{\rho\sigma} \quad (2.1.2)$$

where  $g_{\mu\nu}$  is the metric tensor,  $\text{diag } g_{\mu\nu} = (-1, 1, 1, 1)$ . Notice that (2.1.2) implies  $\det \Lambda^2 = 1$ .

Transformation (2.1.1) leaves invariant the space-time interval  $ds^2 = g_{\mu\nu} dx^{\mu} dx^{\nu}$ ,

$$g_{\mu\nu} dx'^{\mu} dx'^{\nu} = g_{\mu\nu} dx^{\mu} dx^{\nu} \quad (2.1.3)$$

These transformations form a group, the *inhomogeneous Lorentz group* or *Poincaré group*. Among Lorentz transformations those which preserve the direction of time are called *orthochronous* (they correspond to the subgroup with  $\Lambda^0_0 \geq +1$ ). Those which preserve spatial orientation are called *proper*, and as linear transformations they have determinant +1. The improper Lorentz transformations have determinant -1.

An important subgroup of the Poincaré group is the *homogeneous Lorentz group* which corresponds to the case  $a_\mu = 0$ . The *proper orthochronous Lorentz group* is the subgroup of transformations with  $\det \Lambda = 1$  and  $\Lambda_0^0 \geq 1$ . Any transformation that can be obtained from the identity by a continuous change of parameters must belong to this last subgroup.

The Lorentz group is a non-compact<sup>1</sup> Lie group that preserves the quadratic form  $x^\mu x_\mu$ . In this sense, it is sometimes denoted as  $O(1, 3)$  or  $O(3, 1)$ . In the same vein, the subgroup of proper Lorentz transformations is denoted as  $SO(1, 3)$ . The subgroup of orthochronous transformations is often denoted as  $O^+(1, 3)$ .

Any Lorentz transformation is either proper and orthochronous or it can be written as a proper orthochronous transformation times a space inversion or/and a time reversal.

One can derive the Poincaré algebra by considering an infinitesimal Lorentz transformation

$$\Lambda^\mu_\nu = \delta^\mu_\nu + \omega^\mu_\nu, \quad a^\mu = \epsilon^\mu \quad (2.1.4)$$

Inserting this expression in (2.1.2) one finds that

$$\omega_{\mu\nu} = -\omega_{\nu\mu} \quad (2.1.5)$$

which then corresponds to 6 independent parameters. This, together with the 4 parameters  $\epsilon^\mu$  gives the 10 parameters of the inhomogeneous Lorentz group.

Let us consider the action of Lorentz transformation on vectors  $\Psi$  in the physical Hilbert space,

$$\Psi \rightarrow \Psi' = U(\Lambda, a)\Psi \quad (2.1.6)$$

Here  $U(\Lambda, a)$  represents a unitary linear transformation. In the infinitesimal case one can write

$$U(\Lambda, a) = 1 + \frac{i}{2}\omega_{\rho\sigma}J^{\rho\sigma} - i\epsilon_\rho P^\rho + \dots \quad (2.1.7)$$

where generators  $J^{\rho\sigma}$  and  $P^\rho$  satisfy

$$J^{\rho\sigma} = -J^{\sigma\rho}, \quad P^{\rho\dagger} = P^\rho \quad (2.1.8)$$

and, if we consider unitary representations then one should have

$$J^{\rho\sigma\dagger} = J^{\rho\sigma} \quad (2.1.9)$$

Let us recall that  $P^\mu$  is the energy-momentum operator and  $J^{ij}$  the components of the angular momentum operator. The remaining generators form the boost 3-vector  $\vec{K} = (J^{10}, J^{20}, J^{30})$

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<sup>1</sup>A non compact group is a topological group whose topology is non-compact. Because the Lorentz group is non-compact, it has no finite-dimensional unitary representations.



They satisfy the algebra

$$\begin{aligned}
[P^\mu, P^\nu] &= 0 \\
i[J^{\mu\nu}, J^{\rho\sigma}] &= g^{\nu\rho} J^{\mu\sigma} - g^{\mu\rho} J^{\nu\sigma} - g^{\sigma\mu} J^{\rho\nu} + g^{\sigma\nu} J^{\rho\mu} \\
i[P^\mu, J^{\rho\sigma}] &= g^{\mu\rho} P^\sigma - g^{\mu\sigma} P^\rho
\end{aligned} \tag{2.1.10}$$

This is the Lie algebra of the Poincaré group. The Casimir operators of this group are the mass-square operator  $P_\mu P^\mu$  and the generalized spin operator  $W^2 = W_\mu W^\mu$  where  $W^\mu$  is the Pauli-Lubanski vector

$$W^\mu = \frac{1}{2} \varepsilon^{\mu\nu\rho\sigma} P_\nu J_{\rho\sigma} \tag{2.1.11}$$

In the rest frame one has in the massive case

$$P^\mu = (m, 0, 0, 0), \quad P^\mu P_\mu = -m^2, \quad W^2 = m^2 \vec{J}^2 \tag{2.1.12}$$

with

$$\vec{J} = (J^{23}, J^{31}, J^{12}) \tag{2.1.13}$$

Let us end by noting that the Lorentz group (and also the Poincaré group) is not compact and, related to this, the  $J_{\mu\nu}$  cannot all be represented by Hermitian matrices and then the finite dimensional representations of the Lorentz group are not unitary. The noncompact part of the Poincaré group are the boosts and translations.

## 2.2 The little group

It was Wigner [4] who has found how to reduce the representations of the inhomogeneous Lorentz group to representations of a “little group”. Here we shall explain (without much mathematical justification) how one constructs such little group leaving for the end of this chapter the construction of its representations.

Since the components of  $P^\mu$  commute with each other, it is natural to express the physical state vectors in terms of eigenvectors of the four momentum,  $\Psi_{p\sigma}$  with eigenvalues  $p_\mu$ ,

$$P^\mu \Psi_{p,\sigma} = p^\mu \Psi_{p,\sigma} \tag{2.2.14}$$

Here  $\sigma$  represents all other degrees of freedom.

Now, a Lorentz transformation not only changes the 4-vector  $x^\mu$  according to (see eq.(2.1.1))

$$x'^\mu = \Lambda^\mu_\nu x^\nu + a^\mu, \quad \mu, \nu = 0, 1, 2, 3 \tag{2.2.15}$$

but also changes the 4-momentum  $p^\mu$  leaving however  $p^\mu p_\mu = p^2$  invariant. So in order to specify a physical state, one can choose a ‘standard’ 4-momentum  $k^\mu$  representing the equivalent class of all momenta  $p^\mu$  satisfying the condition

$$p^\mu p_\mu = k^\mu k_\mu = p^2 \quad (2.2.16)$$

Once a  $k^\mu$  is chosen, any  $p^\mu$  satisfying (2.2.16) can be obtained by an appropriate Lorentz transformation

$$p^\mu = L^\mu{}_\nu(p) k^\nu \quad (2.2.17)$$

where  $L$  depends on  $p^\mu$  and, implicitly on  $k^\mu$ .

Concerning vector states one can define

$$\Psi_{p,\sigma} = N(p) U(L(p)) \Psi_{k,\sigma} \quad (2.2.18)$$

with  $N(p)$  a normalization to be specified later. If one applies an arbitrary Lorentz transformation to this last expression, one obtains

$$U(\Lambda) \Psi_{p,\sigma} = N(p) U(\Lambda) U(L(p)) \Psi_{k,\sigma} = N(p) U(\Lambda L(p)) \Psi_{k,\sigma} = U(L(\Lambda p)) U(L^{-1}(\Lambda p) \Lambda L(p)) \Psi_{k,\sigma} \quad (2.2.19)$$

Now the transformation Lorentz  $L^{-1}(\Lambda p) \Lambda L(p)$  takes  $k$  to  $L(p)k = p$  and then to  $\Lambda p$  and then back to  $k$ . Then, it belongs to a subgroup  $W$  of the homogeneous Lorentz group that leaves invariant  $k^\mu$ .

This subgroup is called the *little group* and we shall write it as  $W$ . If a Lorentz group element  $W^\mu{}_\nu$  is such that

$$W^\mu{}_\nu k^\nu = k^\mu \quad (2.2.20)$$

then  $W^\mu{}_\nu \in W$ .

One can prove that if one has a matrix representation  $D_{\sigma\sigma'}(W)$  for the little group then one can write for a general Lorentz transformation  $\Lambda$ . That is, if one knows that

$$U(W) \Psi_{k,\sigma} = \sum_{\sigma'} D_{\sigma'\sigma}(W) \Psi_{k,\sigma'} \quad (2.2.21)$$

then for a general  $U(\Lambda)$  one will have

$$U(\Lambda) \Psi_{p,\sigma} = N'(p, \Lambda) \sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda p)) \Psi_{\Lambda p, \sigma'} \quad (2.2.22)$$

Hence, finding the representations of the little group one finds the general representation of the Lorentz group. The proof is rather simple. One applies (2.2.21) to the case of

$$W(\Lambda p) = L^{-1}(\Lambda p) \Lambda L(p) \quad (2.2.23)$$

and then uses (2.2.19) and (2.2.18) ending with (2.2.22).

A convenient choice of the standard momentum and the corresponding little group for the cases of physical interest is given below

|                                 | Standard $k^\mu$ | <i>Little group</i> |          |
|---------------------------------|------------------|---------------------|----------|
| $p^2 = -m^2 < 0, \quad p^0 > 0$ | $(m, 0, 0, 0)$   | $SO(3)$             | (2.2.24) |
| $p^2 = 0, \quad p^0 > 0$        | $(E, 0, 0, E)$   | $ISO(2)$            |          |
| $p^\mu = 0,$                    | $(0, 0, 0, 0)$   | $SO(1, 3)$          |          |

One can understand while little groups are those listed above as follows:  $SO(3)$  is the ordinary rotation group in 3 dimensions (excluding space inversions). They are the only proper ( $\det \Lambda = +1$ ) orthochronous ( $\Lambda^0_0 > 1$ ) Lorentz transformations that leave at rest a particle with  $p^i = 0, i = 1, 2, 3$ . The  $ISO(2)$  group is the group of rotations and translations in 2 dimensions, in the present case the one that acts on spatial directions (1, 2). Finally  $SO(1, 3)$  is the whole proper orthochronous Lorentz group leaving invariant the null 4-vector (Some authors write  $SO^+(1, 3)$  with the + indicating this last sign condition).

## 2.3 More on the Lorentz group

We have seen that in 4 dimensions the Lorentz group has 6 generators, 3 of them corresponds to the generators of the group of rotations in 3 dimensions,  $SO(3)$  which we called  $J_i$ ,

$$\vec{J} = (J^{23}, J^{31}, J^{12}) \quad (2.3.25)$$

in terms of which a rotation of angle  $\varphi$  with respect to a direction fixed by a unit vector  $\check{e}$  is written as

$$R(\check{e}, \varphi) = \exp \left( i\varphi \check{e} \cdot \vec{J} \right) \quad (2.3.26)$$

The other three generators of the Lorentz group are associated with boosts  $\vec{K}_i$

$$\vec{K} = (J^{10}, J^{20}, J^{30}) \quad (2.3.27)$$

along the three spatial directions. A boost with rapidity  $\lambda$

$$\lambda = \left( \operatorname{arctanh} \frac{V}{c} \right) \quad (2.3.28)$$

along a direction  $\check{u}$  is given by

$$B(\check{u}, \lambda) = \exp \left( i \lambda \check{u} \cdot \vec{K} \right) \quad (2.3.29)$$

(but notice that the  $K_i$  are anti-hermitian, see below).

In terms of these 6 generators together with those associated with energy-momentum the algebra (2.1.10) takes the form

$$[J_i, J_j] = i \varepsilon_{ijk} J_k \quad (2.3.30)$$

$$[J_i, K_j] = i \varepsilon_{ijk} K_k \quad (2.3.31)$$

$$[K_i, K_j] = -i \varepsilon_{ijk} J_k \quad (2.3.32)$$

$$[J_i, P_j] = i \varepsilon_{ijk} P_k \quad (2.3.33)$$

$$[K_i, P_j] = i H \delta_{ij} \quad (2.3.34)$$

$$[J_i, H] = [P_i, H] = [H, H] = 0 \quad (2.3.35)$$

$$[K_i, H] = i P_i \quad (2.3.36)$$

where we have rewritten the energy operator as  $P_0 = H$ . The first line corresponds to commutation relations of the rotation group  $SO(3)$ , the second line shows that the  $K_i$  generators transform like vectors under rotations.

Concerning the commutation relations between  $K_i$ 's, note the minus sign in eq.(2.3.32) compared to (2.3.30). The origin of this can be traced back to the fact that  $g_{00} = -1$ . As a result, while  $\vec{J}$  is hermitian, boost generators  $\vec{K}$  are anti-hermitian<sup>2</sup>. For example, for a boost along the  $x^1$  direction one has, for Dirac fermions,

$$B(\check{i}, \lambda) = \exp \left( -\frac{\lambda}{2} \alpha_1 \right) = B(\check{i}, \lambda)^\dagger \neq B(\check{i}, \lambda)^{-1} \quad (2.3.37)$$

with

$$\alpha_1 = \begin{pmatrix} 0 & \sigma_1 \\ \sigma_1 & 0 \end{pmatrix} \quad (2.3.38)$$

Evidently, momentum and angular momentum are conserved as of course the energy itself is ((2.3.35)). In contrast, eq.(2.3.36) shows that  $\vec{K}$  is not conserved and this, together with its non-hermiticity, explains why its eigenvalues are not used to label physical states

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<sup>2</sup>This non-hermiticity of  $\vec{K}$  is related to the fact that the homogeneous Lorentz group  $SO(1,3)$  is non-compact and does not have finite dimensional unitary representations. This represents no problem since we are not concerned here with wave-functions but with fields which do not need to have a Lorentz invariant positive norm.

while one uses the conserved eigenvalues of the four-momentum and the angular momentum to this end.

In order to find representations of such algebra it is convenient to define

$$J_i^\pm = \frac{1}{2}(J_i \pm iK_i) \quad (2.3.39)$$

which satisfy the following algebra

$$\begin{aligned} [J_i^\pm, J_j^\pm] &= i\varepsilon_{ijk}J_k^\pm \\ [J_i^+, J_j^-] &= 0 \end{aligned} \quad (2.3.40)$$

Note that in the way we have defined  $\vec{J}$  and  $\vec{K}$ , we needed to introduce a factor  $i$  in front of  $\vec{K}$  in definition (2.3.39) so that if  $\vec{J}^\pm$  is to be hermitian then  $K_i$  should be antihermitian.

From (2.3.40) it becomes evident that the the Lorentz algebra is equivalent to 2 copies of the algebra of  $SU(2) \approx SO(3)$ . More precisely, the Lorentz homogeneous (proper or-tochronous) group is  $SL(2, C)$ , the group of  $2 \times 2$  complex matrices  $M$  such that  $\det M = 1$  a complexified version of  $SU(2) \times SU(2)$ .

In view of the discussion above, the irreducible representations of the homogeneous Lorentz group can be obtained from those of  $SU(2)$ . These last are label by the spin  $s = 1/2 + k$  with  $k \in \mathbb{N}$  and then any representation of the Lorentz group can be identified by a the pair  $(s_-, s_+)$ , the spins of each one of the copies of  $SU(2)$ . The dimension of the representation is  $(2s_+ + 1) \times (2s_- + 1)$ .

The basis vectors spanning the representation space  $(s_-, s_+)$  and  $(s_+, s_-)$  can be chosen so that the representation matrices  $M$  satisfy

$$M^{(s_-, s_+)} = M^{(s_+, s_-)*} \quad (2.3.41)$$

where  $M^*$  is the complex conjugate of  $M$

- A quantity that transforms under the  $(s_+, s_-) = (0, 0)$  representation is a singlet under  $J^\pm$  and hence under rotations and boosts. It is called a scalar.
- One which transforms under the  $(1/2, 0)$  is a (two-component) (left-handed) Weyl spinor.
- A quantity transforming under the  $(0, 1/2)$  representation is a conjugate Weyl (right-handed) spinor.
- The Dirac representation is constructed from the direct sum  $(1/2, 0) \oplus (0, 1/2)$ . In this way on has Dirac fermions.

- The  $(1/2, 1/2)$  representation is a vector representation and the objects transforming according to it are called 4-vectors.

Since a Weyl fermion transforming according to the  $(1/2, 0)$  representation is a left-handed 2-component fermion  $\psi_L$  and that transforming according to the  $(0, 1/2)$  representation is a right handed fermion  $\psi_R$ , a Dirac fermion can then be written as

$$\psi_D = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \quad (2.3.42)$$

Let us recall that the chiral representation of Dirac matrices learnt in the Relativistic Quantum Mechanics course leads to block-diagonal generators for boosts and rotations of 4-component Dirac spinors, this making apparent that the Dirac representation of the Lorentz group is *reducible*. The chiral representation is one in which the Dirac matrices take the form

$$\gamma^0 = \begin{pmatrix} 0 & -I_{2 \times 2} \\ -I_{2 \times 2} & 0 \end{pmatrix} ; \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} ; \quad \gamma^5 = \gamma^0 \gamma^1 \gamma^2 \gamma^3 = i \begin{pmatrix} -I_{2 \times 2} & 0 \\ 0 & I_{2 \times 2} \end{pmatrix} \quad (2.3.43)$$

In the chiral representation the boost and rotation generators are

$$J^{0i} = \frac{i}{4} [\gamma^0, \gamma^i] = - \begin{pmatrix} \sigma^i & 0 \\ 0 & \sigma^i \end{pmatrix} \quad (2.3.44)$$

$$J^{ij} = \frac{i}{4} [\gamma^i, \gamma^j] = \frac{1}{2} \epsilon^{ijk} \begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix} = \frac{1}{2} \epsilon^{ijk} \Sigma^k \quad (2.3.45)$$

One easily finds that

$$[\gamma^5, J^{\mu\nu}] = 0 \quad (2.3.46)$$

Thus, according to Schur Lemma, the Dirac representation must be reducible since eigenvectors of  $\gamma_5$  whose eigenvalues are different ( $\gamma_5^2 = 1 \Rightarrow \lambda = \pm 1$ ) transform without mixing.

A Dirac spinor with only left ( $\psi_L$ ) or right-handed ( $\psi_R$ ) components are eigenvectors of  $\gamma_5$  with  $\lambda = -1$  and  $\lambda = 1$  eigenvalues respectively.

Under Lorentz transformations left and right-handed Weyl spinors transform according to

$$\begin{aligned} \psi_L &\rightarrow \psi' = \left(1 - \frac{i}{2} \theta^i \sigma^i + \frac{1}{2} \beta^i \sigma^i\right) \psi \\ \psi_R &\rightarrow \psi' = \left(1 - \frac{i}{2} \theta^i \sigma^i - \frac{1}{2} \beta^i \sigma^i\right) \psi \end{aligned} \quad (2.3.47)$$

with  $\theta^i$  and  $\beta^i$  infinitesimal rotation and boost parameters.

## 2.4 Weyl spinors in the dotted (Van der Waerden) notation

We want to discuss here how the  $SL(2, C)$  matrices represent the action of the Lorentz group on two-component Weyl spinors. We teach and learn in the Quantum mechanics courses about the connection between the  $SU(2)$  and the rotation group and the role played there by the 3 Pauli matrices  $\sigma^i$ . In analogy, there is a connection between  $SL(2, C)$  and the Lorentz group established through the so-called  $\sigma^\mu$ -matrices ( $\mu = 1, \dots, 4$ ), defined as

$$\sigma^0 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \quad \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (2.4.48)$$

These matrices, which we write  $(\sigma^\mu) = (\sigma^0, \vec{\sigma})$  form a basis for  $2 \times 2$  complex matrices

$$P = P_\mu \sigma^\mu = \begin{pmatrix} -P_0 + P_3 & P_1 - iP_2 \\ P_1 + iP_2 & -P_0 - P_3 \end{pmatrix}, \quad \mu = 0, 1, 2, 3 \quad (2.4.49)$$

Now, given any hermitian matrix  $P$ , we can transform it in the form

$$P' = M P M^\dagger \quad (2.4.50)$$

with  $M$  unimodular (i.e.,  $\det M = 1$ ), i.e.,  $M \in SL(2, C)$ . We can also write

$$\sigma^\mu P'_\mu = M \sigma^\mu P_\mu M^\dagger \quad (2.4.51)$$

Taking det on both sides

$$\det(\sigma^\mu P'_\mu) = \det(\sigma^\mu P_\mu) \quad (2.4.52)$$

or

$$P'^2_0 - \vec{P}'^2 = P^2_0 - \vec{P}^2 \quad (2.4.53)$$

So the coefficients  $(P_0, \vec{P})$  transform as the components of a 4-vector under Lorentz transformations. Of course we can construct matrices analogous to  $P$  for the cuadvivector  $x^\mu$ , etc, arriving to invariance equations under  $M$  transformations of the form (2.4.53), so that we can use  $SL(2, C)$  to represent Lorentz trasnformations.

A 2-component Weyl spinor transforming under the  $(1/2, 0)$  representation of the Lorentz group (a left-handed fermion) will be written with undotted greek indexes  $\alpha = 1, 2$  in the form  $\psi_\alpha$ . Those transforming under the  $(0, 1/2)$  representation (a right-handed fermion) will be written with dotted greek indexes  $\dot{\alpha} = \dot{1}, \dot{2}$ .

Given  $M \in SL(2, C)$ , its action on spinors is written as

$$\psi'_\alpha = M_\alpha{}^\beta \psi_\beta, \quad \bar{\psi}'_{\dot{\alpha}} = M^*_{\dot{\alpha}}{}^{\dot{\beta}} \bar{\psi}_{\dot{\beta}} \quad (2.4.54)$$

Spinors with upper and lower indices are related through the  $\varepsilon$  tensor:

$$\begin{aligned} \varepsilon_{11} = \varepsilon_{22} = \varepsilon^{11} = \varepsilon^{22} &= 0 \\ \epsilon^{12} = -\epsilon^{21} &= 1, \quad \epsilon_{12} = -\epsilon_{21} = -1 \\ \psi^\alpha &= \varepsilon^{\alpha\beta} \psi_\beta, \quad \psi_\beta = \varepsilon_{\alpha\beta} \psi^\alpha \\ \bar{\psi}^{\dot{\alpha}} &= \varepsilon^{\dot{\alpha}\dot{\beta}} \bar{\psi}_{\dot{\beta}}, \quad \bar{\psi}_{\dot{\beta}} = \varepsilon_{\dot{\alpha}\dot{\beta}} \bar{\psi}^{\dot{\alpha}} \end{aligned} \quad (2.4.55)$$

One can then prove

$$\psi'^\alpha = (M^{-1})^\alpha{}_\beta \psi^\beta, \quad \bar{\psi}'^{\dot{\alpha}} = (M^{*-1})^{\dot{\alpha}}{}_{\dot{\beta}} \bar{\psi}^{\dot{\beta}} \quad (2.4.56)$$

We can compactly describe the notation as follows:

$$\begin{aligned} \left( \frac{1}{2}, 0 \right) &: \psi_\alpha \\ \left( 0, \frac{1}{2} \right) &: \bar{\psi}_{\dot{\alpha}} \end{aligned} \quad (2.4.57)$$

and we can identify

$$\bar{\psi}_{\dot{\alpha}} \equiv (\psi_\alpha)^* \quad (2.4.58)$$

Concerning the  $\sigma^\mu$  matrices, from eqs.(2.4.54) we see that they have the following index structure

$$(\sigma^\mu) = \sigma^\mu_{\alpha\dot{\alpha}} \quad (2.4.59)$$

so that

$$\bar{\sigma}^{\mu\dot{\alpha}\alpha} = \varepsilon^{\dot{\alpha}\dot{\beta}} \varepsilon^{\alpha\beta} \sigma^\mu_{\beta\dot{\beta}} \quad (2.4.60)$$

this meaning that

$$\bar{\sigma}^0 = \sigma^0, \quad \bar{\sigma}^i = -\sigma^i \quad (2.4.61)$$

With this notation, a Dirac spinor can be written in the form

$$\psi_D = \begin{pmatrix} \chi_\alpha \\ \bar{\psi}^{\dot{\alpha}} \end{pmatrix} \quad (2.4.62)$$

while a Majorana spinor in the Weyl basis is just

$$\psi_M = \begin{pmatrix} \chi_\alpha \\ \bar{\chi}^{\dot{\alpha}} \end{pmatrix} \xrightarrow{\text{Majorana basis}} \begin{pmatrix} \text{Re } \chi_\alpha \\ \text{Im } \chi_\alpha \end{pmatrix} \quad (2.4.63)$$



- Spinor summation convention

$$\begin{aligned}\psi\chi &= \psi^\alpha\chi_\alpha = -\psi_\alpha\chi^\alpha = \chi^\alpha\psi_\alpha = \chi\psi \\ \bar{\psi}\bar{\chi} &= \bar{\psi}_{\dot{\alpha}}\bar{\chi}^{\dot{\alpha}} = -\bar{\psi}^{\dot{\alpha}}\bar{\chi}_{\dot{\alpha}} = \bar{\chi}_{\dot{\alpha}}\bar{\psi}^{\dot{\alpha}} = \bar{\chi}\bar{\psi}\end{aligned}\tag{2.4.64}$$

These quantities are Lorentz scalars. Note that  $\psi^\alpha\chi_\alpha$  anticommute but  $\psi\chi$  commute (and the same for  $\bar{\psi}, \bar{\chi}$ )

•

$$\begin{aligned}\chi\sigma^\mu\bar{\psi} &= -\bar{\psi}\sigma^\mu\chi & (\chi\sigma^\mu\bar{\psi})^+ &= \psi\sigma^\mu\bar{\chi} \\ \chi\sigma^\mu\bar{\sigma}^\nu\psi &= -\psi\sigma^\nu\bar{\sigma}^\mu\chi & (\chi\sigma^\mu\bar{\sigma}^\nu\psi)^+ &= \bar{\psi}\bar{\sigma}^\nu\sigma^\mu\bar{\chi}\end{aligned}\tag{2.4.65}$$

We end this list introducing matrices  $\sigma^{\mu\nu}$  and  $\bar{\sigma}^{\mu\nu}$  which are the generators of the Lorentz group in the spinor representation

•

$$\begin{aligned}\sigma^{\mu\nu}{}_\alpha{}^\beta &= \frac{1}{4}(\sigma_{\alpha\dot{\alpha}}{}^\mu\bar{\sigma}^{\nu\dot{\alpha}\beta} - \sigma_{\alpha\dot{\alpha}}{}^\nu\bar{\sigma}^{\mu\dot{\alpha}\beta}) \\ \bar{\sigma}^{\mu\nu\dot{\alpha}}{}_{\dot{\beta}} &= \frac{1}{4}(\bar{\sigma}^{\mu\dot{\alpha}\alpha}\sigma_{\alpha\dot{\beta}}{}^\nu - \bar{\sigma}^{\nu\dot{\alpha}\alpha}\sigma_{\alpha\dot{\beta}}{}^\mu)\end{aligned}\tag{2.4.66}$$

It is occasionally convenient to do a “fake” conversion of an undotted to a dotted index or vice versa using the fact that  $\sigma^0$  and  $\bar{\sigma}^0$  are just the identity matrix:

$$\psi^\alpha = (\bar{\psi}_{\dot{\beta}})^*(\bar{\sigma}^0)^{\dot{\beta}\alpha}, \quad \bar{\psi}^{\dot{\alpha}} = (\psi_\beta)^*(\sigma^0)^{\beta\dot{\alpha}}\tag{2.4.67}$$

## 2.5 The Coleman-Mandula No-Go theorem in brief

Adopting the following (reasonable) assumptions

1. finiteness of the number of particle types below any given mass
2. the existence of scattering at almost all energies (i.e. all energies except perhaps an isolated set)
3. analyticity of the  $S$  matrix

Coleman and Mandula proved in 1967 [1] that the most general Lie algebra of symmetry operators that commute with the  $S$  matrix, that take single-particle states into single-particle states and that act on multiparticle states as the direct sum of the action of single-particle states consist of the generators  $P_\mu$  and  $J_{\mu\nu}$  of the Poincaré group, plus ordinary internal symmetry generators that act on one-particle states with matrices that are diagonal in and independent of both momentum and spin.

Both the Poincaré algebra and the Lie algebras associated to internal symmetries imply *commutator relations*. Although historically supersymmetry arose in other way, we can see that it is natural today to note that a way to evade Coleman-Mandula theorem is to consider *anti-commutation relations* which would imply considering transformations that take bosons and fermions into each other. Supersymmetry is precisely such kind of superalgebra. This was realized in 1975, when Haag, Lopuszanski, and Sohnius [2] published their proof that weakening the assumptions of the ColemanMandula theorem by allowing both commuting and anticommuting symmetry generators, there is a nontrivial extension of the Poincar algebra, namely the supersymmetry algebra.

## 2.6 Graded algebras

Let us consider generators  $B_a$  and  $F_\alpha$  satisfying the following algebraic relations

$$\begin{aligned} [B_a, B_b] &= ic_{ab}{}^c B_c \\ \{F_\alpha, F_\beta\} &= \gamma_{\alpha\beta}{}^i B_i \\ [F_\alpha, B_a] &= is_{\alpha a}{}^\beta F_\beta \end{aligned} \tag{2.6.68}$$

where

$$\begin{aligned} [B_a, B_b] &= B_a B_b - B_b B_a \\ \{F_\alpha, F_\beta\} &= F_\alpha F_\beta + F_\beta F_\alpha \end{aligned} \tag{2.6.69}$$

Relations (2.6.68) define a graded algebra in which the structure constants satisfy

$$c_{ab}{}^c = -c_{ba}{}^c, \quad \gamma_{\alpha\beta}{}^i = \gamma_{\beta\alpha}{}^i \tag{2.6.70}$$

A *graded Jacobi identity* holds:

$$[[G_1, G_2], G_3] + \text{graded cyclic} = 0 \tag{2.6.71}$$

where  $G = (B, F)$  and “graded cyclic” coincides with the cyclic sum except when two anticommuting generators  $F$  are interchanged

$$F_\alpha F_\beta B_a + \text{graded cyclic} = F_\alpha F_\beta B_a + B_a F_\alpha F_\beta - F_\beta B_a F_\alpha \quad (2.6.72)$$

The  $B_a$  are *bosonic* generators while the  $F_\alpha$  are fermionic ones.

## 2.7 The $N = 1$ supersymmetry algebra

The supersymmetry algebra is a graded algebra like the one defined by eqs.(2.6.68). The  $F$  generators will be the supersymmetry generators, that from here on will be called  $Q_\alpha$  and  $\bar{Q}_{\dot{\alpha}}$ , their hermitian conjugate. Generators  $Q_\alpha$  are chosen so that they transform in the  $(1/2, 0)$  representation of the Lorentz group while  $\bar{Q}_{\dot{\alpha}}$  transforms in the  $(0, 1/2)$ . They are hence fermionic generators acting in a Hilbert space which one assumes has a positive definite metric. The supersymmetry algebra takes the following form

$$[P^\mu, P^\nu] = 0 \quad (2.7.73)$$

$$i[J^{\mu\nu}, J^{\rho\sigma}] = g^{\nu\rho} J^{\mu\sigma} - g^{\mu\rho} J^{\nu\sigma} - g^{\sigma\mu} J^{\rho\nu} + g^{\sigma\nu} J^{\rho\mu} \quad (2.7.74)$$

$$i[P^\mu, J^{\rho\sigma}] = g^{\mu\rho} P^\sigma - g^{\mu\sigma} P^\rho \quad (2.7.75)$$

$$\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = 2\sigma_{\alpha\dot{\beta}}^\mu P_\mu \quad (2.7.76)$$

$$\{Q_\alpha, Q_\beta\} = \{\bar{Q}_{\dot{\alpha}}, \bar{Q}_{\dot{\beta}}\} = 0 \quad (2.7.77)$$

$$[P_\mu, Q_\alpha] = [P_\mu, \bar{Q}_{\dot{\beta}}] = 0 \quad (2.7.78)$$

$$[Q_\alpha, J_{\mu\nu}] = \frac{1}{2}(\sigma_{\mu\nu})_\alpha^\beta Q_\beta \quad (2.7.79)$$

$$[\bar{Q}_{\dot{\alpha}}, J_{\mu\nu}] = -\frac{1}{2}\bar{Q}_{\dot{\beta}}(\sigma_{\mu\nu})^{\dot{\beta}}_{\dot{\alpha}} \quad (2.7.80)$$

In view of the fact that the  $Q$ ,  $\bar{Q}$  generators transform in the  $(1/2, 0)$  and the  $(0, 1/2)$  representation respectively, the  $\{Q_\alpha, \bar{Q}_{\dot{\beta}}\}$  anticommutator should transform in the  $(1/2, 1/2)$ , i.e. as a 4-vector, in this case  $P_\mu$ , as explicitly seen in (5.3.50). In view of what we have learned concerning Weyl fermions, we can write (see (2.4.58))

$$(Q_\alpha)^\dagger = \bar{Q}_{\dot{\alpha}} \quad (2.7.81)$$

## 2.8 Representations of the supersymmetry algebra

One defines a supermultiplet as a set of quantum states (or, in a different context, of quantum fields) which can be transformed into one another by one or more supersymmetry transformations.

tions. This is exactly analogous to the concept of “multiplet” known from atomic, nuclear and elementary particle physics where, e.g., the proton and the neutron form an iso-spin doublet and can be transformed into each other by an iso-spin rotation.

We have already noticed that  $P^2 = P^\mu P_\mu$  is a Casimir operator so that irreducible representations of the supersymmetry algebra are of equal mass: (i.e. fermions and bosons in each multiplet have all the same mass).

One can also prove that every representation of the supersymmetry algebra contains an equal number of fermions and bosons. To see this, let us define a fermion number operator  $N_F$  such that

$$\begin{aligned} N_F |\text{fermion}\rangle &= |\text{fermion}\rangle \\ N_F |\text{boson}\rangle &= 0 \end{aligned} \quad (2.8.82)$$

Then,  $(-1)^{N_F}$  has eigenvalue  $+1$  for bosons and  $-1$  for fermions. Note that  $(-1)^{N_F}$  anti-commutes with  $Q_\alpha$

$$\{(-1)^{N_F}, Q_\alpha\} = (-1)^{N_F} Q_\alpha + Q_\alpha (-1)^{N_F} = 0 \quad (2.8.83)$$

For any finite-dimensional representation of the algebra (such that the trace is well-defined), we then find

$$\begin{aligned} \text{Tr} \left( (-1)^{N_F} \{Q_\alpha, \bar{Q}_{\dot{\beta}}\} \right) &= \text{Tr} \left( (-1)^{N_F} (Q_\alpha \bar{Q}_{\dot{\beta}} + \bar{Q}_{\dot{\beta}} Q_\alpha) \right) \\ &= \text{Tr} \left( -Q_\alpha (-1)^{N_F} \bar{Q}_{\dot{\beta}} + Q_\alpha (-1)^{N_F} \bar{Q}_{\dot{\beta}} \right) = 0 \end{aligned} \quad (2.8.84)$$

Now, in view of the anticommutation relation (5.3.50) we can write

$$2\sigma^\mu_{\alpha\dot{\beta}} \text{Tr} \left( (-1)^{N_F} P_\mu \right) = 0 \quad (2.8.85)$$

so that for fixed non-zero momentum  $P_\mu$  this reduces to

$$\text{Tr}(-1)^{N_F} = 0 \quad (2.8.86)$$

which means that supersymmetry representations contain equal number of fermions and bosons.

Being the supersymmetry algebra a graded algebra, the vector space on which a representation acts should be graded. What we have learnt is that:

- The fermionic and bosonic subspaces of the representation space of a supersymmetry algebra have the same dimensions.

## 2.9 Irreducible representations of massive and massless particle states

### 2.9.1 The Lorentz group, helicity and massless particles

According to (2.7.73), the components of the energy-momentum 4-vector  $P^\mu$  all commute with each other so that it is natural to express physical state-vectors in terms of eigenvectors of  $P^\mu$ ,

$$P^\mu \Psi_{p,\sigma} = p^\mu \Psi_{p,\sigma} \quad (2.9.87)$$

where  $\sigma$  labels all other degrees of freedom. We shall consider the relevant case (see below) in which  $\sigma$  is discrete.

Let us specialize to the zero mass case. We have seen (see table (2.2.24)) that the normalized standard 4-momentum for this cases is  $(k^\mu) = (1, 0, 0, 1)$ . One can then show that the states  $\Psi_{k,\sigma}$  are just distinguished by the eigenvalues of  $J_3$ ,

$$J_3 \Psi_{k,\sigma} = \sigma \Psi_{k,\sigma} \quad (2.9.88)$$

(The proof can be found in [5]) Since the momentum  $k$  is in the 3-direction, then  $\sigma$  is just the eigenvalue of the component of the angular momentum along the direction of motion. Such projection is called the helicity of the state

$$\lambda = \vec{J} \cdot \check{k} \quad (2.9.89)$$

where

$$\check{k} = \frac{\vec{k}}{|\vec{k}|} \quad (2.9.90)$$

is the direction of the (standard) momentum of the particle.

Because the eigenvalues of  $J_3$  with respect to an axis 3 have discrete values, the eigenvalues of helicity are also discrete. For a particle of angular momentum  $j_3$  the eigenvalues of helicity are  $\lambda = j_3, j_3 - 1, \dots, -j_3$ . Note that the helicity can also be written as  $\lambda = \vec{S} \cdot \check{k}$ , with  $\vec{S}$  the spin, since  $\vec{L} \cdot \vec{k} = 0$ . Of course the intrinsic spin of a massless particle cannot be measured separately from its orbital angular momentum since the particle cannot be put at rest in any Lorentz frame.

Although helicity can be defined both for massive and massless particles, it is in the massless case where it finds its true relevance, as first noted by Jacob and Wick [6]. In fact for massless particles (as for example photons) it is an invariant quantity (a pseudoscalar) in contrast with the case of massive particles.

We know that photons are bosons carrying momentum  $\vec{k}$  and energy  $E = |\vec{k}|$  (we take  $\hbar = c = 1$ ). Moreover, being the field representing photons a vector ( $A_\mu$  with only 3 independent componets), it is natural to assign the photon a spin 1 and then we have for the photon

$$\lambda = \pm 1 \quad (2.9.91)$$

One eliminates the possibility of  $\lambda = 0$  because having zero mass, the photon cannot be brought to a rest reference frame to measure its spin for zero orbital momentum.

In summary, one particle states for massless particles can be seen to be labeled by their momentum  $p$  and their helicity  $\lambda$ ,

$$\begin{aligned} \hat{P}^\mu \Psi_{k,\lambda} &= \hbar k^\mu \Psi_{k,\lambda} \\ \hat{\lambda} \Psi_{k,\lambda} &= \lambda \Psi_{k,\lambda} \end{aligned} \quad (2.9.92)$$

Under a Lorentz transformation the massless state  $\Psi_{k,\lambda}$  will transform according to

$$U(\Lambda) \Psi_{k,\lambda} = \sqrt{\frac{(\Lambda k^0)}{k^0}} \exp(ih\theta(\Lambda, k)) \Psi_{k,\lambda} \quad (2.9.93)$$

where  $\theta$  is a certain rotation around the 3-axis, depending on the Lorentz transformation and on  $k$ .

A general one-photon state can be written as

$$\Psi_{k,h} = \alpha_+ \Psi_{k,+1} + \alpha_- \Psi_{k,-1} \quad (2.9.94)$$

where

$$|\alpha_+|^2 + |\alpha_-|^2 = 1 \quad (2.9.95)$$

The generic case is one of *elliptic* polarization where  $\alpha_+ \neq \alpha_-$  and they are unequal. *Circular* polarization corresponds to the case where either  $\alpha_+$  or  $\alpha_-$  vanish and *linear* polarization corresponds to the case in which  $\alpha_+ = \alpha_-^*$ . We then see that even the helicity of a massless state is a Lorentz invariant, the state itself is not because of the helicity dependent phase in (2.9.93)

## 2.9.2 Space inversion and time reversal

Space inversion  $\mathcal{P}$  and time reversal  $\mathcal{T}$  are discrete transformations defined as

$$\mathcal{P}^\mu_\nu = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad \mathcal{T}^\mu_\nu = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (2.9.96)$$

such that

$$\begin{aligned}\mathcal{P}^\mu_\nu x^\nu &= y^\mu, & (y^\mu) &= (x^0, -x^1, -x^2, -x^3) \\ \mathcal{T}^\mu_\nu x^\nu &= z^\mu, & (z^\mu) &= (-x^0, x^1, x^2, x^3)\end{aligned}\tag{2.9.97}$$

Concerning the Poincaré generators, their transformation under space inversion and time reversal are implemented by operators  $P$  and  $T$  so that

$$\begin{aligned}P\vec{J}P^{-1} &= +\vec{J} \\ P\vec{P}P^{-1} &= -\vec{P} \\ T\vec{J}T^{-1} &= -\vec{J} \\ T\vec{P}T^{-1} &= -\vec{P}\end{aligned}\tag{2.9.98}$$

while for  $P^0 \equiv H$  one has

$$PHP^{-1} = THT^{-1} = H\tag{2.9.99}$$

Concerning the “boost operators”  $\vec{K} = (J^{10}, J^{20}, J^{30})$ , they change according to

$$\begin{aligned}P\vec{K}P^{-1} &= -\vec{K} \\ T\vec{K}T^{-1} &= +\vec{K}\end{aligned}\tag{2.9.100}$$

Acting on a massless state  $\Psi_{k,\sigma}$  with momentum eigenvalue  $(k^\mu) = (E, 0, 0, E)$  and  $J_3$  eigenvalue  $\sigma$ , we know that

$$\begin{aligned}\mathcal{P}^\mu_\nu k^\nu &= (E, 0, 0, -E) \\ PJ_3P^{-1} &= J_3 \xrightarrow{\mathcal{P}} \sigma \rightarrow \sigma\end{aligned}\tag{2.9.101}$$

and then the helicity  $\lambda = k_3 J_3$  changes sign. Then, if there exists space inversion symmetry a particle with helicity  $\lambda$  should be accompanied by one of helicity  $-\lambda$ .

Of course, in 1956-1957 it became understood that  $\mathcal{P}$  symmetry is not conserved by weak interactions such as those that produce nuclear beta decay. Later (in 1964) it was discovered that  $\mathcal{T}$  was also not conserved. One should then keep in mind that to demand for  $\mathcal{P}$  or  $\mathcal{T}$  invariance is only an approximation.

### 2.9.3 Representation of massive particles

Consider 1-particle states with  $p^2 = -m^2$ . Choosing the rest frame,  $(p^\mu) = (-m, 0, 0, 0)$  (the standard momentum, see the table (2.2.24)) the algebra (5.3.50)-(2.7.77) takes the simple

form

$$\begin{aligned}\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} &= 2m\delta_{\alpha\dot{\beta}} \\ \{Q_\alpha, Q_\beta\} &= \{\bar{Q}_{\dot{\alpha}}, \bar{Q}_{\dot{\beta}}\} = 0\end{aligned}\tag{2.9.102}$$

which can be seen to coincide with a creation-annihilation algebra (apart from a normalization  $1/\sqrt{2m}$ ):

$$\begin{aligned}a_\alpha &= \frac{1}{\sqrt{2m}}Q_\alpha \\ a_\alpha^\dagger &= \frac{1}{\sqrt{2m}}\bar{Q}_{\dot{\alpha}}\end{aligned}\tag{2.9.103}$$

$$\{a_\alpha, a_\beta^\dagger\} = \delta_\alpha^\beta, \quad \{a_\alpha, a_\beta\} = \{a_{\dot{\alpha}}, a_{\dot{\beta}}\} = 0\tag{2.9.104}$$

One can then use the Wigner method of induced representations which starts by defining a ‘‘Clifford’’ vacuum  $|m, J, J_3\rangle$  where  $J, J_3$  are the eigenvalues of the angular momentum (Recall that the little group is in this case  $SO(3)$ ). Such state represents one particle at rest and satisfies

$$Q_\alpha|m, J, J_3\rangle = 0\tag{2.9.105}$$

Now, in contrast to the usual case, being  $P^2$  a Casimir, one has

$$P^2|m, J, J_3\rangle = -m^2|m, J, J_3\rangle\tag{2.9.106}$$

One can build up a 4-dimensional Fock space with positive metric by applying the creation operator  $1/\sqrt{2m}\bar{Q}_{\dot{\alpha}}$ . The resulting (fundamental) representation consist of states of the form

$$|m, J, J_3; n_1, n_2\rangle = \bar{Q}_1^{n_1}\bar{Q}_2^{n_2}|m, J, J_3\rangle\tag{2.9.107}$$

in which  $(n_1, n_2) = (0, 0), (0, 1), (1, 0), (1, 1)$ . These states span an irreducible  $4(2J + 1)$  irreducible of the  $Q_\alpha, \bar{Q}_{\dot{\alpha}}$  algebra representation. The representations of lower dimensionality are those with  $J = 0, 1/2, 1, 3/2$ .

For  $J = 0$  the particle content is

|                                                                                    |                                                                |           |
|------------------------------------------------------------------------------------|----------------------------------------------------------------|-----------|
| $ m, J, J_3\rangle$                                                                | state of spin 0, scalar                                        |           |
| $\frac{1}{\sqrt{2m}}\bar{Q}_{\dot{\alpha}} m, J, J_3\rangle$                       | states of spin $\frac{1}{2}$ with projection $\pm \frac{1}{2}$ | (2.9.108) |
| $\frac{1}{\sqrt{22m}}\bar{Q}_{\dot{\alpha}}\bar{Q}_{\dot{\beta}} m, J, J_3\rangle$ | state of spin 0, pseudoscalar                                  |           |



### 2.9.4 Representation of massless particles

In this case  $P^2 = 0$  and one should rotate the massless one-particle state into a light-like reference frame,  $P_\mu = (-E, 0, 0, E)$  in which the movement is in the  $z$ -direction. Such a  $P_\mu$  is nothing but the standard 4-momentum for the case of massless particles (see table (2.2.24)).

The space-time properties of the state are then determined by its energy and its helicity  $\lambda$ . As we have seen, the helicity is the projection of the spin onto the direction of motion, i.e. the eigenvalue of the operator

$$\lambda = \frac{1}{E} \vec{J} \cdot \vec{P} \quad (2.9.109)$$

From the definition (2.1.11) of the Pauli-Lubanski vector one has that

$$W_0 = \vec{J} \cdot \vec{P} = E\lambda \quad (2.9.110)$$

Then, on massless eigenstates  $|E, \lambda\rangle$  of a given energy and helicity one has

$$W_0|E, \lambda\rangle = \lambda E|E, \lambda\rangle \quad (2.9.111)$$

or, from Lorentz covariance

$$W_\mu|E, \lambda\rangle = \lambda P_\mu|E, \lambda\rangle \quad (2.9.112)$$

Let us consider

$$W_0 Q_\alpha|E, \lambda\rangle = Q_\alpha W_0|E, \lambda_0\rangle + [W_0, Q_\alpha]|E, \lambda_0\rangle Q_\alpha|E, \lambda_0\rangle = E(\lambda I - \frac{1}{2}\sigma^3)_\alpha^\beta Q_\beta|E, \lambda_0\rangle \quad (2.9.113)$$

Using the explicit form of the Pauli matrix  $\sigma^3$  we see that  $Q_1$  lowers the helicity of the state by  $1/2$  while the  $Q_2$  raises it. A similar calculation shows that  $\bar{Q}_{\dot{\alpha}}$  raises the helicity by  $1/2$  for  $\dot{\alpha} = 1$  and lowers it for  $\dot{\beta} = 2$ .

The algebra (5.3.50)-(2.7.77) takes the form

$$\begin{aligned} \{Q_\alpha, \bar{Q}_{\dot{\beta}}\} &= 2 \begin{pmatrix} 2E & 0 \\ 0 & 0 \end{pmatrix} \\ \{Q_\alpha, Q_\beta\} &= \{\bar{Q}_{\dot{\alpha}}, \bar{Q}_{\dot{\beta}}\} = 0 \end{aligned} \quad (2.9.114)$$

Again one recognizes (after rescaling, this time with a  $1/2\sqrt{E}$  factor) a creation-annihilation algebra  $a, a^\dagger$

$$\begin{aligned} a &= \frac{1}{2\sqrt{E}} Q_1 \\ a^\dagger &= \frac{1}{2\sqrt{E}} \bar{Q}_{\dot{1}} \end{aligned} \quad (2.9.115)$$

so that

$$\{a, a^\dagger\} = 1, \quad \{a, a\} = \{a^\dagger, a^\dagger\} = 0 \quad (2.9.116)$$

From the point of view of the angular momentum algebra,  $a$  is a lowering operator and  $a^\dagger$  a raising operator for the helicity of massless states

$$J_3(a^\dagger|E, \lambda\rangle) = \left(\lambda - \frac{1}{2}\right)(a^\dagger|E, \lambda\rangle) \quad (2.9.117)$$

One then defines a Clifford ground state  $|E, \lambda_0\rangle$ , such that

$$a|E, \lambda_0\rangle = 0, \quad J_3|E, \lambda_0\rangle = \lambda_0|E, \lambda_0\rangle \quad (2.9.118)$$

and then one generates a state of helicity  $\lambda_0 - \frac{1}{2}$  by applying  $a^\dagger$

$$a^\dagger|E, \lambda_0\rangle = |E, \lambda_0 - \frac{1}{2}\rangle \quad (2.9.119)$$

For the case of angular momentum, we see that  $a$  is a raising operator and  $a^\dagger$  a lowering operator. If one considers the case  $\lambda_0 = 1$ , a photon, then there will be a state with  $\lambda = 1/2$ , a Weyl spinor.

In CPT invariant theories, the number of states must in general be doubled since CPT reverses the sign of the helicity

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## Problems

1. Prove eq.(2.1.10)

*Hint: use that  $U(\bar{\Lambda}, \bar{a})U(\Lambda, a) = U(\bar{\Lambda}\Lambda, \bar{\Lambda}a + \bar{a})$  and calculate  $U(\Lambda, a)U((1+\omega, \epsilon)U^{-1}(\Lambda, a))$ .*

2. Discuss the group  $ISO(2)$

# Chapter 3

## Superfields

15/5/10

### 3.1 Grassmann variables and superspace

Let us introduce 4 Grassmann variables  $\theta_\alpha, \bar{\theta}_{\dot{\alpha}}$  ( $\alpha, \dot{\alpha} = 1, 2$ ) through the anticommutation relations

$$\{\theta_\alpha, \theta_\beta\} = \{\bar{\theta}_{\dot{\alpha}}, \bar{\theta}_{\dot{\beta}}\} = \{\theta_\alpha, \bar{\theta}_{\dot{\beta}}\} = 0 \quad (3.1.1)$$

More explicitly,

$$\theta_1^2 = \theta_2^2 = 0, \quad \theta_1 \theta_2 = -\theta_2 \theta_1 \quad (3.1.2)$$

and the same for  $\bar{\theta}_{\dot{\alpha}}$ .

These anticommuting coordinates, together with the 4 ordinary space-time coordinates  $x_\mu$  ( $\mu = 0, 1, 2, 3$ ) form superspace.

The  $\theta_\alpha$  coordinates transform as 2-component Weyl spinors while the  $\bar{\theta}_{\dot{\alpha}}$  are their conjugates. Again we have  $\bar{\theta}^{\dot{\alpha}} = (\theta^\alpha)^*$ . All the rules we learnt in the previous class when we introduced Weyl fermions hold for  $\theta, \bar{\theta}$ . The following identities, which can be derived from eqs.(2.4.55) will be useful in what follows

$$\begin{aligned} \theta^\alpha \theta^\beta &= -\frac{1}{2} \varepsilon^{\alpha\beta} \theta\theta \\ \theta_\alpha \theta_\beta &= \frac{1}{2} \varepsilon_{\alpha\beta} \theta\theta \\ \bar{\theta}^{\dot{\alpha}} \bar{\theta}^{\dot{\beta}} &= \frac{1}{2} \varepsilon^{\dot{\alpha}\dot{\beta}} \bar{\theta}\bar{\theta} \\ \bar{\theta}_{\dot{\alpha}} \bar{\theta}_{\dot{\beta}} &= -\frac{1}{2} \varepsilon_{\dot{\alpha}\dot{\beta}} \bar{\theta}\bar{\theta} \end{aligned} \quad (3.1.3)$$

$$\theta\sigma^\mu\bar{\theta}\theta\sigma^\nu\bar{\theta} = -\frac{1}{2}\theta\theta\bar{\theta}\bar{\theta}g^{\mu\nu} \quad (3.1.4)$$

In view of these properties, any scalar complex function  $f = f(\theta)$  can be written as the following second order polynomial

$$f(\theta) = \phi + \theta_\alpha\psi^\alpha + \theta_\alpha\theta_\beta f^{\alpha\beta} = \phi + \theta\psi + \theta\theta F \quad (3.1.5)$$

where  $F = f^{\alpha\beta}\varepsilon_{\alpha\beta}/2$ . Here  $\phi$  and  $F$  are complex numbers and  $\psi$  a Weyl spinor.

Analogous formulæ hold for any function  $g = g(\bar{\theta})$ . Finally, functions  $h = h(\theta, \bar{\theta})$  have an expansion which can be written in the form

$$h(\theta, \bar{\theta}) = \phi + \theta\psi + \bar{\theta}\bar{\chi} + \theta\theta F + \bar{\theta}\bar{\theta}G + \theta\sigma^\mu\bar{\theta}v_\mu + \theta\theta\bar{\theta}\bar{\lambda} + \bar{\theta}\bar{\theta}\theta\eta + \theta\theta\bar{\theta}\bar{\theta}D \quad (3.1.6)$$

We shall also need the rules for integration over  $\theta_\alpha$  and  $\bar{\theta}_{\dot{\alpha}}$ . They are just the Berezin rules popularized through the definition of fermionic path-integrals:

$$\begin{aligned} \int d^2\theta &= \int d^2\theta \theta_\alpha = 0, & \int d^2\theta \theta\theta &= 1 \\ \int d^2\bar{\theta} &= \int d^2\bar{\theta} \bar{\theta}_\alpha = 0, & \int d^2\bar{\theta} \bar{\theta}\bar{\theta} &= 1 \end{aligned} \quad (3.1.7)$$

Note the following property which will be important when constructing supersymmetric actions

$$\begin{aligned} \int d^2\theta f(\theta) &= F \\ \int d^2\bar{\theta} d^2\theta h(\bar{\theta}, \theta) &= D \end{aligned} \quad (3.1.8)$$

That is, integrals over Grassmann variables select the highest term in the polynomial defining each function.

We can now consider a scalar complex superfield  $\Phi(x, \theta, \bar{\theta})$ , a function of all the superspace variables, spatial and Grassmann ones. It will have an expansion as that in (3.1.9) except that the coefficients in the  $\theta, \bar{\theta}$  polynomial are now functions of the space-time coordinates  $x_\mu$ ,

$$\Phi(x, \theta, \bar{\theta}) = \phi(x) + \theta\psi(x) + \bar{\theta}\bar{\chi}(x) + \theta\theta F(x) + \bar{\theta}\bar{\theta}G(x) + \theta\sigma^\mu\bar{\theta}v_\mu(x) + \theta\theta\bar{\theta}\bar{\lambda}(x) + \bar{\theta}\bar{\theta}\theta\eta(x) + \theta\theta\bar{\theta}\bar{\theta}D(x) \quad (3.1.9)$$

We shall call  $\Phi$  a scalar superfield. Its components  $\phi(x), F(x), G(x), D(x)$  are complex scalar fields.  $\psi(x)$  and  $\eta(x)$  Weyl spinors,  $\chi(x)$  and  $\lambda(x)$  conjugate Weyl spinors and  $v_\mu(x)$  a complex vector field.

## 3.2 The graded Lie group

We know that an element of the group of space-time translations when acting on a scalar field  $\phi(x)$  can be written as

$$T(a) = \exp(ia^\mu p_\mu) \quad (3.2.10)$$

with  $p_\mu = -i\partial_\mu$  the space-time translations generator and  $a^\mu$  the corresponding parameter of the transformation. We then have

$$\phi(x + a) = T(a)\phi(x) \quad (3.2.11)$$

Let us generalize this object so as to consider translations in superspace. The group elements that we are looking for can be guessed to be:

$$G(a, \epsilon, \bar{\epsilon}) = \exp(ia^\mu P_\mu + \epsilon Q + \bar{\epsilon} \bar{Q}) \quad (3.2.12)$$

where  $\epsilon$  and  $\bar{\epsilon}$  are the (anticommuting) parameters of the supersymmetry transformation and, as established previously (see eqs.(2.4.64))

$$\epsilon Q = \epsilon^\alpha Q_\alpha, \quad \bar{\epsilon} \bar{Q} = \bar{\epsilon}_{\dot{\alpha}} \bar{Q}^{\dot{\alpha}} \quad (3.2.13)$$

Using Baker-Campbell-Hausdorff formula

$$\exp A \exp B = \exp \left( A + B + \frac{1}{2}[A, B] + \dots \right) \quad (3.2.14)$$

and the supersymmetry algebra eqs.(2.7.73)-(2.7.80) one can easily calculate the product of two  $G$ 's since all higher commutators included in the ellipsis vanish leading to

$$G(0, \epsilon, \bar{\epsilon})G(a^\mu, \delta, \bar{\delta}) = G(a^\mu + i\delta\sigma^\mu\bar{\epsilon} - i\epsilon\sigma^\mu\bar{\delta}, \delta + \epsilon, \bar{\delta} + \bar{\epsilon}) \quad (3.2.15)$$

The multiplication of group elements induces a motion in the parameter space

$$(a^\mu, \delta, \bar{\delta}) \xrightarrow{G(0, \epsilon, \bar{\epsilon})} a^\mu + i\delta\sigma^\mu\bar{\epsilon} - i\epsilon\sigma^\mu\bar{\delta}, \delta + \epsilon, \bar{\delta} + \bar{\epsilon} \quad (3.2.16)$$

That is,  $Q$  produces a translation  $-i\epsilon\sigma^\mu\bar{\delta}$  in  $x^\mu$  space and a translation  $\epsilon$  in the anticommuting coordinates  $\theta_\alpha$ . Analogously  $\bar{Q}$  produces a translation  $i\delta\sigma^\mu\bar{\epsilon}$  in  $x^\mu$  space and a translation  $\bar{\epsilon}$  in the anticommuting coordinates  $\bar{\theta}_{\dot{\alpha}}$ . In analogy with ordinary translations, the translation in anticommuting coordinates  $\theta$  and  $\bar{\theta}$  can be easily seen to be produced by a derivative operator

$$\theta_\alpha \xrightarrow{\partial/\partial\theta^\alpha} \theta_\alpha + \epsilon_\alpha, \quad \bar{\theta}_{\dot{\alpha}} \xrightarrow{\partial/\partial\bar{\theta}^{\dot{\alpha}}} \bar{\theta}_{\dot{\alpha}} + \bar{\epsilon}_{\dot{\alpha}} \quad (3.2.17)$$

As for the space-time translations, they are of course generated by the  $\partial_\mu$  operator so that we can write

$$\epsilon Q + \bar{\epsilon} \bar{Q} = \epsilon^\alpha \left( \frac{\partial}{\partial \theta^\alpha} - i \sigma_{\alpha\dot{\alpha}}^\mu \bar{\theta}^{\dot{\alpha}} \partial_\mu \right) + \bar{\epsilon}_{\dot{\alpha}} \left( \frac{\partial}{\partial \bar{\theta}_{\dot{\alpha}}} - i \theta^\alpha \sigma_{\alpha\dot{\beta}}^\mu \bar{\epsilon}^{\dot{\beta}} \partial_\mu \right) \quad (3.2.18)$$

One has then a realization of the supersymmetry generators in terms of differential operators

$$Q_\alpha = \frac{\partial}{\partial \theta^\alpha} - i \sigma_{\alpha\dot{\alpha}}^\mu \bar{\theta}^{\dot{\alpha}} \partial_\mu \quad (3.2.19)$$

$$\bar{Q}^{\dot{\alpha}} = \frac{\partial}{\partial \bar{\theta}_{\dot{\alpha}}} - i \theta^\alpha \sigma_{\alpha\dot{\beta}}^\mu \bar{\epsilon}^{\dot{\beta}} \partial_\mu \quad (3.2.20)$$

or, lowering the index in the second line through multiplication by an  $\varepsilon_{\dot{\alpha}'\dot{\alpha}}$  and noting that

$$\varepsilon_{\dot{\alpha}\dot{\beta}} \frac{\partial}{\partial \bar{\theta}_{\dot{\beta}}} = - \frac{\partial}{\partial \bar{\theta}^{\dot{\alpha}}} \quad (3.2.21)$$

one can write

$$Q_\alpha = \frac{\partial}{\partial \theta^\alpha} - i \sigma_{\alpha\dot{\alpha}}^\mu \bar{\theta}^{\dot{\alpha}} \partial_\mu \quad (3.2.22)$$

$$\bar{Q}_{\dot{\alpha}} = - \frac{\partial}{\partial \bar{\theta}^{\dot{\alpha}}} + i \theta^\alpha \sigma_{\alpha\dot{\alpha}}^\mu \partial_\mu \quad (3.2.23)$$

which can be seen to satisfy the supersymmetry algebra

$$\begin{aligned} \{Q_\alpha, \bar{Q}_{\dot{\alpha}}\} &= 2i \sigma_{\alpha\dot{\alpha}}^\mu \partial_\mu \\ \{Q_\alpha, Q_\beta\} &= \{\bar{Q}_{\dot{\alpha}}, \bar{Q}_{\dot{\beta}}\} = 0 \end{aligned} \quad (3.2.24)$$

An infinitesimal supersymmetry transformation acting on the superfield (3.1.9) can be now calculated using the differential equations (3.2.22)-(3.2.23)

$$\delta_{\epsilon, \bar{\epsilon}} \Phi = (\epsilon Q + \bar{\epsilon} \bar{Q}) \Phi(\theta, \bar{\theta}) \quad (3.2.25)$$

Writing

$$\begin{aligned} \delta_{\epsilon, \bar{\epsilon}} \Phi &= \delta\phi(x) + \theta\delta\psi(x) + \bar{\theta}\delta\bar{\chi}(x) + \theta\theta\delta F(x) + \bar{\theta}\bar{\theta}\delta G(x) + \theta\sigma^\mu\bar{\theta}\delta v_\mu(x) + \theta\theta\bar{\theta}\bar{\theta}\delta\lambda(x) \\ &\quad + \bar{\theta}\bar{\theta}\theta\delta\eta(x) + \theta\theta\bar{\theta}\bar{\theta}\delta D(x) \end{aligned} \quad (3.2.26)$$



and matching powers of  $\theta$  and  $\bar{\theta}$  one can get the transformation laws for each component field in the superfield. For example, from the  $0^{th}$  powers in  $\theta, \bar{\theta}$  resulting from derivatives with respect to these variables

$$\delta\phi = \epsilon\psi + \bar{\epsilon}\bar{\chi} \quad (3.2.27)$$

We shall not complete the list of transformation laws because a general superfield like (3.1.9) has much more components than necessary. Note that in view of relations (3.2.24)  $Q$  and  $\bar{Q}$  have mass dimension  $1/2$ . Then, fields of mass dimensions  $d$  should transform into fields of mass dimensions  $d + 1/2$ . For example, if the scalar field has mass dimension 1 then the fermion  $\psi$  will have mass dimension  $3/2$ .

To see this in detail let us stress that since  $Q$  and  $\bar{Q}$  are linear differential operators that superfields form linear representations of the supersymmetry algebra. Hence, linear combinations of superfields and product of superfields are superfields and also space-time derivatives or complex conjugation of superfields also yield to superfields.

Suppose we want to get from (3.1.9) to a superfield corresponding to an irreducible representation by applying some constraint. That is, we are looking for operators that reduce the number of degrees of freedom (projects any function in superspace into a sub-superspace). Constraints should (anti)commute with the generators  $Q_\alpha$  and  $\bar{Q}_{\dot{\alpha}}$  so that they are preserved during the supersymmetry transformations (3.2.25).

Although as we stated, space-time derivatives of superfields yield to superfields, derivatives with respect to  $\theta$  or  $\bar{\theta}$  do not. So, if the constraint were to include such derivatives, one should consider appropriate combinations. Consider in particular the following differential operators (alternatively called covariant deperivatives or superderivatives)

$$D_\alpha = \frac{\partial}{\partial\theta^\alpha} + i\sigma_{\alpha\dot{\alpha}}^\mu \bar{\theta}^{\dot{\alpha}} \partial_\mu \quad (3.2.28)$$

$$\bar{D}_{\dot{\alpha}} = -\frac{\partial}{\partial\bar{\theta}^{\dot{\alpha}}} - i\theta^\alpha \sigma_{\alpha\dot{\alpha}}^\mu \partial_\mu \quad (3.2.29)$$

Note that the change in relative signs in  $D_\alpha$  and  $\bar{D}_{\dot{\alpha}}$  with respect to  $Q_\alpha$  and  $\bar{Q}_{\dot{\alpha}}$  guarantees that  $D$  and  $Q$  anticommute

$$\{D_\alpha, Q_\beta\} = \{\bar{D}_{\dot{\alpha}}, Q_\beta\} = \{\bar{D}_{\dot{\alpha}}, Q_\beta\} = \{\bar{D}_{\dot{\alpha}}, \bar{Q}_{\dot{\beta}}\} = 0 \quad (3.2.30)$$

and satisfy the following anticommutation relations

$$\begin{aligned} \{D_\alpha, \bar{D}_{\dot{\alpha}}\} &= -2i\sigma_{\alpha\dot{\alpha}}^\mu \partial_\mu \\ \{D_\alpha, D_\beta\} &= \{\bar{D}_{\dot{\alpha}}, \bar{D}_{\dot{\beta}}\} = 0 \end{aligned} \quad (3.2.31)$$

which again differ from the  $Q, \bar{Q}$  algebra in the sign on the sign in the first line.

Relations (3.2.30) imply that  $D$  and  $\bar{D}$  transform covariantly under supersymmetry since they guarantee that (for simplicity we take  $\Phi = \Phi(x, \theta)$ )

$$D_\alpha \delta \Phi(x; \theta) = D_\alpha \epsilon^\beta Q_\beta \Phi = -\epsilon^\beta D_\alpha Q_\beta \phi = \epsilon^\beta Q_\beta D_\alpha \Phi = \epsilon Q D_\alpha \Phi = \delta (D_\alpha \Phi) \quad (3.2.32)$$

### 3.3 Chiral fields

We are now in conditions to apply a constraint to a general superfield  $\Phi(x, \theta, \bar{\theta})$  of the form (3.1.9) to one in an irreducible representation. Indeed, consider the constraint

$$\bar{D}_{\dot{\alpha}} \Phi = 0 \quad (3.3.33)$$

To understand its meaning, note that

$$\bar{D}_{\dot{\alpha}}(x^\mu + i\theta\sigma^\mu\bar{\theta}) = 0, \quad \text{and} \quad \bar{D}_{\dot{\alpha}}\theta = 0 \quad (3.3.34)$$

In view of this, consider the change of variables

$$x^\mu \rightarrow y^\mu = x^\mu + i\theta\sigma^\mu\bar{\theta}, \quad (3.3.35)$$

$$\theta \rightarrow \theta, \quad \bar{\theta} \rightarrow \bar{\theta} \quad (3.3.36)$$

In terms of the new variables the covariant derivatives read

$$D_\alpha = \frac{\partial}{\partial \theta^\alpha} + 2i\sigma_{\alpha\dot{\alpha}}^\mu \bar{\theta}^{\dot{\alpha}} \partial_\mu \quad (3.3.37)$$

$$\bar{D}_{\dot{\alpha}} = -\frac{\partial}{\partial \bar{\theta}^{\dot{\alpha}}} \quad (3.3.38)$$

and then it is clear that condition (3.3.33) constraints  $\Phi$  not to depend on  $\bar{\theta}$  but just on  $y$  and  $\theta$

$$\Phi(y, \theta) = \phi(y) + \sqrt{2}\theta\psi(y) + \theta\theta F(y) \quad (3.3.39)$$

satisfies the constraint (3.3.33) and this is its more general solution.

A field satisfying condition (3.3.33) is called a chiral field. Note that the fermion content of  $\Phi$  consists of 1 Weyl fermion (i.e., a spinor with one chirality, left-handed in the present case) and that is the origin of the name one uses (although Wess and Zumino called them scalar superfields when they introduced these fields in [1])

The multiplet  $\Phi = (\phi, \psi, F)$  is called the scalar or chiral multiplet. If  $\phi$  is assigned a mass dimension 1 then  $\psi$  has dimension 3/2 and  $F$  has dimension 2. We see that  $F$  transforms into a spacetime derivative. This will always be the case of the component of

the highest dimension in a multiplet and will be a very important property for constructing supersymmetric invariant Lagrangians.

Let us count the number of degrees of freedom (each real unconstrained component counting for 1). Concerning bosonic fields, we have 2 degrees of freedom associated with  $\phi$  and 2 with  $F$ , so, a total of 4. The Weyl fermion has 2 components,  $\psi_1$  and  $\psi_2$ , each one with real and imaginary parts so there are also 4 degrees of freedom for fermions. The multiplet has then  $4 + 4$  degrees of freedom and this is the smallest possible number in 4 space-time dimensions since each multiplet should have at least 1 fermion and a fermion with the smallest number of components (Weyl or Majorana) has necessarily 4 degrees of freedom. The chiral multiplet that we have found is therefore irreducible.

By expanding each superfield component in (3.3.39) one can easily write the chiral field in terms of the original variables,

$$\Phi = \phi(x) + i\theta\sigma^\mu\bar{\theta}\partial_\mu\phi(x) + \frac{1}{4}\theta\theta\bar{\theta}\bar{\theta}\square\phi(x) + \sqrt{2}\theta\psi(x) - \frac{i}{\sqrt{2}}\theta\theta\partial_\mu\psi(x)\sigma^\mu\bar{\theta} + \theta\theta F(x) \quad (3.3.40)$$

Using (3.2.25) one can derive the change of the component fields of a chiral field under a supersymmetry transformation. To do this it is easier to first write  $Q, \bar{Q}$  in terms of  $y$ . First, observe that

$$\begin{aligned} Q_\alpha y^\mu &= \bar{Q}_{\dot{\alpha}} \bar{y}^\mu = 0 \\ Q_\alpha \bar{y}^\mu &= -2i(\sigma^\mu \bar{\theta})_{\dot{\alpha}} \\ \bar{Q}_{\dot{\alpha}} y^\mu &= 2i(\theta \sigma^\mu)_{\dot{\alpha}} \end{aligned} \quad (3.3.41)$$

Then, acting on general superfields  $\Phi(y, \theta, \bar{\theta})$  where we have made the change  $x \rightarrow y$  one can prove that

$$\begin{aligned} Q_\alpha \Phi(y, \theta, \bar{\theta}) &= -i \frac{\partial}{\partial \theta^\alpha} \Phi(y, \theta, \bar{\theta}) \\ \bar{Q}_{\dot{\alpha}} \Phi(y, \theta, \bar{\theta}) &= -i \left( -\frac{\partial}{\partial \bar{\theta}^{\dot{\alpha}}} - 2i(\theta \sigma^\mu)_{\dot{\alpha}} \frac{\partial}{\partial y^\mu} \right) \Phi(y, \theta, \bar{\theta}) \end{aligned} \quad (3.3.42)$$

Analogously, acting on general superfields  $\Phi(\bar{y}, \theta, \bar{\theta})$  where we have made the change  $x^\mu \rightarrow y^{\mu\dagger} = x^\mu - i\theta\sigma^\mu\bar{\theta}$  one ends with

$$\begin{aligned} Q_\alpha \Phi(y^\dagger, \theta, \bar{\theta}) &= -i \left( -\frac{\partial}{\partial \theta^\alpha} + 2i(\sigma^\mu \bar{\theta})_\alpha \frac{\partial}{\partial y^{\mu\dagger}} \right) \\ \bar{Q}_{\dot{\alpha}} \Phi(y^\dagger, \theta, \bar{\theta}) &= -i \frac{\partial}{\partial \bar{\theta}^{\dot{\alpha}}} \Phi(y^\dagger, \theta, \bar{\theta}) \end{aligned} \quad (3.3.43)$$

Applying eq.(3.3.42) to a chiral field one easily gets

$$\begin{aligned}\delta\phi &= \sqrt{2}\epsilon\psi, & \delta\phi^* &= \sqrt{2}\bar{\psi}\bar{\epsilon} \\ \delta\psi &= i\sqrt{2}\sigma^\mu\bar{\epsilon}\partial_\mu\phi + \sqrt{2}\epsilon F, & \delta\bar{\psi} &= -i\sqrt{2}\epsilon\sigma^\mu\partial_\mu\phi^* + \sqrt{2}\bar{\epsilon}F^* \\ \delta F &= i\sqrt{2}\bar{\epsilon}\bar{\sigma}^\mu\partial_\mu\psi, & \delta F^* &= -i\sqrt{2}\partial_\mu\bar{\psi}\bar{\sigma}^\mu\epsilon\end{aligned}\tag{3.3.44}$$

Note that derivatives are taken with respect to  $y$ . The point is that since  $\theta$  and  $y$  are independent variables, when working with components (after integrating by parts) one can easily change  $y \rightarrow x$ .

Given (3.3.44) we can consider two consecutive supersymmetry transformations, say with transformation parameters  $\epsilon_1$  and  $\epsilon_2$ . For example, take  $\delta_1\delta_2\phi$

$$\delta_1\delta_2\phi = \delta_1\sqrt{2}\epsilon_2\psi = -\sqrt{2}\epsilon_2\delta_1\psi = -2i\epsilon_2\sigma^\mu\bar{\epsilon}_1\partial_\mu\phi - 2\epsilon_2\epsilon_1F\tag{3.3.45}$$

so that we can write

$$[\delta_1, \delta_2]\phi = -2i(\epsilon_2\sigma^\mu\bar{\epsilon}_1 - \epsilon_1\sigma^\mu\bar{\epsilon}_2)\partial_\mu\phi\tag{3.3.46}$$

Note that this result implies that a trivial representation, that is, one where  $\delta\phi = 0$  is forbidden since this would imply  $\partial_\mu\phi = 0$ , that is, a constant scalar field. We conclude that

- The only trivial representation of supersymmetry is given by constant fields

We can construct an antichiral field  $\Phi^\dagger$  by imposing on it, instead of condition (3.3.33)

$$D_{\dot{\alpha}}\Phi^\dagger = 0\tag{3.3.47}$$

The natural variable to work with will be, instead of  $y^\mu$  as defined in (3.3.39), a variable  $y^{\dagger\mu}$ ,

$$y^{\dagger\mu} = x^\mu - i\theta\sigma^\mu\bar{\theta}\tag{3.3.48}$$

which is nothing but the hermitian conjugate of  $y^\mu$  ( $i \rightarrow -i; \theta \rightarrow \bar{\theta}; \bar{\theta} \rightarrow \theta, \psi \rightarrow \bar{\psi}$ , etc). Its power series expansion in terms of  $\theta$  and  $\bar{\theta}$  is obtained from (3.3.40) by hermitian conjugation

$$\begin{aligned}\Phi^\dagger &= \phi^*(y^\dagger) + \sqrt{2}\bar{\theta}\bar{\psi} + \bar{\theta}\bar{\theta}F^*(y^\dagger) \\ &= \phi^*(x) + i\theta\sigma^\mu\bar{\theta}\partial_\mu\phi^*(x) + \frac{1}{4}\theta\theta\bar{\theta}\bar{\theta}\square\phi^*(x) + \sqrt{2}\bar{\theta}\bar{\psi}(x) + \frac{i}{\sqrt{2}}\bar{\theta}\bar{\theta}\partial_\mu\sigma^\mu\bar{\psi}(x) + \bar{\theta}\bar{\theta}F^*(x)\end{aligned}$$

The antichiral multiplet  $\Phi^\dagger = (\phi^*, \bar{\psi}, F^*)$  contains the same  $4 + 4$  real field components as the chiral one.

We see that the bosonic component fields of the antichiral field are just the complex conjugate of those in the chiral field. As for the fermionic component, our conventions are such that

$$\bar{\theta}\bar{\psi} = \bar{\psi}\bar{\theta} = \bar{\psi}_{\dot{\alpha}}\theta^{\dot{\alpha}} = (\theta^\alpha\psi_\alpha)^\dagger = (\theta\psi)^\dagger\tag{3.3.49}$$

That is, the fermionic component of the antichiral field appear as the hermitian conjugate of the chiral one.

Further chiral multiplets can be obtained by allowing each  $\Phi$  or  $\Phi^\dagger$  to have additional Lorentz indices. Such multiplets are not irreducible. One should note that if one adds a spinor index the statistic of each component is inverted.

The product of chiral superfields is again a chiral superfield:

$$\begin{aligned}
\Phi_1 \Phi_2 &= \left( \phi_1(y) + \sqrt{2}\theta\psi_1(y) + \theta\theta F_1(y) \right) \left( \phi_2(y) + \sqrt{2}\theta\psi_2(y) + \theta\theta F_2(y) \right) \\
&= \phi_1(y)\phi_2(y) + \sqrt{2}\theta (\psi_1(y)\phi_2(y) + \psi_2(y)\phi_1(y)) + \theta\theta (\phi_1(y)F_2(y) + F_1(y)\phi_2(y) \\
&\quad - \psi_1(y)\psi_2(y)) \\
\Phi_1 \Phi_2 \Phi_3 &= \phi_1(y)\phi_2(y)\phi_3(y) + \sqrt{2}\theta (\psi_1\phi_2\phi_3 + \psi_2\phi_3\phi_1 + \psi_3\phi_1\phi_2) \\
&\quad + \theta\theta (F_1\phi_2\phi_3 + F_2\phi_3\phi_1 + F_3\phi_1\phi_2 - \psi_1\psi_2\phi_3 - \psi_2\psi_3\phi_1 - \psi_3\psi_1\phi_2)
\end{aligned} \tag{3.3.50}$$

The same happens for the product of antichiral fields. In contrast, the product of a chiral and an antichiral field is neither chiral nor antichiral,

$$\begin{aligned}
\Phi_1^\dagger \Phi_2 &= \phi_1^*(x)\phi_2(x) + \sqrt{2}\theta\psi_2(x)\phi_1(x) + \sqrt{2}\bar{\theta}\bar{\psi}_1(x)\phi_2(x) \\
&\quad + \theta\theta\phi_1(x)F_2^*(x) + \bar{\theta}\bar{\theta}\phi_2(x)F_1^*(x) \\
&\quad + \theta^\alpha\bar{\theta}^{\dot{\alpha}} (i\sigma_{\alpha\dot{\alpha}}^\mu (\phi_1^*(x)\partial_\mu\phi_2(x) - \phi_2(x)\partial_\mu\phi_1(x)) - 2\bar{\psi}_{1\dot{\alpha}}\psi_{2\alpha}) \\
&\quad + \theta\theta\bar{\theta}^{\dot{\alpha}} \left( \frac{i}{\sqrt{2}}\sigma_{\alpha\dot{\alpha}}^\mu (\phi_1(x)\partial_\mu\psi_2^\alpha(x) - \partial_\mu\phi_1(x)\psi_2^\alpha(x)) - \sqrt{2}F_2(x)\bar{\psi}_{1\dot{\alpha}} \right) \\
&\quad + \bar{\theta}\bar{\theta}\theta^\alpha \left( -\frac{i}{\sqrt{2}}\sigma_{\alpha\dot{\alpha}}^\mu (\bar{\psi}_1^{\dot{\alpha}}(x)\partial_\mu\bar{\phi}_2(x) - \partial_\mu\bar{\psi}_1^{\dot{\alpha}}(x)\phi_2(x)) - \sqrt{2}F_1^*(x)\psi_{2\alpha} \right) \\
&\quad + \theta\theta\bar{\theta}\bar{\theta} \left( F_1^*F_2 + \frac{1}{4}\phi_1^*\square\phi_2 + \frac{1}{4}(\square\phi_1^*)\phi_2 - \frac{1}{2}\partial_\mu\phi_1^*\partial^\mu\phi_2^* \right. \\
&\quad \left. + \frac{i}{2}\partial_\mu\bar{\psi}_1\bar{\sigma}^\mu\psi_2 - \frac{i}{2}\bar{\psi}_1\bar{\sigma}^\mu\partial_\mu\psi_2 \right)
\end{aligned} \tag{3.3.51}$$

At this point we can discuss the construction of supersymmetry invariants from products of chiral fields. Let us start by recalling that the dimension of fields in the multiplet defined by the chiral field

$$\Phi = \phi(y) + \sqrt{2}\theta\psi + \theta\theta F(y) \tag{3.3.52}$$

is assigned as follows. In  $d = 4$  dimensions, the scalar and fermion dimensions, in mass units, are

$$[\phi] = m, \quad [\psi] = m^{3/2} \tag{3.3.53}$$

Then one necessarily has

$$[\Phi] = m^1, \quad [\theta] = m^{-1/2} \quad (3.3.54)$$

and hence

$$[F] = m^2 \quad (3.3.55)$$

The same can be done for an antichiral field so that one also has

$$[\bar{\theta}] = m^{-1/2} \quad (3.3.56)$$

Let us consider the supersymmetry transformation law for the “highest” power in the  $\theta$  components, which in this case is  $F$ . Being a boson,  $\delta F$  should contain  $\psi$ , the sole fermion in the game. This fermion should be accompanied by an  $\bar{\epsilon}$  having the dimension  $[\bar{\epsilon}] = m^{-1/2}$ . The product has then dimension  $[\bar{\epsilon}\psi] = m^1$ . Since the dimension of  $F$  is 2,  $[\delta F] = 2$  and the only way to attain this dimension in a linear transformation is through a derivative. This of course is consistent with what we have found in (3.3.44),

$$\delta F = i\sqrt{2}\bar{\epsilon}\bar{\sigma}^\mu\partial_\mu\psi \quad (3.3.57)$$

Since any product of chiral fields is a chiral field, such products are candidates to construct supersymmetry invariants leading to supersymmetric Lagrangians. The same idea holds for antichiral fields and their product.

In the case of products of chiral and antichiral fields like (3.3.51) one can also prove that its highest component changes under a supersymmetry transformation as a total derivative. The easiest way to see this is to observe that formula (64) indicates that a general superfield  $\Sigma$ ,

$$\Sigma = \sigma + \dots + \theta\theta\bar{\theta}P + \theta\bar{\theta}\bar{\theta}Q + \theta\theta\bar{\theta}\bar{\theta}D \quad (3.3.58)$$

(we omit lower powers in  $\theta$  and  $\bar{\theta}$ ) changes according to

$$\delta_{\epsilon,\bar{\epsilon}}\Sigma = (\epsilon Q + \bar{\epsilon}\bar{Q})\Sigma(\theta, \bar{\theta}) \quad (3.3.59)$$

Now, all terms in generators  $Q$  and  $\bar{Q}$  contain, according to (3.2.22)-(3.2.23) derivatives with respect to  $x^\mu$ ,  $\theta$  and  $\bar{\theta}$

$$Q_\alpha = \frac{\partial}{\partial\theta^\alpha} - i\sigma_{\alpha\dot{\alpha}}^\mu\bar{\theta}^{\dot{\alpha}}\partial_\mu \quad (3.3.60)$$

$$\bar{Q}_{\dot{\alpha}} = -\frac{\partial}{\partial\bar{\theta}^{\dot{\alpha}}} + i\theta^\alpha\sigma_{\alpha\dot{\alpha}}^\mu\partial_\mu \quad (3.3.61)$$

So, the highest term in  $\delta\Sigma$  can only come from the action of  $\partial/\partial x^\mu$  on the  $P$  and  $Q$  factors in (3.3.58) since derivatives with respect to  $\theta$  or  $\bar{\theta}$  will lower the order of the power. Hence the highest power term in a general superfield changes as a total derivative.

We can now analyze different products of chiral fields to see how their highest component contributes to supersymmetric interaction Lagrangians. One can see from (3.3.50) that the highest components of the product of two and three superfields contain no derivatives and are then candidates to supersymmetric Lagrangians. In contrast, the  $\theta^2\bar{\theta}^2$  term in (3.3.51) contains typical kinetic energy terms for  $\phi$  and  $\psi$  and a (non-dynamical) term for the  $F$  field which will then become an auxiliary field which can be eliminated through its algebraic equation of motion.

## Problems 3

1. Problem 3.1: Prove eq.(3.1.4)
2. Problem 3.2: Prove eq.(3.2.15)
3. Problem 3.3: Prove eq.(3.3.40)
4. Problem 3.4: Prove eq.(3.3.44)
5. Problem 3.5: Prove eqs.(3.3.42)-(3.3.43)



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# Chapter 4

## The Wess-Zumino model

### 4.1 The Wess-Zumino Lagrangian

The Wess-Zumino model is the most general supersymmetric renormalizable Lagrangian involving only chiral and antichiral superfields  $\Phi_i, \Phi_i^\dagger$  ( $i = 1, 2, \dots, n$ ). It is constructed by selecting the highest component of certain products of superfields. This last is done by integrating over appropriate superspace coordinates  $\theta$  and  $\bar{\theta}$ . The space-time Lagrangian is defined as

$$L = \int d^2\theta^2 d^2\bar{\theta} \Phi_i^\dagger \Phi_i + \left( \int d^2\theta \left( \frac{1}{2} m_{ij} \Phi_i \Phi_j + \frac{1}{3} g_{ijk} \Phi_i \Phi_j \Phi_k + \lambda_i \Phi_i \right) + \text{h.c.} \right) \quad (4.1.1)$$

We have included linear terms although they can be eliminated by appropriate field redefinitions. Couplings  $m_{ij}$  and  $g_{ijk}$  are symmetric in their indices (since  $\Phi_1 \Phi_2 = \Phi_2 \Phi_1$ , etc). As we shall see, quartic or higher powers of chiral and antichiral fields will lead to non-renormalizable interactions. Supersymmetry is guaranteed since the integrals are such that they select the highest component in each term and as we have seen, highest components change as a total derivative under supersymmetry transformations.

From definition (4.1.1) one can obtain the Lagrangian in terms of component fields. As explained before, the kinetic terms all arise from the  $\Phi_i^\dagger \Phi_i$  highest term. The quadratic and cubic products of chiral fields give just the interactions and can be read from eqs.(3.3.50). In fact, after integration over  $\theta_\alpha$  the component fields appear as functions of the independent variables  $y^\mu$  but since the Lagrangian is to be read from the superaction where these variables are integrated, this is just a notation detail. One then gets

$$\begin{aligned} L = & i \partial_\mu \bar{\psi}_i \bar{\sigma}^\mu \psi_i + \phi_i^* \square \phi_i + F_i^* F_i \\ & + \left( m_{ij} \left( \phi_i F_j - \frac{1}{2} \psi_i \psi_j \right) + g_{ijk} (\phi_i \phi_j F_k - \psi_i \psi_j \phi_k) + \lambda_i F_i + \text{h.c.} \right) \end{aligned} \quad (4.1.2)$$

Here total derivatives have been dropped since they do not affect the equations of motion.

We see that  $F$  is an auxiliary field in the sense it has not kinetic energy term so that its equations of motion are just algebraic equations that can be used to eliminate it,

$$\begin{aligned} 0 &= \frac{\partial L}{\partial F_i^*} = F_i + \lambda_i^* + m_{ij}^* \phi_j^* + g_{ijk}^* \phi_j^* \phi_k^* \\ 0 &= \frac{\partial L}{\partial F_i} = F_i^* + \lambda_i + m_{ij} \phi_j + g_{ijk} \phi_j \phi_k \end{aligned} \quad (4.1.3)$$

Replacing the expressions for  $F_i$  and  $F_i^*$  given by (4.1.3) in Lagrangian (4.1.2) we get the *on-shell Lagrangian*. Note that the term  $F_i^* F_i$  in (4.1.2), together with the terms linear in  $F_i$  and  $F_i^*$  will give rise to the scalar potential through which scalars  $\phi_i$  interact,

$$F_i^* F_i + (F_i (\lambda_i + m_{ij} \phi_j + g_{ijk} \phi_j \phi_k) + \text{h.c.})|_{\text{on-shell}} = -|\lambda_i + m_{ij} \phi_j + g_{ijk} \phi_j \phi_k|^2 \quad (4.1.4)$$

so that calling

$$V(\phi_i, \phi_j) = |\lambda_i + m_{ij} \phi_j + g_{ijk} \phi_j \phi_k|^2 \quad (4.1.5)$$

we finally have

$$L = i\partial_\mu \bar{\psi}_i \bar{\sigma}^\mu \psi_i + \phi_i^* \square \phi_i - V(\phi_i, \phi_j) - \frac{1}{2} m_{ij} \psi_i \psi_j - \frac{1}{2} m_{ij}^* \bar{\psi}_i \bar{\psi}_j - g_{ijk} \psi_i \psi_j \phi_k - g_{ijk}^* \bar{\psi}_i \bar{\psi}_j \phi_k^* \quad (4.1.6)$$

Note that  $V$  in (4.1.5) can be written as

$$V(\phi_i, \phi_j^*) = F_i^* F_i|_{\text{on-shell}} \geq 0 \quad (4.1.7)$$

and, being semidefinite positive, points  $F_i = 0$  are absolute minima of the potential since at those points  $V = 0$ .

## 4.2 The on-shell Lagrangian and supersymmetry

Let us consider for simplicity the case of just one chiral field  $\Phi$  so that the Wess-Zumino on-shell Lagrangian reads

$$L = i\partial_\mu \bar{\psi} \bar{\sigma}^\mu \psi + \phi^* \square \phi - V(\phi) - \frac{1}{2} m \psi \psi - \frac{1}{2} m^* \bar{\psi} \bar{\psi} - g \psi \psi \phi - g^* \bar{\psi} \bar{\psi} \phi^* \quad (4.2.8)$$

With no loss of generality we have put  $\lambda = 0$  since the linear term can always be eliminated by an appropriate shift in the fields.

Let us start the analysis rewriting the off-shell supersymmetry transformation laws for chiral fields(3.3.44)

$$\begin{aligned}\delta\phi &= \sqrt{2}\epsilon\psi \\ \delta\psi &= i\sqrt{2}\sigma^\mu\bar{\epsilon}\partial_\mu\phi + \sqrt{2}\epsilon F \\ \delta F &= i\sqrt{2}\bar{\epsilon}\bar{\sigma}^\mu\partial_\mu\psi\end{aligned}\tag{4.2.9}$$

which are independent of the dynamics of the particular model one is to consider, that is, of the Lagrangian. Now, once a Lagrangian like the Wess-Zumino one is given and the auxiliary field equations used to write it on-shell, one should also eliminate the auxiliary field from the supersymmetry transformation laws and in this way one gets the *on-shell supersymmetry transformations*.

In the present case we have

$$F + m^*\phi^* + g^*\phi^*\phi^* = 0\tag{4.2.10}$$

so that the on-shell laws become

$$\begin{aligned}\delta\phi &= \sqrt{2}\epsilon\psi \\ \delta\psi &= i\sqrt{2}\sigma^\mu\bar{\epsilon}\partial_\mu\phi - \sqrt{2}\epsilon(m^*\phi^* + g^*\phi^*\phi^*)\end{aligned}\tag{4.2.11}$$

which are evidently non-linear.

### 4.3 The Noether supercurrent

In order to show how one derives the Noether current associated to supersymmetry, let us consider the simple case of the Wess-Zumino Lagrangian for one chiral field, with  $g = \lambda = 0$ . In term of superfields one has

$$\begin{aligned}L &= L_0 + L_m \\ L_0 &= \int d^2\theta d^2\bar{\theta} \Phi^\dagger\Phi \\ L_m &= \frac{m}{2} \int d^2\theta \Phi^2 + \text{h.c.}\end{aligned}\tag{4.3.12}$$

while in terms of components fields one has

$$\begin{aligned}L_0 &= \frac{i}{2}\partial_\mu\bar{\psi}\bar{\sigma}^\mu\psi - \frac{i}{2}\bar{\psi}\bar{\sigma}^\mu\partial_\mu\psi - \partial_\mu\phi^*\partial^\mu\phi + F^*F \\ L_m &= -\frac{m}{2}(\psi\psi + \bar{\psi}\bar{\psi})\end{aligned}\tag{4.3.13}$$

According to the Noether theorem, the conserved supercurrent  $\mathcal{J}^\mu$  is given by

$$\mathcal{J}_\mu = \tilde{J}_\mu - K_\mu = \epsilon J_\mu + \bar{\epsilon} \bar{J}_\mu \quad (4.3.14)$$

where

$$\tilde{J}_\mu = \delta\bar{\psi} \frac{\partial L}{\partial \partial^\mu \psi} + \frac{\partial L}{\partial \partial^\mu \psi} \delta\psi + \frac{\partial L}{\partial \partial^\mu \phi} \delta\phi + \frac{\partial L}{\partial \partial^\mu \phi^*} \delta\phi^* \quad (4.3.15)$$

and  $K_\mu$  is related to the variation  $\delta L$  of the Lagrangian under the supersymmetry transformation,

$$\delta L = \partial^\mu \mathcal{K}_\mu \quad (4.3.16)$$

From (4.3.13) one easily computes the explicit form of  $\tilde{J}_\mu$

$$\begin{aligned} \tilde{J}_\mu = & \epsilon \left( \frac{1}{\sqrt{2}} \sigma^\nu \bar{\sigma}_\mu \psi \partial_\nu \phi^* + i \bar{\psi} \bar{\sigma}_\mu F - \sqrt{2} \psi \partial_\mu \phi \right) \\ & + \bar{\epsilon} \left( -\frac{1}{\sqrt{2}} \bar{\psi} \bar{\sigma}_\mu \sigma^\nu \partial_\nu \phi - i F^* \sigma_\mu \psi - \sqrt{2} \bar{\psi} \partial_\mu \phi^* \right) \end{aligned} \quad (4.3.17)$$

Concerning  $K_\mu$  the calculation is rather tedious. One way to compute it is using our previous observation that applying the supersymmetry generators as given in (3.3.60)-(3.3.61) the resulting highest  $\theta, \bar{\theta}$  term will result from the space-time derivative term when applied to the cubic  $\theta, \bar{\theta}$  terms in  $\Phi^+ \Phi$  or linear terms in  $\Phi^2$  and  $\Phi^{+2}$ .

We will just present the final result without including the “improvement terms” that one can always add to a Noether current (that is, divergenceless second order tensors)

$$\begin{aligned} J_\mu^\alpha = & \left( \frac{1}{\sqrt{2}} \sigma^\nu \bar{\sigma}_\mu \psi \partial_\nu \phi^* + i \bar{\psi} \bar{\sigma}_\mu F - \sqrt{2} \psi \partial_\mu \phi - i \sqrt{2} m \phi^* \bar{\psi} \bar{\sigma}_\mu \right)^\alpha \\ & - \left( \frac{1}{\sqrt{2}} \sigma_\mu \sigma^\nu (\phi \partial_\nu \psi - \partial_\nu \phi \psi) + i \sqrt{2} F \sigma_\mu \psi \right)^\alpha \end{aligned} \quad (4.3.18)$$

and an analogous formula for  $\bar{J}_\mu$ .

## 4.4 Spontaneous supersymmetry breaking

The important inequality (4.1.7) has a deep origin related to supersymmetry. To see this, let us start from the supersymmetry algebra relation (5.3.50)

$$\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = 2\sigma_{\alpha\dot{\beta}}^\mu P_\mu \quad (4.4.19)$$

and observe that the only  $\sigma$  matrices with elements in the diagonal are  $\sigma^0$  and  $\sigma^3$  (see (2.4.48))

$$\sigma^0 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (4.4.20)$$

Then, if we consider

$$\begin{aligned} \{Q_1, \bar{Q}_1\} &= 2\sigma_{11}^\mu P_\mu = -2P_0 + 2P_3 \\ \{Q_2, \bar{Q}_2\} &= 2\sigma_{22}^\mu P_\mu = -2P_0 - 2P_3 \end{aligned} \quad (4.4.21)$$

we have

$$\{Q_1, \bar{Q}_1\} + \{Q_2, \bar{Q}_2\} = -4P_0 = 4P^0 \quad (4.4.22)$$

Identifying  $P^0$  with the energy density (Hamiltonian), we have

$$H = \frac{1}{4} (\bar{Q}_1 Q_1 + Q_1 \bar{Q}_1 + \bar{Q}_2 Q_2 + Q_2 \bar{Q}_2) \quad (4.4.23)$$

All four terms in (4.4.23) are positive (remember that  $\bar{Q} = Q^+$ ) and hence for every state  $|\Psi\rangle$  one has

$$\langle \Psi | H | \Psi \rangle \geq 0 \quad (4.4.24)$$

Thus, the spectrum of the Hamiltonian in a theory with supersymmetry contains no negative eigenvalues.

Eq.(4.4.24) implies that states  $|0\rangle$  with vanishing energy density are supersymmetric ground states of the theory. Indeed

$$\langle 0 | H | 0 \rangle = 0 \Rightarrow Q_\alpha |0\rangle = 0 \text{ and } \bar{Q}_{\dot{\alpha}} |0\rangle = 0 \quad (4.4.25)$$

That is, states  $|0\rangle$  are annihilated by the supersymmetry generators. They are ground states because there could not be a state of energy lower than zero.

Any state whose energy is not zero, e.g. any one-particle state, cannot be invariant under supersymmetry transformations. This means that there must be one (or more) superpartner state  $Q|1\rangle$  or  $\bar{Q}|1\rangle$  for every one-particle state  $|1\rangle$ . The spin of these partner states will be different by  $1/2$  from that of  $|1\rangle$ . Thus each supermultiplet must contain at least one boson and one fermion whose spins differ by  $1/2$ .

Ground states of zero energy thus preserve supersymmetry while states of positive energy break it spontaneously<sup>1</sup>. Different situations are sketched in figures 1-3. Figure 1 corresponds

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<sup>1</sup>The term “spontaneously broken symmetry” is used when the interaction potentials in a theory, and therefore the basic dynamics, are symmetric but states are not.

to a potential with one supersymmetric ground state while Figure 2 shows a potential leading to spontaneous breaking of supersymmetry. Finally in Figure 3 we represent the case of a potential with a metastable non-supersymmetric vacuum (a local minimum sometimes called a *false vacuum*) and a supersymmetric vacuum which corresponds to a global minimum. As we shall see it is rather easy to construct models exhibiting this last behavior which, for different reasons, is phenomenologically welcome.

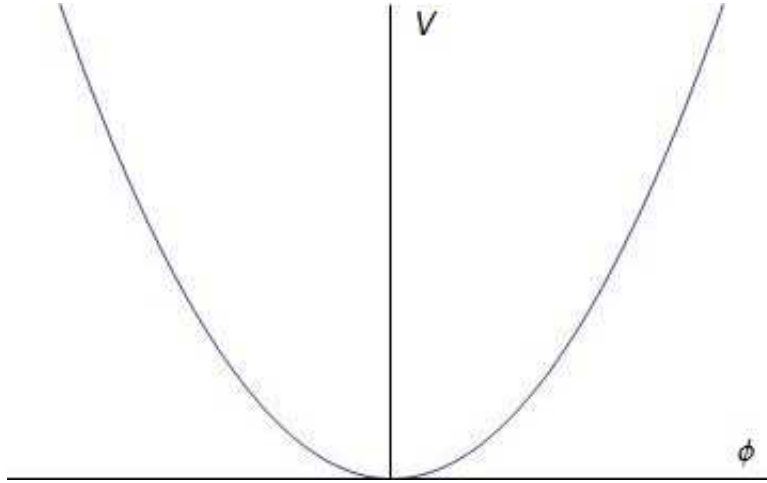


Figure 1: A potential with a ground state that preserves supersymmetry

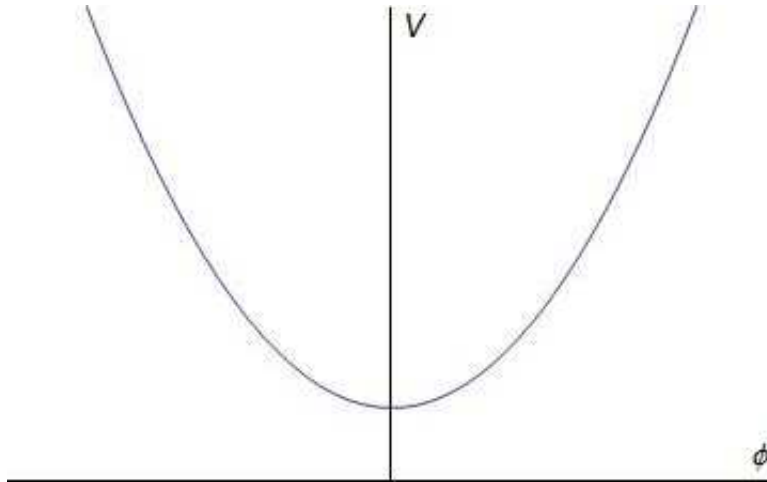


Figure 2: A potential with a ground state that spontaneously breaks supersymmetry



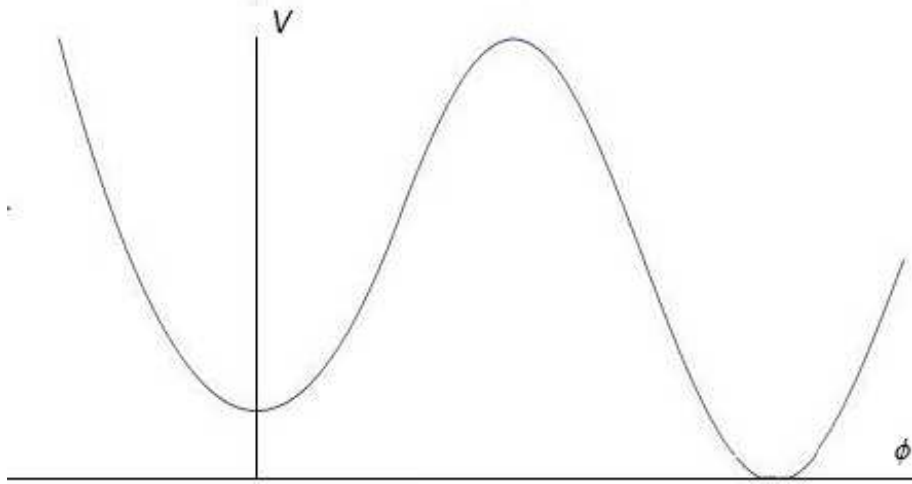


Figure 3: A potential with a metastable ground state that spontaneously breaks supersymmetry (left) and a supersymmetric one (right)



# Chapter 5

## Supersymmetry breaking in detail

3/6/10

### 5.1 Superpotentials and Kähler potentials

When we introduced the Wess-Zumino model, we considered two kinds of products. On the one hand we multiplied a chiral field times an antichiral field and from that product (which was neither chiral nor antichiral) we got the kinetic energy terms. On the other hand we considered products of chiral fields up to third order (and the corresponding products of antichiral terms, necessary to end up with a real Lagrangian. Then we proceed by integration on the Grassmann coordinates of superspace) to select just the highest  $\theta, \bar{\theta}$  term in each product so as to ensuring the supersymmetry of the resulting action (the highest term in a superfield always change as a total derivative).

In general, any analytic function of a chiral superfield,  $W[\Phi_i]$  is also a chiral superfield referred to as a superpotential. Together with its Hermitian conjugate  $W[\Phi_i]^\dagger$  it will give rise to general interaction terms,

$$L_W = \int d^2\theta W[\Phi_i] + \int d^2\bar{\theta} W[\Phi_i]^\dagger \quad (5.1.1)$$

Using the  $y$  variables defined as (see (3.3.39))

$$y^\mu = x^\mu + i\theta\sigma^\mu\bar{\theta} \ , \quad (5.1.2)$$

one can easily expand  $W[\Phi_i]$  in Taylor series,

$$W[\Phi_i] = W[\phi_i(y)] + \sqrt{2}\frac{\partial W}{\partial\phi_i}\theta\psi_i(y) - \theta\theta\left(\frac{\partial W}{\partial\phi_i}F_i(y) + \frac{1}{2}\frac{\partial^2 W}{\partial\phi_i\partial\phi_j}\right)\psi_i(y)\psi_j(y) \quad (5.1.3)$$

The expansion leading to the formula was done as follows. We write for a given superfield (we avoid indices  $i$ )

$$\Phi = \phi + \sqrt{2}\theta^\rho\psi_\rho + \theta^\rho\theta^\sigma\varepsilon_{\rho\sigma}F \quad (5.1.4)$$

Then

$$\begin{aligned} \frac{\partial\Phi}{\partial\theta^\alpha} &= \sqrt{2}\psi_\alpha + \varepsilon_{\alpha\sigma}\theta^\sigma F - \varepsilon_{\rho\alpha}\theta^\rho F = \sqrt{2}\psi_\alpha + 2\varepsilon_{\alpha\sigma}\theta^\sigma F \\ \frac{\partial^2\Phi}{\partial\theta^\beta\partial\theta^\alpha} &= 2\varepsilon_{\alpha\beta}F \end{aligned} \quad (5.1.5)$$

Now

$$\begin{aligned} \frac{\partial W}{\partial\theta^\alpha} &= \frac{\partial W}{\partial\Phi} \frac{\partial\Phi}{\partial\theta^\alpha} \\ \frac{\partial^2 W}{\partial\theta^\beta\partial\theta^\alpha} &= \frac{\partial}{\partial\theta^\beta} \left( \frac{\partial W}{\partial\Phi} \frac{\partial\Phi}{\partial\theta^\alpha} \right) = \frac{\partial^2 W}{\partial\Phi\partial\theta^\beta} \frac{\partial\Phi}{\partial\theta^\alpha} + \frac{\partial W}{\partial\Phi} \frac{\partial^2\Phi}{\partial\theta^\beta\partial\theta^\alpha} \end{aligned} \quad (5.1.6)$$

Then

$$\begin{aligned} \left. \frac{\partial W}{\partial\theta^\alpha} \right|_{\theta=0} &= \sqrt{2} \frac{\partial W}{\partial\phi} \psi_\alpha \\ \left. \frac{\partial^2 W}{\partial\theta^\beta\partial\theta^\alpha} \right|_{\theta=0} &= 2 \frac{\partial^2 W}{\partial\phi\partial\phi} \psi_\beta \psi_\alpha + 2 \frac{\partial W}{\partial\phi} \varepsilon_{\alpha\beta} F \end{aligned} \quad (5.1.7)$$

so that the linear and quadratic term in the expansion take the form

$$\begin{aligned} \left. \frac{\partial W}{\partial\theta^\alpha} \right|_{\theta=0} \theta^\alpha &= \sqrt{2} \frac{\partial W}{\partial\phi} \psi\theta \\ \frac{1}{2} \left. \frac{\partial^2 W}{\partial\theta^\beta\partial\theta^\alpha} \right|_{\theta=0} \theta^\beta \theta^\alpha &= - \frac{\partial^2 W}{\partial\phi\partial\phi} \theta^\alpha \theta^\beta \psi_\alpha \psi_\beta - \frac{\partial W}{\partial\phi} \theta^\alpha \theta^\beta \varepsilon_{\alpha\beta} F = -\theta\theta \left( \frac{1}{2} \frac{\partial^2 W}{\partial\phi\partial\phi} \psi\psi + \frac{\partial W}{\partial\phi} F \right) \end{aligned} \quad (5.1.8)$$

In the same way we have generalized the chiral fields products used to construct the interaction Wess-Zumino Lagrangian introducing a general analytic function  $W[\Phi_i]$ , we can generalize the product  $\Phi_i^\dagger \Phi_i$  that yields to the kinetic part of the Wess-Zumino Lagrangian to a real function  $K[\Phi_i, \Phi_j^\dagger]$ . In general these generalization will correspond to non-renormalizable quantum field theories but there are several circumstances in which non-renormalizable Lagrangians are necessary and they are accepted as describing effective field theories valid below certain scale.

The non-holomorphic function  $K[\Phi_i, \Phi_j^\dagger]$  is called a Kähler potential for the following reason. Suppose that the space of the chiral fields has a non-trivial metric so that the scalar kinetic term takes the form  $g_{ij}[\phi_k] \partial_\mu \phi_i \partial^\mu \phi_j$ , with appropriate modifications for other terms. Then, the product  $\Phi_i^\dagger \Phi_i$  should be replaced precisely by  $K[\Phi, \Phi^\dagger]$  so that

$$g_{ij}[\phi_k] = \frac{\partial^2 K}{\partial \phi_i \partial \phi_j^*} \quad (5.1.9)$$

This is the form taken by the metric in what is called a Kähler manifold. A Kähler manifold is a manifold with a Hermitian metric that satisfies an integrability condition, which has several equivalent formulations. Kähler manifolds can be characterized in many ways: they are often defined as a complex manifold with an additional structure. The metric in a Kähler manifold locally satisfies (5.1.9). In other words, the Lagrangian for the chiral field turns out to be a nonlinear sigma model for which the target space is a complex manifold with metric (5.1.9). It is remarkable that wherever in ordinary quantum field theory we find a general structure from real analysis, the supersymmetric version of the theory has a corresponding complex analytic structure. More details on Kähler geometry are presented in an appendix.

We can then write a general supersymmetric Lagrangian in the form

$$L_{Susy} = \int d^2\bar{\theta} d^2\theta K[\Phi_i, \Phi_i^\dagger] + \int d^2\theta W[\Phi_i] + \int d^2\bar{\theta} \bar{W}[\Phi_i]^\dagger \quad (5.1.10)$$

and specify in each case the actual form of the Kähler potential and superpotential that one is going to consider.

In order to obtain the component field contribution of the Kähler term to the Lagrangian, one has to make a Taylor expansion as the one we did in eq.(5.1.3). We shall not give the tedious intermediate process but simply quote the result of integrating the Kähler potential over the Grassmann variables. One obtains, in component fields,

$$\begin{aligned} \int d^2\bar{\theta} d^2\theta K[\Phi_i, \Phi_i^\dagger] &= g_{ij} \left( \frac{i}{2} \partial_\mu \bar{\psi}_i \bar{\sigma}^\mu \psi_j - \frac{i}{2} \bar{\psi}_i \bar{\sigma}^\mu \partial_\mu \psi_j - \partial_\mu \phi_i^* \partial^\mu \phi_j + F_i^* F_j \right) \\ &+ \frac{1}{2} \left( \frac{\partial g_{ij}}{\partial \phi_k} (\psi_k \psi_i F^{*j} - i \bar{\psi}_j \sigma^\mu \partial_\mu \phi_k \psi_i) \right) \\ &+ \frac{1}{2} \left( \frac{\partial g_{ij}}{\partial \phi_k^*} (\bar{\psi}_k \bar{\psi}_i F^j + i \bar{\psi}_j \sigma^\mu \partial_\mu \phi_k^* \psi_i) \right) \\ &+ \frac{1}{4} \frac{\partial^2 g_{ij}}{\partial \phi_k \partial \phi_l^*} \bar{\psi}_l \bar{\psi}_j \psi_i \psi_k \end{aligned} \quad (5.1.11)$$

## 5.2 O’Raifeartaigh models: F-term spontaneous supersymmetry breaking

If supersymmetric theories are to find realistic applications in high energy physics supersymmetry should be broken since, as we have seen, all particles in a given representation of supersymmetry must have the same mass, this following from the relation between the four dimensional translations and the anticommutators of the supersymmetry generators. Hence, for each one of the particles we know today exist one should have detected a supersymmetric partner. The only way to escape to the prediction of massless ”photinos” (the supersymmetric massless fermion companions of photons), “s-electrons” (supersymmetric boson companions of electrons, i.e. having the same mass) etc, is to find a way to break supersymmetry, in a similar way one does with gauge symmetry when constructing the Salam-Weinberg electroweak model.

As mentioned in the introduction and will be discussed in detail later spontaneous breaking of supersymmetry is preferable and hence this will be the subject of this class.

Since the first works on supersymmetric models it was recognized that spontaneous breakdown of supersymmetry does not occur for Lagrangians with only one chiral fields. It was O’Raifeartaigh, in 1975, the first to construct a family of models with more than one chiral fields showing that it is possible to construct, at least at the classical level, models with spontaneously broken supersymmetry models provided the number of chiral fields is  $N \geq 3$ .

In order to analyze spontaneous symmetry breaking one starts by considering a background configuration which is by itself a solution of the classical equations of motion and then studies the quantum dynamics of the system by using perturbation theory around such configuration.

The standard simplest choice of such background, usually called *vacuum*, corresponds to constant, Lorentz invariant fields minimizing the classical potential. In the case of chiral fields, only scalar fields  $\phi_i$  are then allowed to have non-zero (vacuum expectation) values since the remaining scalar are auxiliary fields which can be eliminated through their algebraic equations of motion.

To determine the classical equations of motion let us consider the case of a canonical Kähler potential  $K = \Phi^\dagger \Phi$  and use the expansion (5.1.3) to write the Lagrangian (5.1.10), one fermion fields have been put to zero in the form

$$L = -V = F_i F_i^* + \frac{\partial W}{\partial \phi_i} F_i + \frac{\partial \bar{W}}{\partial \phi_i^*} F_i^* \quad (5.2.12)$$

The first term comes from the  $\Phi^\dagger \Phi$  part of the superaction; the other two terms arise from the  $\theta^2$  term in the expansion (5.1.3) of the superpotential and the  $\bar{\theta}^2$  one in the corresponding

expansion of  $\bar{W}[\Phi^\dagger]$  with fermion (No derivative terms for  $\phi$  appear since we are considering constant configurations. The auxiliary field eqs. read

$$\begin{aligned} F_i &= -\frac{\partial \bar{W}}{\partial \phi_i^*} \\ F_i^* &= -\frac{\partial W}{\partial \phi_i} \end{aligned} \quad (5.2.13)$$

so that

$$V = F_i^* F_i = \frac{\partial \bar{W}}{\partial \phi_i^*} \frac{\partial W}{\partial \phi_i} \quad (5.2.14)$$

The equations of motion take then the simple form

$$\frac{\partial V}{\partial \phi_i} = 0 \quad (5.2.15)$$

and the condition of local stability is then that the matrix

$$M = \begin{pmatrix} \frac{\partial^2 V}{\partial \phi_i \partial \phi_j^*} & \frac{\partial^2 V}{\partial \phi_i \partial \phi_j} \\ \frac{\partial^2 V}{\partial \phi_i^* \partial \phi_j^*} & \frac{\partial^2 V}{\partial \phi_i^* \partial \phi_j} \end{pmatrix} \quad (5.2.16)$$

has only positive or zero eigenvalues. In the language of field theory the  $M^2$  matrix is nothing but the mass matrix of scalar fields  $\phi_i$  and local stability is just equivalent to the absence of scalar fields with negative mass squared (absence of tachyon states).

It is not necessary to solve (5.2.15) to determine whether supersymmetry is spontaneously broken. At the light of what we learnt at the end of class 4 existence of a supersymmetric vacuum requires  $V = 0$  or, according to (5.2.14)

$$F_i^* = -\frac{\partial W}{\partial \phi_i} = 0, \quad i = 1, 2, \dots, N \quad (5.2.17)$$

Supersymmetry will be then spontaneously broken whenever the system of  $N$  homogeneous equations (5.2.17) has no non-trivial solution. The idea of O’Raifeartaigh was to chose a minimal set of scalar fields and superpotential such that the system has no such non-trivial solution. Let us write as in (4.1.1)

$$W = \lambda_i \Phi_i + \frac{1}{2} m_{ij} \Phi_i \Phi_j + \frac{1}{3} g_{ijk} \Phi_i \Phi_j \Phi_k \quad (5.2.18)$$

If the superpotential has no non-linear term,  $\phi_i = 0$  will always be a solution and a supersymmetry will not be broken. So, let us consider that linear terms are present and take

the simplest case in which only one chiral field contributes to this linear part. We call the corresponding superfield  $X$ . The idea is that  $X$  will force another superfield, say  $\Phi_1$ , to have a vacuum expectation value. The third superfield  $\Phi_2$  will serve to force the vacuum expectation value of  $X$  to vanish. Consider then

$$W = \lambda X + m\Phi_1\Phi_2 + gX\Phi_1^2 \quad (5.2.19)$$

The scalar potential for this O’Raifeartaigh model can be easily calculated from eq.(5.2.14),

$$V = |F_X|^2 + |F_{\Phi_1}|^2 + |F_{\Phi_2}|^2 \quad (5.2.20)$$

using the equations of motion for the auxiliary fields

$$\begin{aligned} -F_X^* &= \left. \frac{\partial W}{\partial X} \right|_{vac} = \lambda + g\phi_1^2 \\ -F_{\Phi_1}^* &= \left. \frac{\partial W}{\partial \Phi_1} \right|_{vac} = m\phi_2 + 2gx\phi_1 \\ -F_{\Phi_2}^* &= \left. \frac{\partial W}{\partial \Phi_2} \right|_{vac} = m\phi_1 \end{aligned} \quad (5.2.21)$$

Here, with the subindex *vac* we indicate that once differentiation is done, one should put  $X = x$  and  $\Phi_i = \phi_i$ , i.e, replace superfields by their scalar component. We then have

$$V = |\lambda + g\phi_1^2|^2 + |m\phi_2 + 2gx\phi_1|^2 + |m\phi_1|^2 \quad (5.2.22)$$

The absolute minimum will correspond to the case in which each one of the three terms vanishes. This is equivalent to demanding that

$$F_i = 0 \quad (5.2.23)$$

As advanced, the condition  $F_X = 0$  (first term in the scalar potential) forces  $\phi_1$  to have a nonzero v.e.v such that  $\phi_1^2 = -\lambda/g$  while the  $F_{\phi_2} = 0$  one forces  $\phi_1$  to vanish. Then the system is incompatible,  $V > 0$  and supersymmetry is spontaneously broken.

Using (5.2.22) one can find the supersymmetry breaking minima (vacua) by solving

$$\frac{\partial V}{\partial \phi_i} = \frac{\partial V}{\partial \phi_i^*} = 0 \quad (5.2.24)$$

where  $(\phi_i) = (x, \phi_1, \phi_2)$ . One can see that the absolute minimum corresponds to the solution

$$\phi_{1m} = \phi_{2m} = 0, \quad x_m : \text{arbitrary} \quad (5.2.25)$$



$$V[\phi_{1m}, \phi_{2m}, x_m] = |\lambda|^2 \quad (5.2.26)$$

Since the minimum conditions do not fix  $x_m$  there is a continuous manifold of vacua, an important feature that will be present in many relevant models of supersymmetry breaking. One refers to this phenomenon by saying that there is a moduli space of non-supersymmetric vacua. As we shall see, this degeneracy at the classical level is often lifted once quantum corrections are taken into account and in that case it is preferable to talk about a pseudomoduli space.

Let now compute second order derivatives at the minimum, the only non-trivial being (for simplicity we take  $\lambda, g, m \in \mathbb{R}$ )

$$\frac{\partial^2 V}{\partial \phi_1^2} = \frac{\partial^2 V}{\partial \phi_1^{*2}} = 2g\lambda, \quad \frac{\partial^2 V}{\partial \phi_1 \partial \phi_1^*} = \frac{\partial^2 V}{\partial \phi_2 \partial \phi_2^*} = m^2 \quad (5.2.27)$$

We then expand around the vacuum

$$V = \lambda^2 + \frac{1}{2} \frac{\partial^2 V}{\partial \phi_i \partial \phi_j} \varphi_i \varphi_j + \frac{1}{2} \frac{\partial^2 V}{\partial \phi_i^* \partial \phi_j^*} \varphi_i^* \varphi_j^* + \frac{\partial^2 V}{\partial \phi_i^* \partial \phi_j} \varphi_i^* \varphi_j \quad (5.2.28)$$

and identify masses of the six real valued scalar fields

$$\varphi_x = x^R + ix^I, \quad \varphi_1 = \varphi_1^R + i\varphi_1^I, \quad \varphi_2 = \varphi_2^R + i\varphi_2^I \quad (5.2.29)$$

resulting

$$m_{x^R} = m_{x^I} = 0, \quad m_{\varphi_1^R} = \sqrt{m^2 + 2\lambda g}, \quad m_{\varphi_1^I} = \sqrt{m^2 - 2\lambda g}, \quad m_{\varphi_2^R} = m_{\varphi_2^I} = m \quad (5.2.30)$$

The fact  $m_{\varphi_1^I} \neq m_{\varphi_1^R}$  is an indication of supersymmetry breaking: 2 boson fields belonging to the same multiplet have different masses. Note that  $m^2 > 2\lambda g$  in order to avoid tachyons.

Concerning fermions, their masses can be read from eq.(5.1.3),

$$L_{fm} = \frac{1}{2} \frac{\partial^2 W}{\partial \phi_i \partial \phi_j} \psi_i(y) \psi_j(y) + c.c. \quad (5.2.31)$$

Using that

$$\frac{\partial W}{\partial \phi_i} = -F_i^* \Rightarrow \frac{\partial^2 W}{\partial \phi_i \partial \phi_j} = -\frac{\partial F_i^*}{\partial \phi_j} \quad (5.2.32)$$

and the explicit form for  $F_i^*$  given in (5.2.21), the only nontrivial terms contributing to  $\partial^2 W / \partial \phi_i \partial \phi_j$  are those corresponding to  $\psi_1$  and  $\psi_2$ ,

$$L_{fm} = -m (\psi_1 \psi_2 + \bar{\psi}_1 \bar{\psi}_2) \quad (5.2.33)$$

which can be arranged in the form

$$L_{fm} = -m(\psi_2 \bar{\psi}_1) \begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix} \begin{pmatrix} \psi_1 \\ \bar{\psi}_2 \end{pmatrix} \quad (5.2.34)$$

So one is left with a Dirac fermion  $\Psi_D$

$$\Psi_D = \begin{pmatrix} \psi_{1\alpha} \\ \bar{\psi}_2^{\dot{\alpha}} \end{pmatrix} \quad (5.2.35)$$

with mass  $m$ , and a massless Weyl fermion  $\psi_X$  which is the Goldstone particle associated with spontaneously supersymmetry breaking and is usually called goldstino. Being the supersymmetric charge fermionic, the spontaneously broken supersymmetry generator is a Grassmann object from which a fermionic zero mode follows.

An important property of the masses is that the sum rule for fermions and boson masses remains valid even if supersymmetry is broken

$$\text{Str}[m^2] = \sum_i m_{bi}^2 - \sum_i m_{fi}^2 = 0 \quad (5.2.36)$$

We shall see the relevance of this fact when studying renormalization of supersymmetric models since the coefficient in the 1-loop quadratic divergency in the vacuum energy is precisely supertrace (5.2.36) so that supersymmetry breaking does not affect the ultraviolet structure of the theory.

Let us end this section by noting that for the case of a non-canonical Kähler potential, formula (5.2.14) relating the scalar potential and the superpotential takes the form

$$V = g^{ij} \frac{\partial W^+}{\partial \phi_i^*} \frac{\partial W}{\partial \phi_j} \quad (5.2.37)$$

where  $g^{ij}$ , the inverse of the Kähler metric defined in eq.(5.1.9),

$$g^{ij} g_{jk} = \delta_k^i, \quad (5.2.38)$$

replaces here the Kronecker delta of the canonical case. This result is obtained as follows. Using Lagrangian (5.1.11) one can see that the auxiliary field equations (5.2.17) take now the form

$$g_{ij} F_j^* = -\frac{\partial W}{\partial \phi_i} \Rightarrow F_i^* = -g^{ij} \frac{\partial W}{\partial \phi_j} \quad (5.2.39)$$

so that after insertion in (5.2.12) one ends with (5.2.37).

### 5.3 Introducing R-symmetry

The form of the general superspace action (5.1.10) for a renormalizable supersymmetric theory suggests that this theory has a natural continuous global symmetry. Consider the case of a Kähler potential in which the superspace Lagrangian takes the form

$$L_{Susy} = \int d^2\bar{\theta}d^2\theta \Phi^\dagger\Phi + \int d^2\theta W[\Phi_i] + \int d^2\bar{\theta}W[\Phi_i]^\dagger \quad (5.3.40)$$

As we have seen, the contribution of the superpotential terms to the component field Lagrangian is

$$L_W = \int d^2\theta W[\Phi_i] = \int d^2\theta (\dots + \theta^2 \mathcal{F}(\phi, \psi, F)) \quad (5.3.41)$$

$$= \mathcal{F}(\phi, \psi, F) \quad (5.3.42)$$

and the same for  $\bar{W}[\Phi^\dagger]$ . Now, suppose one performs the following change in Grassman variables:

$$\begin{aligned} \theta &= \exp(-i\alpha)\theta', & \bar{\theta} &= \exp(i\alpha)\bar{\theta}' \\ d\theta &= \exp(i\alpha)d\theta', & d\bar{\theta} &= \exp(-i\alpha)d\bar{\theta}' \end{aligned} \quad (5.3.43)$$

where we have used that if  $\theta \rightarrow a\theta$  then  $d\theta \rightarrow a^{-1}d\theta$ . We take  $\alpha$  as a real parameter defined on  $0 \leq \alpha < 2\pi$  so that we are considering an  $U(1)$  transformation.

It is obvious that the integral in the r.h.s. of eq.(5.3.41) remains invariant so that the Lagrangian has not changed. Now, at the level of the chiral superfield the change implies

$$\Phi = \phi + \sqrt{2}\theta\psi + \theta^2 F \rightarrow \Phi' = \phi + \sqrt{2}\exp(-i\alpha)\theta'\psi + \exp(-2i\alpha)\theta'^2 F \quad (5.3.44)$$

so that we can think that in component fields the change implies

$$\begin{aligned} \phi &\rightarrow \phi' = \phi \\ \psi &\rightarrow \psi' = \exp(-i\alpha)\psi \\ F &\rightarrow \psi' = \exp(-2i\alpha)F \end{aligned} \quad (5.3.45)$$

We can then state the invariance at the component Lagrangian level by saying that transformation (5.3.45) leaves it invariant.

We shall now generalize a little this transformation. Consider the particular case in which the superpotential  $W[\Phi_i]$  for a system with  $N$  chiral superfields  $\Phi_i$  contains just cubic products (that is, there are only dimensionless parameters  $g_{ijk}$ ). If one performs the same

global change of the Grassmann variables as before and impose chiral superfields to change according to

$$\Phi_i(x, \theta) \rightarrow \exp\left(-\frac{2i\alpha}{3}\right) \Phi_i(x, \theta') \quad (5.3.46)$$

Lagrangian (5.3.40) will be invariant. Indeed, under the change (5.3.43) the measure compensates the overall factor coming from the change in the product of 3 superfields since the factor  $2/3$  in the argument of the exponential in (5.3.46) cancels out the  $(\theta, \bar{\theta})$  Jacobian.

This symmetry is called *R-symmetry* and in component fields it takes the form (remember that  $\theta' = \exp(i\alpha)\theta$ )

$$\begin{aligned} \phi &\rightarrow \phi' = \exp\left(-\frac{2i\alpha}{3}\right) \phi \\ \psi &\rightarrow \psi' = \exp\left(\frac{i\alpha}{3}\right) \psi \\ F &\rightarrow F' = \exp\left(\frac{4i\alpha}{3}\right) F \end{aligned} \quad (5.3.47)$$

We see that the R-charge of fermions and bosons differs in 1 unit in such a way that the superpotential vertices have zero net R charge (Think in the  $\phi_i\psi_j\psi_k$  vertex).

Suppose one now considers superpotential also including linear and quadratic terms. It is often possible to define a  $U(1)$  R symmetry by replacing (5.3.46) by

$$\Phi_i(x, \theta) \rightarrow \exp(-i\beta_i) \Phi_i(x, \theta') \quad (5.3.48)$$

and demanding that the  $\beta_i$  are such that make the superpotential change according to

$$W[\Phi(x, \theta)] \rightarrow \exp(2i\alpha) W[\Phi(x, \theta')] \quad (5.3.49)$$

That is, the R-symmetric superpotential needs to have charge  $R_W = 2$ .

Originally R-symmetry was introduced in connection with the problem of fermion number conservation in supersymmetric theories constructed from chiral superfields [1]-[2]. As we shall see, the continuous  $U(1)$  R-symmetry should be necessarily broken so as to allow for the gravitino and gluinos to acquire masses. There is a discrete remnant of this broken R-symmetry which is called R-parity whose possible violation opens the question of conservation or non-conservation of baryon and lepton numbers in supersymmetric theories.

R-symmetry can also be seen at the level of supersymmetry generators. Indeed, in the supersymmetry algebra involving the spinor generators  $Q_\alpha$  and  $\bar{Q}_{\dot{\alpha}}$  (eqs.(2.7.77)-(2.7.80))

$$\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = 2\sigma_{\alpha\dot{\beta}}{}^\mu P_\mu \quad (5.3.50)$$

$$\{Q_\alpha, Q_\beta\} = \{\bar{Q}_{\dot{\alpha}}, \bar{Q}_{\dot{\beta}}\} = 0 \quad (5.3.51)$$

$$[P_\mu, Q_\alpha] = [P_\mu, \bar{Q}_{\dot{\beta}}] = 0 \quad (5.3.52)$$

$$[Q_\alpha, J_{\mu\nu}] = \frac{1}{2}(\sigma_{\mu\nu})_\alpha^\beta Q_\beta \quad (5.3.53)$$

$$[\bar{Q}_{\dot{\alpha}}, J_{\mu\nu}] = -\frac{1}{2}\bar{Q}_{\dot{\beta}}(\sigma_{\mu\nu})^{\dot{\beta}}_{\dot{\alpha}} \quad (5.3.54)$$

there is the  $U(1)_R$  symmetry we have introduced above,

$$\begin{aligned} Q &\rightarrow Q' = \exp(i\alpha)Q \\ \bar{Q} &\rightarrow \bar{Q}' = \exp(-i\alpha)\bar{Q} \end{aligned} \quad (5.3.55)$$

which can be confirmed just by implementing transformations (5.3.43) in the realization of the supersymmetry generators in terms of differential operators, eqs.(3.2.22)-(3.2.23).

This can be seen to be related to the fact that the R-symmetry generator  $R$  satisfies

$$[R, Q] = -Q, \quad [R, \bar{Q}] = \bar{Q} \quad (5.3.56)$$

We shall prove this as follows. Consider just one chiral superfield  $\Phi$  transforming according to (5.3.48)

$$\mathcal{R}\Phi(x, \theta) = \exp(-i\beta)\Phi_i(x, \exp(i\alpha\theta)) \quad (5.3.57)$$

this meaning that each component transform according to

$$\mathcal{R}\phi = \exp(-i\beta)\phi \Rightarrow \delta_R\phi = -i\beta\phi \quad (5.3.58)$$

$$\mathcal{R}\psi = \exp(-i(\beta + \alpha))\psi \Rightarrow \delta_R\psi = -i(\beta + \alpha)\psi \quad (5.3.59)$$

From (5.3.58) one then has, after applying an infinitesimal supersymmetry transformation

$$\delta_\epsilon\delta_R\phi = -i\beta\delta_\epsilon\phi = -i\sqrt{2}\beta\epsilon\psi \quad (5.3.60)$$

while applying the transformations in the opposite order one has

$$\delta_R\delta_\epsilon\phi = \sqrt{2}\epsilon\delta_R\psi = -i\sqrt{2}\epsilon(\beta + \alpha)\psi \quad (5.3.61)$$

Then, subtracting both identities one finds

$$\delta_R\delta_\epsilon\phi - \delta_\epsilon\delta_R\phi = -i\sqrt{2}\alpha\epsilon\psi = -i\alpha\delta_\epsilon\phi \quad (5.3.62)$$

The same can be seen for the fermionic component (and the auxiliary field if it has not been replaced) so that one can write in terms of generators

$$[\mathcal{R}, Q]\Phi = Q\Phi \quad (5.3.63)$$

There is a very important problem, connected to supersymmetry breaking in which R symmetry plays a central role. We shall discuss it in the next section.

## 5.4 The Nelson-Seiberg “Theorem”

In 1993 Nelson and Seiberg published a celebrated paper [3] titled “R symmetry breaking versus supersymmetry breaking,” showing a kind of tension or contradiction between the necessity of supersymmetry breaking and R-symmetry. One can summarize their result with the following statements:

- Supersymmetry breaking needs exact R symmetry (otherwise, fine tuning is needed)
- R-symmetry should be broken (otherwise one should have detected massless gauginos, the fermionic partners of gauge fields like the photon)
- If R-symmetry is broken supersymmetric vacua arise...

The solution to such chain of contradictions was postponed for more than 10 years until Intriligator, Seiberg and Shih [4] showed, using variants of the O’Raifeartaigh model, that a broken R-symmetry and the consequence of supersymmetric vacua can be accepted because it is rather easy to also have metastable non-supersymmetric vacua sufficiently long-lived as to guarantee that the present state of the Universe corresponds to one of them. In fact, their work generated many investigations that suggest that metastable supersymmetry breaking is generic in field theory (and string theory).

Let us present the Nelson-Seiberg analysis restricted to the case of models with just chiral fields (no gauge symmetry) and a canonical Kähler potential (but the analysis can be extended in particular if the metric defined by a complicated Kähler potentials non singular. Nelson and Seiberg have developed their analysis including quantum corrections and for that reason they talk about *effective* theories (and *effective* superpotentials). At this stages, we shall just reproduce the analysis for the classical theory and come back to the problem after discussing quantum corrections in class X.

As we have seen, the criterium for unbroken supersymmetry is (see (5.2.17))

$$\frac{\partial W}{\partial \phi_i} = 0, \quad i = 1, 2, \dots, N \quad (5.4.64)$$

We now examine whether or not there is a supersymmetric minimum for various possibilities related to possible global symmetries of the superpotential:

### 1. No symmetries

If there are no global symmetries the conditions (5.4.64) represent N equations for N unknowns, which are soluble for generic  $W[\phi]$ <sup>1</sup>. Hence supersymmetry should be

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<sup>1</sup>The concept of genericity, which will become important when considering quantum corrections means, at this level, that no fine tuning of the coefficient is implied.

unbroken. It may be possible to find special superpotentials with no global symmetries for which the conditions (2.2) are not soluble, however these do not satisfy the genericity assumption. They are unstable under small variations in the couplings.

## 2. Continuous global symmetry which commutes with supersymmetry

Constraining the theory by means of a non-R global symmetry does not allow one to evade the conclusion that supersymmetry is unbroken. Because  $W[\Phi]$  is holomorphic, if there are  $n$  generators of global symmetries and hence it may be written as a function of  $N - n$  variables.

To see this, consider the case of just one global symmetry, i.e. there is a  $U(1)$  global charge for which the fields  $\Phi_i$  have charges  $q_i$ . Invariance under such transformation will establish a (linear) relation among the fields which can be used to eliminate one of them. So, let us suppose that the field  $\phi_N$  has charge  $q_N \neq 0$  and eliminate it. The superpotential, which necessarily has zero net charge in order to lead to an invariant Lagrangian can be then written in terms of the new variables carrying each one zero charge,

$$X_i = \frac{\phi_i}{\phi_N^{q_i/q_N}}, \quad i = 1, 2, \dots, N - 1 \quad (5.4.65)$$

Written in the form  $W = W[X_i]$  we explicitly see that the superpotential is independent of  $\phi_N$ . The minimum equation corresponding to  $\phi_N$  will then be automatically satisfied. We are then left with a system of  $N - 1$  equations with  $N - 1$  unknowns and applying the arguments in point 1 we conclude that supersymmetry is unbroken for a generic potential.

If in general  $W$  is independent of  $n$  complex variables,  $n$  of the equations (5.4.64) will be automatically satisfied. The other equations give  $N - n$  constraints for  $N - n$  unknowns, which typically have solutions. The counting of equations and unknowns in the case of a global discrete symmetry is as in case 1 above (no symmetries).

## 3. Spontaneously broken $U(1)$ R-symmetry

The superpotential needs in this case to carry charge 2 under the R-symmetry, and so if  $\Phi_n$  receives an expectation value and carries R-charge  $q_N$ ,  $W[\Phi]$  may be written as

$$W = \Phi_N^{2/q_N} f(X_i) \quad (5.4.66)$$

with  $X_i$  defined as in (5.4.65) and  $f$  a generic function

Now equation (5.4.64) for supersymmetry breaking can be written as the following differential equations that should be obeyed by  $f$ ,

$$\frac{\partial f}{\partial X_i} = 0 \quad \text{for } i = 1, 2, \dots, N-1 \quad (5.4.67)$$

$$f = 0 \quad (5.4.68)$$

Eq. (5.4.68), resulting from (5.4.64) in the case  $i = N$  holds because we have chosen to call  $\Phi_n$  the field that is non-zero and finite. We have  $N-1$  **unknowns** and  $N$  **equations** to satisfy and this is not possible in a generic case. We therefore conclude that

**In this case, generically, there is no supersymmetric solution**

From this result we conclude that in order to have supersymmetry breaking we need the existence of an R-symmetry. As we shall see after introducing supersymmetric gauge theories, if R-symmetry is not broken the fermionic partners of gauge fields (the gauginos), should be massless (they necessarily have a non zero R charge and a mass term will then explicitly break R-symmetry. The conundrum can be bypassed if one accepts a small breaking of R symmetry (which then leads to massive gauginos). Of course the price one should pay is to accept the appearance of a supersymmetric vacuum but if the non-supersymmetric one exists (as a metastable vacuum since the symmetric one has the lowest possible energy) and they are separated by a very large and high barrier, there is the possibility of making contact with realistic models.

Let us remark that the important point in the demonstration is that the superpotential has zero charge for symmetries of the type 2 and non zero charge for R-symmetry. This makes the superpotential depend only on zero charge fields in point 2 and to have the crucial overall factor depending on  $\phi_N$  in point 3.

## 5.5 A toy model with a metastable vacuum

Let us modify the original basic O’Raifeartaigh model (5.2.19) by a small correction

$$W = \lambda X + m\Phi_1\Phi_2 + gX\Phi_1^2 + \frac{1}{2}\epsilon m\Phi_2^2 \quad (5.5.69)$$

with  $|\epsilon| \ll 1$ . This added term breaks the  $U(1)_R$  symmetry as can be seen from the following analysis.



To ensure R-symmetry superpotential  $W$  should have R-charge  $R_W = 2$ . This forces the fields in the r.h.s. of (5.2.19), to have the following charges:

$$\begin{aligned} R_X &= 2 \text{ because of the first term} \\ R_{\Phi_1} &= 0 \text{ because of the third term} \end{aligned} \tag{5.5.70}$$

$$R_{\Phi_2} = 2 \text{ because of the second term} \tag{5.5.71}$$

In view of this assignment, the added term has R-charge 4 thus so that it explicitly breaks R-symmetry. As before, the scalar potential is given by

$$V = |F_X|^2 + |F_{\phi_1}|^2 + |F_{\phi_2}|^2 \tag{5.5.72}$$

but instead of eqs.(5.2.21) we now have for the system corresponding to a supersymmetric vacuum

$$\begin{aligned} -F_X^* &= \left. \frac{\partial W}{\partial X} \right|_{vac} = \lambda + g\phi_1^2 = 0 \\ -F_1^* &= \left. \frac{\partial W}{\partial \Phi_1} \right|_{vac} = m\phi_2 + 2gX\phi_1 = 0 \\ -F_2^* &= \left. \frac{\partial W}{\partial \Phi_2} \right|_{vac} = m\phi_1 + \epsilon m\phi_2 = 0 \end{aligned} \tag{5.5.73}$$

where only the third equation has changed but in a way that allows for two supersymmetric vacuum solutions

$$\phi_{1S} = \pm \sqrt{-\lambda/g}, \quad \phi_{2S} = \mp \frac{1}{\epsilon} \sqrt{-\lambda/g}, \quad X_S = \frac{m}{2g\epsilon} \tag{5.5.74}$$

Concerning non-supersymmetric vacua, the one existing when  $\epsilon = 0$  which corresponded to  $\phi_1 = \phi_2 = 0$  will continue to exist with a small modification provided  $|\epsilon| \ll 1$  and  $m^2 > 2\lambda g$  (a no-tachyon condition, see (5.2.30)).

As  $\epsilon \rightarrow 0$  the supersymmetric preserving vacua (5.5.74) are pushed to infinity and at  $\epsilon = 0$  it ceases to exist and we are left with the pseudomoduli space of the non-supersymmetric vacuum

## 5.6 Appendix: Kähler geometry

A Kähler manifold is analytic. It can be parametrized in terms of coordinates  $a^i$  and  $a^{*j}$  ( $i = 1, 2, \dots, n$ ). Under the analytic transformation

$$a^i = a^i(a') \quad a^{*i} = a^{*i}(a^{*'}) \tag{5.6.75}$$

differentials and derivatives transform as usual

$$\begin{aligned} da'^i &= \frac{\partial a'^i}{\partial a^j} da^j & a^{*i} &= \frac{a^{*i}}{\partial a^{*j}} da^{*j} \\ \frac{\partial}{\partial a'^i} &= \frac{\partial a^j}{\partial a'^i} \frac{\partial}{\partial a^j} & \frac{\partial}{\partial a^{*i}} &= \frac{\partial a^{*j}}{\partial a^{*i}} \frac{\partial}{\partial a^{*j}} \end{aligned} \quad (5.6.76)$$

The same happens for vector fields. For example, for covariant ones

$$\begin{aligned} V'_i(a', a^{*'}) &= \frac{\partial a^j}{\partial a'^i} V_j(a, a^*) \\ V'_{i^*}(a', a^{*'}) &= \frac{\partial a^{*j}}{\partial a'^{i^*}} V_{j^*}(a, a^*) \end{aligned} \quad (5.6.77)$$

and analogously for contravariant ones.

The first condition on a manifold to be a Kähler manifold is that it has to be endowed with a positive definite and invertible hermitian metric  $g_{ij^*}$ ,

$$\begin{aligned} V_i &= g_{ij^*} V^{j^*} & V_{j^*} &= g_{ij^*} V^i \\ V^i &= g^{ij^*} V_{j^*} & V^{j^*} &= g^{ij^*} V_i \end{aligned} \quad (5.6.78)$$

The second requirement is that the covariant derivative must respect the analytic structure, this implying that the connection should satisfy

$$\Gamma_{ij}^{k^*} = \Gamma_{i^*j}^{k^*} = 0 \quad (5.6.79)$$

$$\begin{aligned} \nabla_i V_j &= \frac{\partial}{\partial a^i} V_j - \Gamma_{ij}^k V_k \\ \nabla_{i^*} V_j &= \frac{\partial}{\partial a^{i^*}} V_j - \Gamma_{i^*j}^k V_k \end{aligned} \quad (5.6.80)$$

The third condition is that the condition should be compatible with the hermitian metric,

$$\nabla_k g_{ij^*} = 0, \quad \nabla_{k^*} g_{ij^*} = 0 \quad (5.6.81)$$

One defines the transformation law for the connection so that covariant derivatives transform as tensors. It is consistent to put torsion to zero so that only the symmetric part of the connection survives,

$$\Gamma_{ij}^k = \Gamma_{ji}^k \quad (5.6.82)$$

and also one can choose

$$\Gamma_{i^*j}^k = 0 \quad (5.6.83)$$

Eqs.(5.6.82) and (5.6.83) are the two remaining conditions that define a Kähler manifold.

With all those conditions one has the following properties

- The only nonzero components are  $\Gamma_{ij}^k$  and its complex conjugate,  $(\Gamma_{ij}^k)^* = \Gamma_{i^*j^*}^{k^*}$
- Integrability conditions:

$$\frac{\partial}{\partial a^k} g_{ij^*} = \frac{\partial}{\partial a^i} g_{kj^*} , \quad \frac{\partial}{\partial a^{*k}} g_{ij^*} = \frac{\partial}{\partial a^{*j}} g_{ik^*} \quad (5.6.84)$$

From integrability conditions one has the important fact that the metric **is a derivative of a scalar function**  $K$ ,

$$g_{ij^*} = \frac{\partial}{\partial a^i} \frac{\partial}{\partial a^{*j}} K(a, a^*) \quad (5.6.85)$$

Function  $K$  is called the Kähler potential. Its derivatives determine the metric and the connection. One can define a Kähler manifold just by stating that the metric takes the form (5.6.85) and then deduce all conditions and properties. Condition (5.6.85) should be understood as a local condition.

We see that the Kähler potential completely specifies the Kähler geometry. Note that the metric is invariant under analytic shifts of  $K$ , known as Kähler transformations,

$$K(a, a^*) \rightarrow K(a, a^*) + F(a) + F^* a^* \quad (5.6.86)$$

As usual, the curvature of a Kähler manifold can be defined from the commutator of two covariant derivatives

$$\begin{aligned} [\nabla_i, \nabla_j] V_k &= R_{ijk}^l V_l \\ [\nabla_i, \nabla_{j^*}] V_k &= R_{ij^*k}^l V_l \end{aligned} \quad (5.6.87)$$

Let us conclude by stressing that a Kähler manifold is a complex manifold, a Riemannian manifold, and a symplectic manifold, with these three structures all mutually compatible.

A manifold with a Hermitian metric is an almost Hermitian manifold; a Kähler manifold is a manifold with a Hermitian metric that satisfies an integrability condition, which has several equivalent formulations.

- on-shell trafo



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# Chapter 6

## Supersymmetric gauge theories

1/7/10

In the construction of supersymmetric gauge models one needs a new kind of constrained superfield, which will be called vector superfields. They can be seen as the supersymmetric generalization of gauge fields since from them one can construct supersymmetric field strengths which will lead to the supersymmetric version of Maxwell and Yang-Mills actions.

It is important to stress that all relevant renormalizable Lagrangians resulting from the supersymmetric extensions of the usual Lagrangians discussed in ordinary Quantum Field Theory can be constructed just from scalar and vector superfields.

We shall then start by defining and studying the properties of vector superfields. Then, we shall explain how does one construct supersymmetric gauge theories.

### 6.1 Vector superfields: the Abelian case

As for chiral superfields, obtained from general superfields by imposing a constraint, one can define another kind of constrained superfields  $V$ , the vector superfields, defined through the condition

$$V = V^\dagger \quad (6.1.1)$$

Recall that a general superfield was defined in eq.(3.1.9) as

$$\Phi(x, \theta, \bar{\theta}) = \phi(x) + \theta\psi(x) + \bar{\theta}\bar{\chi}(x) + \theta\theta F(x) + \bar{\theta}\bar{\theta} G(x) + \theta\sigma^\mu\bar{\theta}v_\mu(x) + \theta\theta\bar{\theta}\bar{\lambda}(x) + \bar{\theta}\bar{\theta}\theta\eta(x) + \theta\theta\bar{\theta}\bar{\theta}D(x) \quad (6.1.2)$$

Now, in terms of its power series expansion,  $V$  will be written in the form

$$\begin{aligned}
V = & C(x) + i\theta\chi(x) - i\bar{\theta}\bar{\chi}(x) + \frac{i}{2}\theta^2(M(x) + iN(x)) - \frac{i}{2}\bar{\theta}^2(M(x) - iN(x)) \\
& -\theta\sigma^\mu\bar{\theta}A_\mu + i\theta\theta\bar{\theta}\left(\bar{\lambda}(x) + \frac{i}{2}\bar{\sigma}^\mu\partial_\mu\chi(x)\right) - i\bar{\theta}\bar{\theta}\theta\left(\lambda(x) + \frac{i}{2}\sigma^\mu\partial_\mu\bar{\chi}(x)\right) \\
& + \frac{1}{2}\theta\theta\bar{\theta}\bar{\theta}\left(D(x) + \frac{1}{2}\square C(x)\right)
\end{aligned} \tag{6.1.3}$$

The choice of certain combinations of fields was chosen in view of condition (6.1.1) and also of future notational simplifications. Indeed, taking  $C, D, M, N, A_\mu$  as real functions condition (6.1.1) will be automatically satisfied. The d'Alembertian in the last term of the superfield is added in view of what will be the definition of gauge invariance. One can see from (6.1.3) that a vector superfield has 8 bosonic degrees of freedom together with the 8 fermionic ones (the two component complex  $\lambda$  and  $\chi$ ).

The choice of condition (6.1.1) was originated by the following observation [1]. Consider the hermitian field  $\Phi + \Phi^\dagger$  resulting from the sum of chiral and antichiral fields,

$$\begin{aligned}
\Phi + \Phi^\dagger = & \Lambda(x) + \Lambda^*(x) + \sqrt{2}(\sqrt{\theta}\psi + \bar{\theta}\bar{\psi}) + \theta\theta F + \bar{\theta}\bar{\theta}F^* + i\theta\sigma^\mu\bar{\theta}\partial_\mu(\Lambda - \Lambda^*) \\
& + \frac{i}{\sqrt{2}}\theta\theta\bar{\theta}\bar{\sigma}^\mu\partial_\mu\psi + \frac{i}{\sqrt{2}}\bar{\theta}\bar{\theta}\theta\sigma^\mu\partial_\mu\bar{\psi} = \frac{1}{4}\theta\theta\bar{\theta}\bar{\theta}\square(\Lambda + \Lambda^*)
\end{aligned} \tag{6.1.4}$$

The gradient appearing as a coefficient of the  $+i\theta\sigma^\mu\bar{\theta}$  factor can be identified with that appearing in the gauge transformation of gauge fields. It is then natural to propose as the supersymmetric generalization of the gauge transformation for the vector superfield the following transformation law

$$V \rightarrow V + \Phi + \Phi^\dagger \tag{6.1.5}$$

From this, one can easily determine the change in component fields

$$\begin{aligned}
C & \rightarrow C + \Lambda + \Lambda^* \\
\chi & \rightarrow \chi - i\sqrt{2}\psi \\
M + iN & \rightarrow M + iN - 2iF \\
A_\mu & \rightarrow A_\mu - i\partial_\mu(\Lambda - \Lambda^*) \\
\lambda & \rightarrow \lambda
\end{aligned} \tag{6.1.6}$$

$$D \rightarrow D \tag{6.1.7}$$

As expected,  $A_\mu$  changes as a  $U(1)$  gauge field while neither the fermion  $\lambda$  (the gaugino) nor the (auxiliary) boson  $D$  change.



At this point it is important to discuss a very useful gauge fixing condition known as the Wess-Zumino gauge. It allows to put  $C, \chi, M$  and  $N$  all to zero just by choosing

$$\Lambda + \Lambda^* = -C, \quad \psi = -\frac{i}{2}\chi, \quad F = -\frac{i}{2}(M + iN) \quad (6.1.8)$$

In this gauge  $V$  takes the very economic form

$$V = -\theta\sigma^\mu\bar{\theta}A_\mu + i\theta\theta\bar{\theta}\bar{\lambda}(x) - i\bar{\theta}\bar{\theta}\theta\lambda(x) + \frac{1}{2}\theta\theta\bar{\theta}\bar{\theta}D(x) \quad (6.1.9)$$

Moreover, by computing powers of  $V$  one gets

$$\begin{aligned} V^2 &= -\frac{1}{2}\theta\theta\bar{\theta}\bar{\theta}A_\mu A^\mu \\ V^3 &= 0 \end{aligned} \quad (6.1.10)$$

One should stress that, up to now, we have just used the real part of  $\Lambda$  to eliminate the scalar field  $C$ . Concerning  $Im\Lambda$ , it can be chosen so as to eliminate one component of the gauge field (see eqs.(6.1.7)) leaving just 3 independent components of the 4-vector. This, together with the auxiliary field  $D$  gives 4 bosonic degrees of freedom, the same number as that corresponding to the the Weyl fermion  $\lambda$ . When one considers the on-shell theory by using the equations of motion the auxiliary field can be eliminated and one is left with two degrees of freedom for bosons and fermions, the two helicities of the massless gauge field and those of the gaugino.

One should notice that the choice of the Wess-Zumino gauge breaks supersymmetry in the sense that

$$\delta_{susy}V_{WZ} \neq V'_{WZ} \quad (6.1.11)$$

As it happens with chiral superfields, the highest component of the vector multiplet (i.e., the auxiliary field  $D$ ) transforms as a total derivative and thus is a good candidate for a supersymmetry invariant Lagrangian. Notice that it does not matter in which gauge the vector multiplet is written down: the difference between it and that arising in the WZ gauge is again proportional to a total spacetime derivative, and so it is irrelevant for the dynamics.

The fact that one just disposes of  $V$  and  $V^2$  and that they both already have non trivial powers of  $\theta$  or  $\bar{\theta}$  implies that there is not enough structure in  $V$  to generate derivatives leading to a nontrivial dynamics. A remedy for this is to apply supercovariant derivatives which introduce automatically the spacetime derivatives needed for kinetic terms.

A natural first possibility would be to consider fields like  $D_\alpha V$  and  $\bar{D}_{\dot{\alpha}}V$ . However, this introduces new complications since these fields are not real, i.e.  $D_\alpha V^\dagger \neq D_\alpha V$ , and not gauge invariant, i.e.  $D_\alpha(\Phi + \Phi^\dagger) \neq 0$ , for any given chiral field  $\Phi$ .

These problems can be overcome if one instead considers the following objects

$$\begin{aligned} W_\alpha &= -\frac{1}{4}\bar{D}\bar{D}D_\alpha V \\ \bar{W}_{\dot{\alpha}} &= -\frac{1}{4}DD\bar{D}_{\dot{\alpha}}V \end{aligned} \quad (6.1.12)$$

These superfields are chiral and gauge-invariant. They are chiral because  $D^3 = 0$  and  $\bar{D}^3 = 0$  (the operator  $D$  is a spinor operator) and hence

$$\begin{aligned} \bar{D}_{\dot{\alpha}}W_\alpha &= 0 \\ D_\alpha\bar{W}_{\dot{\alpha}} &= 0 \end{aligned} \quad (6.1.13)$$

As for gauge invariance, if  $V \rightarrow V' = V + \Phi + \Phi^\dagger$  one has

$$\bar{D}\bar{D}D_\alpha(\Phi + \Phi^\dagger) = \varepsilon^{\dot{\alpha}\dot{\beta}}\bar{D}_{\dot{\alpha}}\bar{D}_{\dot{\beta}}D_\alpha\Phi = -2i\varepsilon^{\dot{\alpha}\dot{\beta}}\bar{D}_{\dot{\alpha}}\sigma_{\alpha\dot{\beta}}^\mu\partial_\mu\Phi = 0 \quad (6.1.14)$$

so that  $W_\alpha$  remains unchanged.

We then conclude that we can interpret  $W_\alpha$  and  $\bar{W}_{\dot{\alpha}}$  as the supersymmetric version of field strengths. Since  $W_\alpha$  and  $\bar{W}_{\dot{\alpha}}$  are chiral fields, it will be convenient to compute their components working (in the Wess-Zumino gauge) with variables  $y = x + i\theta\sigma\bar{\theta}$  and  $y^\dagger = x - i\theta\sigma\bar{\theta}$ .

$$\begin{aligned} V &= -\theta\sigma^\mu\bar{\theta}A_\mu(y) + i\theta\theta\bar{\theta}\bar{\lambda}(y) - i\bar{\theta}\bar{\theta}\theta\lambda(y) + \frac{1}{2}\theta\theta\bar{\theta}\bar{\theta}(D(y) - i\partial_\mu A^\mu(y)) \\ &= -\theta\sigma^\mu\bar{\theta}A_\mu(y^\dagger) - i\bar{\theta}\bar{\theta}\theta\lambda(y^\dagger) + i\theta\theta\bar{\theta}\bar{\lambda}(y^\dagger) + \frac{1}{2}\theta\theta\bar{\theta}\bar{\theta}(D(y^\dagger) + i\partial_\mu A^\mu(y^\dagger)) \end{aligned} \quad (6.1.15)$$

The result is

$$\begin{aligned} W_\alpha &= -i\lambda_\alpha(y) + \left( \delta_\alpha^\beta D(y) - \frac{i}{2}(\sigma^\mu\sigma^\nu)_\alpha^\beta(\partial_\mu A_\nu(y) - \partial_\nu A_\mu(y)) \right) \theta_\beta + \theta\theta\sigma_{\alpha\dot{\alpha}}^\mu\partial_\mu\bar{\lambda}^{\dot{\alpha}}(y) \\ \bar{W}_{\dot{\alpha}} &= i\bar{\lambda}_{\dot{\alpha}}(y^\dagger) + \left( \varepsilon_{\dot{\alpha}\dot{\beta}}D(y^\dagger) + \frac{i}{2}\varepsilon_{\dot{\alpha}\dot{\gamma}}(\bar{\sigma}^\mu\bar{\sigma}^\nu)^{\dot{\gamma}\dot{\beta}}(\partial_\mu A_\nu(y^\dagger) - \partial_\nu A_\mu(y^\dagger)) \right) \bar{\theta}^{\dot{\beta}} \\ &\quad - \varepsilon_{\dot{\alpha}\dot{\beta}}\bar{\theta}^{\dot{\beta}}\bar{\sigma}^{\mu\dot{\beta}\alpha}\partial_\mu\bar{\lambda}_\alpha(y^\dagger) \end{aligned} \quad (6.1.16)$$

The superfields  $W_\alpha, \bar{W}_{\dot{\alpha}}$  contain only the gauge invariant fields  $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$  (the gauge field strength),  $\lambda$  (the gaugino) and  $D$  (the auxiliary field). In this sense it can be considered a ( $U(1)$  in this case) superfield strength. They are chiral fields and satisfy the additional constraint

$$D^\alpha W_\alpha = \bar{D}^{\dot{\alpha}}\bar{W}_{\dot{\alpha}} \quad (6.1.17)$$

For the lowest component this relation simply express the fact that  $D$  is a real field.

Consider now the highest component of the product  $W_\alpha W^\alpha$ . One obtains, up to a total derivative

$$W^\alpha W_\alpha|_{\theta\theta} = -\frac{1}{2}F_{\mu\nu}F^{\mu\nu} - 2i\lambda\sigma^\mu\partial_\mu\bar{\lambda} + D^2 + \frac{i}{4}F^{\mu\nu}F^{\rho\sigma}\varepsilon_{\mu\nu\rho\sigma} \quad (6.1.18)$$

For  $\bar{W}_{\dot{\alpha}}\bar{W}^{\dot{\alpha}}$  one has

$$\bar{W}_{\dot{\alpha}}\bar{W}^{\dot{\alpha}}|_{\theta\theta} = -\frac{1}{2}F_{\mu\nu}F^{\mu\nu} - 2i\lambda\sigma^\mu\partial_\mu\bar{\lambda} + D^2 - \frac{i}{4}F^{\mu\nu}F^{\rho\sigma}\varepsilon_{\mu\nu\rho\sigma} \quad (6.1.19)$$

To accommodate the gaugino term where we have used that

$$(\chi\sigma^\mu\bar{\psi})^\dagger = \psi\sigma^\mu\bar{\chi} \quad (6.1.20)$$

and ignored a total derivative.

Being both products chiral superfields their last component transforms under supersymmetry as a total derivative so that

$$L = \frac{1}{4} \int d^2\theta W^\alpha W_\alpha + \frac{1}{4} \int d^2\bar{\theta} \bar{W}_{\dot{\alpha}} \bar{W}^{\dot{\alpha}} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} - i\lambda\sigma^\mu\partial_\mu\bar{\lambda} + \frac{1}{2}D^2 \quad (6.1.21)$$

is the supersymmetric extension of the Maxwell theory. Only the gauge field,  $A_\mu$ , its supersymmetric partner, the gaugino  $\lambda$  and the auxiliary field  $D$  appear in the supersymmetric Abelian gauge theory Lagrangian.

One can easily find the supersymmetry transformations of the gauge invariant fields of the multiplet,

$$\begin{aligned} \delta F_{\mu\nu} &= -i\epsilon(\sigma^\mu\partial_\nu\bar{\lambda} - \sigma^\nu\partial_\mu\bar{\lambda}) - i\bar{\epsilon}(\bar{\sigma}^\mu\partial_\nu\lambda - \bar{\sigma}^\nu\partial_\mu\lambda) \\ \delta\lambda &= i\epsilon D + \sigma^{\mu\nu}\epsilon F_{\mu\nu} \\ \delta D &= \bar{\epsilon}\bar{\sigma}^\mu\partial_\mu\lambda - \epsilon\sigma^\mu\partial_\mu\bar{\lambda} \end{aligned} \quad (6.1.22)$$

## 6.2 Coupling to a chiral superfield

In order to couple the  $U(1)$  vector superfield to a charged chiral superfield one could be tempted to impose to this last field a transformation law of the form

$$\Phi'(y, \theta) = \exp(i\Lambda(x, \theta, \bar{\theta})) \Phi(y, \theta) \quad (6.2.23)$$

but it happens that the resulting transformed fields  $\Phi'$  is not a chiral superfield since in general  $\bar{D}\Lambda \neq 0$ . This can be obviously solved if  $\Lambda$  gets promoted to a chiral superfield  $\Lambda = \Lambda(y, \theta)$  so that  $\bar{D}\Lambda = 0$ . If this is so one has

$$\Phi(y, \theta) \rightarrow \Phi'(y, \theta) = \exp(i\Lambda(y, \theta)) \Phi(y, \theta) \quad (6.2.24)$$

But a new problem arises:  $\Lambda \neq \Lambda^\dagger$  and then the product  $\Phi^\dagger \Phi$ , which is the appropriate product to generate kinetic terms is not gauge invariant. This in turn can be solved if we define an invariant product of the form

$$K = \Phi^\dagger \exp(V) \Phi \quad (6.2.25)$$

Indeed, if we write the transformation law (6.1.5) for  $V$  in the form

$$V' = V + i(\Lambda^\dagger - \Lambda) \quad (6.2.26)$$

together with formula (6.2.24) for the chiral field transformation law one ends with

$$K \rightarrow K' = \Phi^\dagger \exp(-i\Lambda^\dagger) \exp(V + i(\Lambda^\dagger - \Lambda)) \exp(i\Lambda) \Phi = K \quad (6.2.27)$$

Being  $K$  gauge invariant, one can evaluate it in an arbitrary gauge and the most convenient one is evidently the Wess-Zumino gauge since then  $V^3 = 0$  and the exponential reduces just to the first three terms in its expansion. In order to compute  $K$  we then need the following three quantities

$$\Phi^\dagger \Phi|_{\theta\theta\bar{\theta}\bar{\theta}} = i\partial_\mu \bar{\psi} \bar{\sigma}^\mu \psi + \phi^* \square \phi + F^* F \quad (6.2.28)$$

$$\Phi^\dagger V \Phi|_{\theta\theta\bar{\theta}\bar{\theta}} = A^\mu \left( \frac{1}{2} \bar{\psi} \bar{\sigma}_\mu + \frac{i}{2} \phi^* \partial_\mu \phi - \frac{i}{2} \partial_\mu \phi^* \phi \right) - \frac{i}{\sqrt{2}} (\phi \bar{\lambda} \bar{\psi} - \phi^* \lambda \psi) + \frac{D}{2} \phi^* \phi \quad (6.2.29)$$

$$\Phi^\dagger V^2 \Phi|_{\theta\theta\bar{\theta}\bar{\theta}} = -\frac{1}{4} A_\mu A^\mu \phi^* \phi \quad (6.2.30)$$

Of course, instead of working with (6.2.25) we can work with a general Kähler potential, an arbitrary functional of such product.

Introducing a gauge coupling constant  $e$  and the usual normalization implied by the change  $\exp(V) \rightarrow \exp(2eV)$  we get from the Kähler part of the Lagrangian, after some rearrangements,

$$\begin{aligned} L_K = \int d^2\theta d^2\bar{\theta} \Phi^\dagger \exp(2eV) \Phi = & -D_\mu \phi^* D^\mu \phi - i\bar{\psi} \bar{\sigma}^\mu D_\mu \psi + F^* F \\ & + eD|\phi|^2 + ie\sqrt{2}\phi^* \lambda \psi - ie\sqrt{2}\bar{\lambda} \bar{\psi} \phi \end{aligned} \quad (6.2.31)$$

with the gauge covariant derivative  $D_\mu$  is defined as

$$D_\mu = \partial_\mu + ieA_\mu \quad (6.2.32)$$

both for boson and fermion fields. Note that the first three terms in (6.2.31) are nothing but the gauge covariant extension of the canonical Kähler potential  $\Phi^\dagger \Phi$  obtained by changing  $\partial_\mu \rightarrow D_\mu$  but there are three other terms which appear in order to ensure supersymmetry invariance.

## The superpotential and gauge invariance

Consider the case of  $N$  chiral superfields  $\Phi_i$  ( $i = 1, 2, \dots, N$ ) which under an Abelian gauge transformation change according to

$$\Phi_i \rightarrow \Phi'_i = \exp(iq_i\Lambda)\Phi_i \quad (6.2.33)$$

Here  $q_i$  represents the charge of each superfield  $\Phi_i$  under such gauge transformation. Note that all components of each superfield transform with the same charge. Since there are no derivative terms in the superpotential, global and local gauge invariance conditions will coincide.

Take the quadratic term in the superpotential,

$$W^{(2)} = \frac{1}{2}m_{ij}\Phi_i\Phi_j \quad (6.2.34)$$

It changes under transformation (6.2.33) according to

$$W^{(2)} \rightarrow W'^{(2)} = \exp(i(q_i + q_j)\Lambda) \frac{1}{2}m_{ij}\Phi_i\Phi_j \quad (6.2.35)$$

Gauge invariance then imposes severe conditions on the charges  $q_i$  and parameters  $m_{ij}$ . Indeed, for a given pair  $i, j$  either  $q_i + q_j = 0$  or  $m_{ij} = 0$ . The same analysis holds for cubic terms:  $q_i + q_j + q_k = 0$  or the corresponding coupling satisfies  $g_{ijk} = 0$ . An analogous analysis can be done for the non-Abelian case.

## D-term supersymmetry breaking

We have seen in the case of the Lagrangian for  $N$  chiral fields that supersymmetry can be broken through a mechanism usually referred to as “F term breaking”. This mechanism was based on the fact that the scalar potential was given by  $V = F_i^*F_i$  so that whenever the equations  $F_i = 0$  had no trivial solution  $F_i = 0$  for all  $i = 1, 2, \dots, N$  then  $V > 0$  and supersymmetry was broken.

When gauge fields are present, the scalar potential also includes a  $D^2$  term, with  $D$  the auxiliary field in the gauge multiplet, so that there is an extra condition for the existence of a supersymmetric vacuum: the  $D$  contribution should also vanish. Precisely Fayet and Iliopoulos [2] proposed a mechanism which leads to supersymmetry breaking through a non-zero  $D$ -term vev (in the  $U(1)$  case).

Before to describe the Fayet-Iliopoulos mechanism, let us consider the two terms in the component Lagrangian (6.2.31) where the field  $D$  appears:

$$L_D = \frac{1}{2}D^2 + eD|\phi|^2 \quad (6.2.36)$$

The equation of motion for  $D$  then trivially reduces to

$$D + e|\phi|^2 = 0 \quad (6.2.37)$$

and can be used to write  $L_D$  in terms of dynamical fields

$$L = -\frac{1}{2}e^4|\phi|^2 \quad (6.2.38)$$

Note that this is a typical “ $\lambda\phi^4$ ” term except that the coupling constant  $\lambda$  is not an independent parameter but is related to the square of the gauge coupling constant. This is a characteristic and important property which will be very important in analyzing (see chapter...) BPS states.

Let us see how the Fayet-Iliopoulos symmetry breaking mechanism works in the case of a model with a vector superfield coupled to two chiral fields which couple to the gauge field with opposite charges  $q_i = \pm e$  (so as to avoid chiral anomalies, this requiring  $\Sigma q_i = 0$ ). The corresponding Lagrangian is

$$L = \frac{1}{4}(W_\alpha W^\alpha + \bar{W}_{\dot{\alpha}} \bar{W}^{\dot{\alpha}} + \Phi_1^\dagger \exp(2eV)\Phi_1 + \Phi_2^\dagger \exp(-2eV)\Phi_2 + m(\Phi_1\Phi_2 + \Phi_1^\dagger\Phi_2^\dagger) + 2\kappa V \quad (6.2.39)$$

We included in this Lagrangian a term proportional to  $V$  (the Fayet-Iliopoulos term) which is supersymmetric invariant and, in the case of a  $U(1)$  gauge symmetry, gauge invariant. As announced, the scalar potential takes in this case the form

$$V_{scalar} = \frac{1}{2}D^2 + F_1^*F_1 + F_2^*F_2 \quad (6.2.40)$$

The  $D^2$  term comes from the gauge field part of the Lagrangian (see eq.(6.1.21)). Now the auxiliary fields equations of motion read

$$\begin{aligned} D + \kappa + e(\phi_1^*\phi_1 - \phi_2^*\phi_2) &= 0 \\ F_1 + m\phi_2^* &= 0 \\ F_2 + m\phi_1^* &= 0 \end{aligned} \quad (6.2.41)$$

an there is no solution making  $V_{scalar} = 0$  so supersymmetry is spontaneously broken. This is due to the presence of the Fayet-Iliopoulos term since if  $\kappa$  were absent from the first equation, a configuration with all fields equal to zero would be a solution of eqs.(6.2.41).)

One can use eqs.(6.2.41) to substitute the auxiliary fields in the potential (6.2.40). The answer is

$$V_{scalar}[\phi_1, \phi_2] = \frac{\kappa^2}{2} + (m^2 + e\kappa) |\phi_1|^2 + (m^2 - e\kappa) |\phi_2|^2 + \frac{e^2}{2} (|\phi_1|^2 - |\phi_2|^2)^2 \quad (6.2.42)$$

For  $m^2 > e\kappa$  both  $\phi_1$  and  $\phi_2$  have real masses. Being all terms positive, the minimum of this potential is attained for

$$D = \kappa, \quad \phi_1 = \phi_2 = 0 \quad (6.2.43)$$

leading to

$$V_{scalar}^{min} = \kappa^2 > 0 \quad (6.2.44)$$

Supersymmetry breaking is then induced because of the non-zero vacuum expectation value of  $D$ .

From the potential we see that the model describes two complex scalar fields with masses  $m_{\pm}^2 = m^2 \pm e\kappa$ , three spinor fields  $\psi_1, \psi_2, \lambda$  and one vector field  $A_\mu$ . The masses of the spinors and gauge field are unchanged by the symmetry breaking: both  $\psi_i$  retain their mass  $m$  while  $\lambda$  and  $A_\mu$  remain massless. Note that  $m_1^2 + m_2^2 = m^2$ .  $A_\mu$  plays the role of the gauge field for the unbroken  $U(1)$  gauge symmetry and  $\lambda$  is the Goldstone fermion arising from the spontaneous breaking of supersymmetry. Indeed, the change under a supersymmetry transformation for  $\lambda$ ,

$$\delta_\epsilon \lambda = i\epsilon D + \sigma^{\mu\nu} \epsilon F_{\mu\nu} \quad (6.2.45)$$

indicates that as soon as  $D$  takes its vacuum expectation value the transformation law becomes inhomogeneous,

$$\delta_\epsilon \lambda = i\epsilon \kappa + \dots \quad (6.2.46)$$

and this identifies  $\lambda$  as the Goldstone fermion.

When  $m^2 < e\kappa$ ,  $\phi_1 = \phi_2 = 0$  no longer minimizes the potential (6.2.42). To see this, we study the conditions

$$\begin{aligned} \frac{\partial V_{scalar}}{\partial \phi_1^*} &= (m^2 + e\kappa) \phi_1 + e^2 (|\phi|^2 - |\phi_2|^2) \phi_1 = 0 \\ \frac{\partial V_{scalar}}{\partial \phi_2^*} &= (m^2 - e\kappa) \phi_1 + e^2 (|\phi|^2 - |\phi_2|^2) \phi_2 = 0 \end{aligned} \quad (6.2.47)$$

One still has an extremum for  $\phi_1 = \phi_2 = 0$  but it does not correspond to a minimum since

$$\frac{\partial^2}{\partial \phi_2 \partial \phi_2^*} V_{scalar}[0, 0] = m^2 - e\kappa < 0 \quad (6.2.48)$$

There are also extrema at

$$\phi_1 = 0, \quad \phi_2 = \pm \phi_2^0 \quad (6.2.49)$$

with  $\phi_2^0$  defined through

$$(\phi_2^0)^2 = \frac{e\kappa - m^2}{e^2} \quad (6.2.50)$$

which do correspond to a minimum

$$V_{scalar}[0, \pm \phi_2^0] = V_0 = \frac{m^2}{2e^2}(2e\kappa - m^2) > 0 \quad (6.2.51)$$

Note that with an appropriate gauge transformation one can choose  $\phi_2^0$  to be real. If we expand the potential (6.2.42) (and the other terms in the Lagrangian) around such vacua,

$$\begin{aligned} \phi_1 &= \varphi_1 \\ \phi_2 &= \phi_2^0 + \varphi_2 \end{aligned} \quad (6.2.52)$$

one finds

$$L = V_0 + 2m^2|\varphi_1|^2 + \frac{e^2}{2}(\phi_2^0)^2(\varphi_2 + \varphi_2^*)^2 + (\phi_2^0)^2 A_\mu A^\mu + \dots \quad (6.2.53)$$

The mass term for the gauge field indicates that also gauge symmetry is broken canonically, with the Goldstone boson  $\text{Im}\varphi_2$  being eaten by the massive gauge field so that the total number of degrees of freedom remains unchanged. The symmetry breaking also modifies the fermion mass terms

$$L_{m \text{ fermion}} = -m(\psi_1\psi_2 + \bar{\psi}_1\bar{\psi}_2) + i\sqrt{2}e^2(\phi_2^0)^2(\bar{\psi}_2\bar{\lambda} - \psi_2\lambda) \quad (6.2.54)$$

Diagonalizing this Lagrangian through the transformation

$$\begin{aligned} \psi &= \psi_2 \\ \tilde{\psi} &= \frac{1}{\sqrt{m^2 + 2e^2(\phi_2^0)^2}} \left( m\psi_1 + ie\sqrt{2}\phi_2^0\lambda \right) \\ \tilde{\lambda} &= \frac{1}{\sqrt{m^2 + 2e^2(\phi_2^0)^2}} \left( m\lambda + ie\sqrt{2}\phi_2^0\psi_1 \right) \end{aligned} \quad (6.2.55)$$

one ends with

$$\tilde{L}_{m \text{ fermion}} = -\sqrt{m^2 + 2e^2(\phi_2^0)^2}(\psi\tilde{\psi} + \bar{\psi}\bar{\tilde{\psi}}) \quad (6.2.56)$$

and the Goldstone field  $\tilde{\lambda}$  remains massless and transforms inhomogeneously

$$\delta_\epsilon \tilde{\lambda} = -im\epsilon\sqrt{2e\kappa - m^2} + \dots \quad (6.2.57)$$

The model then describes two spinor fields of mass  $\sqrt{m^2 + 2e^2(\phi_2^0)^2}$ , one vector field and one scalar field, both with mass  $\sqrt{2}e\phi_2^0$ , one complex scalar of mass  $\sqrt{2}m$  and one massless Goldstone spinor.



### 6.3 Supersymmetric Electrodynamics [3]

In order to construct the supersymmetric extension of electrodynamics one needs two chiral superfields so that the corresponding two Weyl fermions can be combined to form one massive Dirac fermion. We shall call them  $\Phi_+$  and  $\Phi_-$ . They transform under gauge transformations according to

$$\Phi'_+ = \exp(-ie\Lambda)\Phi_+, \quad \Phi'_- = \exp(ie\Lambda)\Phi_- \quad (6.3.58)$$

so that they have opposite charges.

The Lagrangian reads

$$\begin{aligned} L_{QED} = & \frac{1}{4} \int d^2\theta d^2\bar{\theta} \left( W^\alpha W_\alpha + \bar{W}_{\dot{\alpha}} \bar{W}^{\dot{\alpha}} + \Phi_+^\dagger \exp(eV) \Phi_+ + \Phi_-^\dagger \exp(-eV) \Phi_- \right) \\ & + m \int d^2\theta \Phi_+ \Phi_- + m \int d^2\bar{\theta} \Phi_+^\dagger \Phi_-^\dagger \end{aligned} \quad (6.3.59)$$

which reduces to

$$\begin{aligned} L_{QED} = & -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} - i\lambda \sigma^\mu \partial_\mu \bar{\lambda} + \frac{1}{2} D^2 + |F_+|^2 + |F_-|^2 - \partial_\mu \phi_+^* \partial^\mu \phi_+ - \partial_\mu \phi_-^* \partial^\mu \phi_- \\ & + i(\partial_\mu \bar{\psi}_+ \bar{\sigma}^\mu \psi_+ + \partial_\mu \bar{\psi}_- \bar{\sigma}^\mu \psi_-) + \frac{e}{2} A^\mu (\bar{\psi}_+ \bar{\sigma}_\mu \psi_+ - \bar{\psi}_- \bar{\sigma}_\mu \psi_- + \phi_+^* \partial_\mu \phi_+ \\ & - \partial_\mu \phi_+^* \phi_- - \phi_-^* \partial_\mu \phi_- + \partial_\mu \phi_-^* \phi_+) - \frac{ie}{\sqrt{2}} (\phi_+ \bar{\psi}_+ \bar{\lambda} - \phi_+^* \psi_+ \lambda - \phi_- \bar{\psi}_- \bar{\lambda} - \phi_-^* \psi_- \lambda) \\ & + \frac{e}{2} D(|\phi_+|^2 - |\phi_-|^2) - \frac{e^2}{4} A_\mu A^\mu (|\phi_+|^2 + |\phi_-|^2) + m(\phi_+ F_- + \phi_- F_+) \\ & + m(\phi_+^* F_-^* + \phi_-^* F_+^* - \psi_+ \psi_- - \bar{\psi}_+ \bar{\psi}_-) \end{aligned} \quad (6.3.60)$$

We can combine two Weyl spinors to form a Dirac spinor

$$\psi_D = \begin{pmatrix} \psi_{+\alpha} \\ \bar{\psi}_{-\dot{\alpha}} \end{pmatrix} \quad (6.3.61)$$

Using

$$\bar{\psi}_{\dot{\alpha}} = \psi_\alpha^* \quad (6.3.62)$$

and

$$\gamma^0 = \begin{pmatrix} 0 & -I_{2 \times 2} \\ -I_{2 \times 2} & 0 \end{pmatrix} \quad (6.3.63)$$

we see that for example the fermionic mass term can be arranged as

$$-m(\psi_+ \psi_- + \bar{\psi}_+ \bar{\psi}_-) = m \bar{\psi}_D \psi_D \quad (6.3.64)$$

and the same for the rest of the Lagrangian. In particular the fermion mass coincides with that of the complex scalars  $\phi_{\pm}$ , its superpartner in the scalar multiplet. Concerning the vector multiplet, the photon and the photino (called the neutrino in the original Wess-Zumino paper [3]) remain massless. All auxiliary fields should be eliminated from the Lagrangian.

## 6.4 Non-Abelian gauge groups

The discussion above can be easily extended to the case in which the gauge group  $G$  is non-Abelian. Let us write the corresponding Lie algebra  $\mathcal{G}$  in the form

$$[t^a, t^b] = if^{abc}t_c, \quad \text{tr} t^a t^b = k\delta^{ab}, \quad k > 0 \quad (6.4.65)$$

The natural extension for the vector superfields  $V$  introduced in section 6.1 is accept that it takes values in  $\mathcal{G}$ , and the same for  $\Lambda$ , the superfield implementing gauge transformations,

$$V = V^a t^a, \quad \Lambda = \Lambda^a t^a \quad (6.4.66)$$

That is,  $V$  and  $\Lambda$  are now matrices,

$$V_{ij} = V^a t_{ij}^a, \quad \Lambda_{ij} = \Lambda^a t_{ij}^a \quad (6.4.67)$$

The vector superfield carries an adjoint representation index (and so will do its component fields  $(A_{\mu}^a, \lambda^a, D^a, a = 1, 2, \dots, \dim \mathcal{G})$ , in the Wess-Zumino gauge, see below). The generators  $(t_{ij}^a)$  will be taken as  $r \times r$  matrices ( $i, j = 1, 2, \dots, r$ ) with  $r$  determined by the representation  $\mathbf{r}$  to which the matter fields belong. In the case of chiral fields, they transform as

$$\Phi' = \exp(-i\Lambda)\Phi, \quad \Phi'^{\dagger} = \Phi^{\dagger} \exp(i\Lambda^{\dagger}) \quad (6.4.68)$$

That is,  $\Phi$  in fact carries an index  $i = 1, 2, \dots, r$  according to the  $\mathbf{r}$ -representation to which it belongs (we shall in general omit that index)

$$\Phi = \begin{pmatrix} \Phi_1 \\ \Phi_2 \\ \vdots \\ \vdots \\ \Phi_r \end{pmatrix} \quad (6.4.69)$$

Correspondingly, the generators  $t^a$  should be represented by  $r \times r$  matrices when acting on  $\Phi$  as in (6.4.68). Each component field in  $\Phi$  will accordingly belong to representation  $\mathbf{r}$  and again we shall in general omit the indices except when they clarify the formulæ.

The natural extension of the Abelian transformation law (6.2.26) to the non-Abelian case is

$$\exp(2gV') = \exp(-i\Lambda^\dagger) \exp(2gV) \exp(i\Lambda) \quad (6.4.70)$$

Expanding this expression one finds

$$2gV' = 2gV + i(\Lambda - \Lambda^\dagger) + \dots \quad (6.4.71)$$

showing that  $\delta V$  starts with a term which is  $V$  independent so that also non-Abelian gauge theories allow for the choice of the Wess-Zumino gauge ensuring that  $V^3 = 0$ .

The supersymmetric field strength  $W^\alpha$  (eq.(6.1.12)) can be readily extended according to

$$W_\alpha = -\frac{1}{4}\bar{D}\bar{D}\exp(-2gV)D_\alpha\exp(2gV) \quad (6.4.72)$$

taking values in the Lie algebra  $\mathcal{G}$ . One can then verify that  $W_\alpha$  transform covariantly,

$$W'_\alpha = \exp(-i\Lambda)W_\alpha\exp(i\Lambda) \quad (6.4.73)$$

and the same for  $\bar{W}_{\dot{\alpha}}$

$$\bar{W}'_{\dot{\alpha}} = \exp(-i\Lambda^\dagger)\bar{W}_{\dot{\alpha}}\exp(i\Lambda^\dagger) \quad (6.4.74)$$

One easily sees that the combination  $\exp(-2gV)D_\alpha\exp(2gV)$  is a gauge covariant superderivative

The most general Lagrangian for the supersymmetric renormalizable interaction of scalar, spinor and vector fields constructed from this objects is

$$L = L_K + L_{int} \quad (6.4.75)$$

$$\begin{aligned} L_K &= \frac{1}{16kg^2}\text{Tr}\left(W^\alpha W_\alpha|_{\theta\theta} + \bar{W}_{\dot{\alpha}}\bar{W}^{\dot{\alpha}}|_{\bar{\theta}\bar{\theta}}\right) + \Phi_i^\dagger \exp(2gV_{ij})\Phi_j \Big|_{\theta\theta\bar{\theta}\bar{\theta}} \\ L_{int} &= \left(\frac{1}{2}m_{ij}\Phi_i\Phi_j + \frac{1}{3}g_{ijk}\Phi_i\Phi_j\Phi_k\right) \Big|_{\theta\theta} + \text{h.c.} \end{aligned} \quad (6.4.76)$$

Gauge invariance requires the mass matrices  $m_{ij}$  and the coupling constants  $g_{ijk}$  to be totally symmetric tensor with respect to the gauge group. After some work one finds for  $L_K$

$$\begin{aligned} L_K &= -\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} - i\lambda^a \bar{\sigma}^\mu (D_\mu \lambda)^a + \frac{1}{2}D^a D^a - D_\mu \phi_i^* D^\mu \phi_i - i\bar{\psi}_i \sigma^\mu (D_\mu \psi)_i + F_i^* F_i \\ &\quad + i\sqrt{2}g(\phi_i^* t_{ij}^a \psi_j \lambda^a - \bar{\lambda}^a t^a \phi \bar{\psi}) + gD^a \phi_i^* t_{ij}^a \phi_j \end{aligned} \quad (6.4.77)$$

with

$$F_{\mu\nu}^a = \partial_\mu A_\nu - \partial_\nu A_\mu - gf^{abc} A_\mu^b A_\nu^c \quad (6.4.78)$$

and the covariant derivatives given by

$$\begin{aligned} D_\mu \phi_i &= \partial_\mu \phi_i + ig A_\mu^a t_{ij}^a \phi_j \\ D_\mu \psi_i &= \partial_\mu \psi_i + ig A_\mu^a t_i^a j \psi_j \\ D_\mu \lambda^a &= \partial_\mu \lambda^a - gf^{abc} A_\mu^b \lambda^c \end{aligned} \quad (6.4.79)$$

Concerning the transformation laws for the component fields, they read

$$\begin{aligned} \delta \phi &= \sqrt{2} \epsilon \psi \\ \delta \psi &= i\sqrt{2} \sigma^\mu \bar{\epsilon} D_\mu \phi + \sqrt{\epsilon} F \\ \delta F &= i\sqrt{2} \bar{\epsilon} \sigma^\mu D_\mu \psi + 2igt^a \phi \bar{\epsilon} \bar{\lambda}^a \\ \delta A_\mu^a &= -i\bar{\lambda}^a \bar{\sigma}_\mu \epsilon + i\bar{\epsilon} \bar{\sigma}_\mu \lambda^a \\ \delta \lambda^a &= \sigma^{\mu\nu} \epsilon F_{\mu\nu}^a \\ \delta D^a &= -\epsilon \sigma^\mu D_\mu \bar{\lambda}^a - D_\mu \lambda^a \sigma^\mu \bar{\epsilon} \end{aligned} \quad (6.4.80)$$

As in the Abelian case the scalar potential is the sum of  $F$  and  $D$  terms. Indeed, the non-Abelian extension of (6.2.40) is

$$V_{scalar} = \sum_{i=1}^r |F_i|^2 + \frac{1}{2} \sum_{a=1}^{\dim \mathcal{G}} D^a D^a = \sum_{i=1}^r \left| \frac{\partial W}{\partial \phi_i} \right|^2 + \frac{g^2}{2} \sum_{i,j=1}^r \sum_{a=1}^{\dim \mathcal{G}} (\phi_i^* t_{ij}^a \phi_j)^2 \quad (6.4.81)$$

We end this section by writing the equations of motion for the auxiliary fields

$$F_i^* = \frac{\partial W}{\partial \phi_i} \quad (6.4.82)$$

$$D^a = -g\phi_i^* t_{ij}^a \phi_j - g\kappa^a \quad (6.4.83)$$

where it is understood that  $\kappa^a = 0$  except for the cases in which  $a$  takes values in the Abelian factors of  $G$ .

## Spontaneous supersymmetry breaking for non-Abelian gauge theories

Since the Fayet-Iliopoulos proposal implies adding to the Lagrangian a term proportional to the auxiliary field  $D$  which in the non-Abelian case takes values in the Lie algebra of  $G$  and

hence is gauge invariant, the  $D$ -term supersymmetry breaking is excluded for non-Abelian gauge theories and spontaneous supersymmetry breaking is solely controlled by  $F$  terms. As in the Abelian case supersymmetry will be spontaneously broken if and only if equations  $F_i^* = 0$  has no solution, i.e., through the O’Raifeartaigh mechanism.

Of course, if the gauge group contains  $U(1)$  factors, Fayet-Iliopoulos terms associated to each factor can be added to break supersymmetry.

## 6.5 Mass formulæ

If supersymmetry is unbroken all particles within a supermultiplet have the same mass. This will no longer be true if supersymmetry is spontaneously broken, but one can still relate the differences of the squared masses to the supersymmetry breaking  $F$  and  $D$  terms.

Let us start with vector fields. In the presence of non-vanishing vevs of the scalars, some or all of the vector gauge fields will become massive by the Higgs mechanism. Indeed the term  $D_\mu \phi^* D^\mu \phi$  present in the Lagrangian (6.4.77) gives rise to mass terms. The corresponding mass matrix takes the form

$$(\mathcal{M}_A^2)^{ab} = 2g^2 \phi^* t^a t^b \phi \quad (6.5.84)$$

Now, from (6.4.83) we have

$$\begin{aligned} \frac{\partial D^a}{\partial \phi_i} &= -g(\phi^* t^a)_i \\ \frac{\partial D^a}{\partial \phi_i^*} &= -g(t^a \phi)_i \end{aligned} \quad (6.5.85)$$

so that we can rewrite (6.5.84) in the form

$$(\mathcal{M}_A^2)^{ab} = 2 \frac{\partial D^a}{\partial \phi_i} \frac{\partial D^b}{\partial \phi_i^*} \quad (6.5.86)$$

We now consider the spin-one-half fermions. Let us recall formula (5.1.3) which allowed us to identify the fermion mass matrix for a pure superchiral field model

$$W[\Phi_i] = W[\phi_i(y)] + \sqrt{2} \frac{\partial W}{\partial \phi_i} \theta \psi_i(y) - \theta \theta \left( \frac{\partial W}{\partial \phi_i} F_i(y) + \frac{1}{2} \frac{\partial^2 W}{\partial \phi_i \partial \phi_j} \right) \psi_i(y) \psi_j(y) \quad (6.5.87)$$

Here, second derivatives of the superpotential  $W$ , which can be written in terms of first derivatives of the auxiliary fields  $F_i$  play a central role,

$$\frac{\partial^2 W}{\partial \phi_i \partial \phi_j} = \frac{\partial F_i^*}{\partial \phi_j}, \quad \frac{\partial^2 \bar{W}}{\partial \phi_i^* \partial \phi_j^*} = \frac{\partial F_i}{\partial \phi_j^*} \quad (6.5.88)$$

One easily finds from (6.4.76),

$$L_{m\text{ fermion}} = -\frac{1}{2}(\psi_i \lambda^a) \mathcal{M}_f \begin{pmatrix} \psi_j \\ \lambda^b \end{pmatrix} + \text{h.c.} \quad (6.5.89)$$

with

$$\mathcal{M}_f = \begin{pmatrix} \frac{\partial F_i^*}{\partial \phi_j} & \sqrt{2}i \frac{\partial D^b}{\partial \phi_i} \\ \sqrt{2}i \frac{\partial D^a}{\partial \phi_j} & 0 \end{pmatrix} \quad (6.5.90)$$

Finally, for the scalars, the mass term can be written as

$$L_{m\text{ scalar}} = -\frac{1}{2}(\phi_i \phi_j^*) \mathcal{M}_s \begin{pmatrix} \phi_k \\ \phi_l^* \end{pmatrix} \quad (6.5.91)$$

with

$$\mathcal{M}_s = \begin{pmatrix} \frac{\partial^2 V_s}{\partial \phi_i \partial \phi_k^*} & \frac{\partial^2 V_s}{\partial \phi_i \partial \phi_l} \\ \frac{\partial^2 V_s}{\partial \phi_j^* \partial \phi_k^*} & \frac{\partial^2 V_s}{\partial \phi_j^* \partial \phi_l} \end{pmatrix} \quad (6.5.92)$$

Again this matrix can be written in terms of derivatives of  $F_i$  and  $D^a$ . We shall not write it explicitly but only when presenting the traces of the square mass matrices

$$\begin{aligned} \text{tr} \mathcal{M}_A^2 &= 2 \frac{\partial D^a}{\partial \phi_i} \frac{\partial D^a}{\partial \phi_i^*} \\ \text{tr} \mathcal{M}_f \mathcal{M}_f^\dagger &= \frac{\partial F_i^*}{\partial \phi_j} \frac{\partial F_i}{\partial \phi_j^*} + 4 \frac{\partial D^a}{\partial \phi_i} \frac{\partial D^a}{\partial \phi_i^*} \\ \text{tr} \mathcal{M}_s^2 &= 2 \frac{\partial F_i^*}{\partial \phi_j} \frac{\partial F_i}{\partial \phi_j^*} + 2 \frac{\partial D^a}{\partial \phi_i} \frac{\partial D^a}{\partial \phi_i^*} - 2g D^a \text{tr} t^a \end{aligned} \quad (6.5.93)$$

We can define a supertrace  $\text{Str}$  by summing the different traces assigning a + sign to bosons and – sign to fermions and a factor counting the number of degrees of freedom (3 for massive vector fields, 2 for fermions<sup>1</sup> and 1 for scalars).

$$\text{Str} \mathcal{M}^2 = 3 \text{tr} \mathcal{M}_A^2 - 2 \text{tr} \mathcal{M}_f \mathcal{M}_f^\dagger + \text{tr} \mathcal{M}_s^2 = -2g D^a \text{tr} t^a \quad (6.5.94)$$

We see that if  $D^a = 0$  or  $\text{tr} t^a = 0$  (no  $U(1)$  factors) this supertrace vanishes, stating that the sum of the squared masses of all bosonic degrees of freedom equals the sum for all

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<sup>1</sup>Originally, a vector multiplet implies 4 bosonic degrees of freedom, one coming from the scalar field and 3 from the vector field. This gives the same number as that of the fermion degrees of freedom, coming from 1 complex 2 component Weyl fermion. After use of the equations of motion we end with 2 degrees of freedom for fermions and 2 for bosons. The counting changes when the vector field gets a mass and gains one more degree of freedom.

fermionic ones. Without supersymmetry breaking this is a trivial result. In the presence of supersymmetry breaking this supertrace formula is still a strong constraint on the mass spectrum. In particular, if supersymmetry is broken only by an  $F$ -term or if all gauge group generators are traceless, one must still have  $\text{Str}\mathcal{M}^2 = 0$ .

Consider as an example Super QCD. The gauge group is  $SU(3)$  and  $\text{tr}t^a = 0$ , and the gauge group must remain unbroken so that  $\mathcal{M}_A^2 = 0$ , this implying that  $\partial D^a / \partial \phi_i = 0$ . Then, the fermion mass reduces to

$$\mathcal{M}_f = \begin{pmatrix} \frac{\partial F_i^*}{\partial \phi_j} & 0 \\ 0 & 0 \end{pmatrix} \quad (6.5.95)$$

We see that gluinos  $\lambda^a$  remain massless, while the supertrace formula tells us that the sum of the masses squared of the scalar quarks must equal (twice) the sum for the quarks. This means that the scalar quarks cannot all be heavier than the heaviest quark, and some must be substantially lighter. Since no massless gluinos and relatively light scalar quarks have been found experimentally, this scenario seems to be ruled out by experiment. However, it would be too quick to conclude that one cannot embed QCD into a susy theory. Indeed, there are two ways out. First, the mass formula derived here only give the tree-level masses and are corrected by loop effects. Typically, one introduces one or several additional chiral multiplets which trigger the susy breaking. Through loop diagrams this susy breaking then propagates to the gauge theory we are interested in and, in principle, one can achieve heavy gauginos and heavy scalar quarks this way, leaving massless gauge fields and light fermions. Second, the susy theory may be part of a supergravity theory which is spontaneously broken, and in this case one rather naturally obtains experimentally reasonable mass relations.

We see from (6.5.95) that in the case in which the gauge symmetry is unbroken the fermion mass squared takes the form

$$\mathcal{M}_f \mathcal{M}_f^\dagger = \begin{pmatrix} \frac{\partial F_i^*}{\partial \phi_l} \frac{\partial F_j}{\partial \phi_l^*} & 0 \\ 0 & 0 \end{pmatrix} \quad (6.5.96)$$

If we assume that there are no Fayet-Iliopoulos terms then  $D^a = 0$  and the scalar mass matrix reduces to

$$\mathcal{M}_s^2 = \begin{pmatrix} \frac{\partial F_i^*}{\partial \phi_l} \frac{\partial F_j}{\partial \phi_l^*} & F^l \frac{\partial^2 F_i^*}{\partial \phi_m \partial \phi_l} \\ F_l^* \frac{\partial^2 F_n}{\partial \phi_j^* \partial \phi_l^*} & \frac{\partial F_m^*}{\partial \phi_l} \frac{\partial F_n}{\partial \phi_l^*} \end{pmatrix} \quad (6.5.97)$$

The block diagonal elements are the same as for the fermions  $\psi_i$  but the block off-diagonal elements give an additional contribution that lifts the degeneracy between the real and the imaginary parts of the scalar fields, splitting the masses in a symmetric way with respect to

the corresponding fermion masses,

$$L_{\delta m} = \frac{1}{2} \left( F^l \frac{\partial^2 F_i^*}{\partial \phi_m \partial \phi_l} \right) \phi_i \phi_m + \text{h.c.} \quad (6.5.98)$$

Of course one still has  $\text{Str} \mathcal{M}^2 = 0$ .

## 6.6 Basics of supersymmetry breaking in generalized Wess-Zumino models

Let us consider a general Wess-Zumino model (or generalized O’Raifeartaigh model) with canonical Kähler potential and superpotential of the form

$$W = f_i \Phi_i + \frac{1}{2} m_{ij} \Phi_i \Phi_j + \frac{1}{6} \lambda_{ijk} \Phi_i \Phi_j \Phi_k \quad (6.6.99)$$

We have seen that the classical scalar potential takes the form

$$V = \sum_i |W_i|^2 \quad (6.6.100)$$

where we have introduced the notation

$$W_i = \frac{\partial W}{\partial \Phi_i} \quad (6.6.101)$$

Suppose that  $V$  has a supersymmetry breaking local minimum at some  $\phi_i = \phi_i^{(0)}$ . This imposes the following constraints to the theory:

1. Supersymmetry breaking requires that at least one of the  $W_i \neq 0$  at the minimum so that  $V \neq 0$ .
2. The extremum condition of  $V$  implies

$$\frac{\partial V}{\partial \phi^i} = 0 \Rightarrow W_{ij} W_j^* = 0 \quad (6.6.102)$$

Now, recalling (6.5.90) (in the case  $D = 0$ ) and the fact that  $W_i = F_i$ , one can write the fermion mass  $\mathcal{M}_F$  in the form

$$(\mathcal{M}_F)_{ij} = W_{ij} \quad (6.6.103)$$



so that (6.6.102) is nothing but a zero mode condition for the fermion sector,

$$(\mathcal{M}_F)_{ij}W_j^* = 0 \quad (6.6.104)$$

This is, at the classical level, the manifestation of the already discussed fact that when supersymmetry is broken, there is a massless Goldston fermion (the goldstino) which corresponds to the direction selected by the non-zero  $F$  terms.

3. The tree level boson mass (6.5.97) can be written as

$$\mathcal{M}_B^2 = \begin{pmatrix} \mathcal{M}_F^* \mathcal{M}_F & \mathcal{F}^* \\ \mathcal{F} & \mathcal{M}_F \mathcal{M}_F^* \end{pmatrix} \quad (6.6.105)$$

where

$$\mathcal{F}_{ij} \equiv W_k^* W_{ijk} \quad (6.6.106)$$

is responsible of supersymmetry breaking. If  $\phi_i^{(0)}$  corresponds to a sensible vacuum one should have  $\mathcal{M}_B^2 \geq 0$  (no tachyons).

Suppose that  $\mathcal{M}_F$  has a zero-mode,

$$\mathcal{M}_F v = 0 \quad (6.6.107)$$

and construct a vector

$$V = \begin{pmatrix} v \\ v^* \end{pmatrix} \quad (6.6.108)$$

Consider now

$$V^\dagger \mathcal{M}_B^2 V = v^\dagger \mathcal{F} v + c.c. \quad (6.6.109)$$

Now this expression should vanish since otherwise an appropriate rotation of  $v$  would guarantee that the r.h.s. is negative, which is not acceptable for a positive semidefinite  $\mathcal{M}_B^2$ . Now, the only possibility that a norm like that in the l.h.s. vanish is that  $V$  is a zero mode of  $\mathcal{M}_B^2$ . We then conclude that

- Lemma

*If there is a massless fermion at the classical level in any supersymmetry breaking vacuum of a generalized O’Raifeartaigh model, then its scalar superpartner must also be classically massless.*

Evidently the validity of this lemma is not restricted to superpotentials of the form (6.6.99) but to any general polynomial  $W$ .

## Existence of pseudo-moduli spaces

We have seen that since the Goldstino is always massless its superpartner must also be massless in the vacuum of any supersymmetry breaking generalized WZ model. the fact that the l.h.s. in (6.6.109) should vanish implies that

$$\mathcal{F}v = 0 \tag{6.6.110}$$

Now, according to (6.6.104)  $v = W_i^*$  satisfies and hence from (6.6.110)

$$\mathcal{F}W_i^* = W_{ijk}W_i^*W_k^* = 0 \tag{6.6.111}$$

That is, for a renormalizable superpotential the superpartner of the Goldstino is not only massless but it can be extended to a whole pseudo-flat direction. Indeed, given  $\Phi_i^{(0)}$ , any  $\Phi_i^{pf}$  of the form

$$\Phi_i^{pf} = \Phi_i^{(0)} + fW_i^* \tag{6.6.112}$$

with  $f \in \mathbb{C}$  leaves the potential unchanged as can be seen by expanding  $V$  (a polynomial of order 3) in powers of  $fW_i^*$  and using (6.6.111).

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# Chapter 7

## Path-integral quantization and effective field theories I

### 7.1 Path-integral formulation of Quantum Field Theories

In the path-integral formulation of Quantum Field Theory one starts by defining the generating functional for quantum correlation functions. Take for simplicity the classical action for a scalar field  $\phi$

$$S = \int_M d^4x \left( -\frac{1}{2} \partial_\mu \phi(x) \partial^\mu \phi(x) - V[\phi] + j(x) \phi(x) \right) \quad (7.1.1)$$

Here  $V[\phi]$  is the potential which can include a quadratic mass term and self-interactions of the type  $\phi^3(x), \phi^4(x) \dots$ . Concerning  $j(x)$ , it is an external source which should be put to zero at the end of the calculations (see below). The generating functional for this theory is defined as

$$Z[j] = \int \mathcal{D}\phi \exp \left( \frac{i}{\hbar} \int_M d^4x (L + j\phi) \right) \quad (7.1.2)$$

where  $L$  is the Lagrangian defining the governing the dynamics of the theory in the absence of sources.

The integral in the r.h.s. of (7.1.2) is the well-honored path-integral introduced by feynman after a proposal of Dirac. It is an integral “over all fields” satisfying appropriate boundary conditions. The meaning of integrating over all fields (in this case over real functions  $\phi(x) : M \rightarrow \mathbb{R}$ ) can be intuitively understood by considering one of the possible definitions of the path-integral measure  $\mathcal{D}\phi$  that we describe below.

Consider consider the eigenvalue problem

$$\hat{A}\varphi_n(x) = \lambda_n\varphi_n(x) \quad (7.1.3)$$

where  $\hat{A}$  is a differential operator providing an orthonormal basis  $\{\varphi_n(x)\}$  to develop any  $\phi(x)$ ,

$$\phi(x) = \sum_n c_n \varphi_n(x) \quad (7.1.4)$$

In general, the choice of operator  $\hat{A}$  is dictated by the quadratic part of the Lagrangian. In the case we are considering, the natural choice would be

$$\hat{A} = \square - m^2 \quad (7.1.5)$$

where  $m$  is the mass of the scalar field (appearing in a quadratic term included in  $V[\phi]$ ). Coefficients  $c_n$  are real numbers,  $c_n \in \mathbb{R}$  and the basis vectors  $\varphi_n(x)$  should satisfy the same conditions one wants to impose to functions  $\phi(x)$ .

In terms of expansion (7.1.4), integrating over all fields means integrating over

$$\mathcal{D}\phi = \prod_n dc_n \quad (7.1.6)$$

where  $c_n$  are the Bessel-Fourier coefficients in the expansion of  $\phi$  in terms of  $\varphi_n$ . In this way one is covering all possible values of the coefficients  $c_1, c_2, \dots, c_n$  thus covering all possible values that the field  $\phi(x)$  can take.

Let us start by considering the case in which the Lagrangian is a “free Lagrangian”, i.e it only contains the kinetic term and the mass term (no interactions, no source term) so that the action reads

$$S_0 = \int_M d^4x \left( -\frac{1}{2} \partial_\mu \phi(x) \partial^\mu \phi(x) - \frac{1}{2} m^2 \phi^2 \right) = \frac{1}{2} \int_M d^4x \phi \hat{A} \phi \quad (7.1.7)$$

We then have

$$Z_0[j=0] = \prod_n \int dc_n \exp \left( \frac{i}{\hbar} \sum_n \lambda_n c_n^2 \right) = \left( \prod_n \lambda_n \right)^{-1/2} \equiv \det^{-1/2} \hat{A} \quad (7.1.8)$$

where we have ignored  $\hbar$  powers that one can recover at the end as well as irrelevant  $\sqrt{\pi}$  powers. Concerning the  $i$  factor in the argument of the exponential, see below.

In order to compute  $Z_0[j]$  when the source linear term is present,

$$Z_0[j] = \int \mathcal{D}\phi \exp \left( \frac{i}{\hbar} \int_M d^4x (L_0 + j(x)\phi(x)) \right) \quad (7.1.9)$$

let us write

$$Z_0[j] = \int D\phi \exp \left( \frac{i}{2\hbar} \int d^4x (\phi - \bar{\phi}) A(\phi - \bar{\phi}) - \frac{i}{2\hbar} \int d^4x C \right) \quad (7.1.10)$$

One easily finds that (7.1.10) coincides with (7.1.9) whenever

$$C = \int d^4x' j(x) A^{-1}(x, x') j(x') \quad (7.1.11)$$

and

$$\bar{\phi}(x) = - \int d^4x' A^{-1}(x, x') j(x') \quad (7.1.12)$$

with  $A^{-1}(x, x')$  the Green function for the operator  $A$ .

Then, one has

$$\begin{aligned} Z_0[j] &= \exp \left( -\frac{i}{2\hbar} \int d^4x d^4x' j(x) A^{-1}(x, x') j(x') \right) \times \\ &\quad \int D\phi \exp \left( \frac{i}{2\hbar} \int d^4x (\phi - \bar{\phi}) A(\phi - \bar{\phi}) \right) \end{aligned} \quad (7.1.13)$$

or

$$Z_0[j] = \exp \left( -\frac{i}{2\hbar} \int d^4x d^4x' j(x) A^{-1}(x, x') j(x') \right) Z_0[j = 0] \quad (7.1.14)$$

We can use formula (7.1.14) in order to compute the vacuum expectation value of the product of two fields at different points in the free theory

$$\begin{aligned} \langle \phi(x_1) \phi(x_2) \rangle_0 &\equiv \frac{1}{Z_0[j]} \int \mathcal{D}\phi \phi(x_1) \phi(x_2) \exp \left( \frac{i}{\hbar} \int_M d^4x (L_0 + j\phi) \right) \Big|_{j=0} \\ &= - \frac{\hbar^2}{Z_0[0]} \frac{\delta^2 Z_0[j]}{\delta j(x_1) \delta j(x_2)} \Big|_{j=0} \end{aligned} \quad (7.1.15)$$

One finds

$$\langle \phi(x_1) \phi(x_2) \rangle_0 = \hbar A^{-1}(x_1, x_2) \quad (7.1.16)$$

which is nothing (apart from a  $\hbar$  factor) but the (two-points) Green function  $G_0^{(2)}(x_1, x_2)$  that one finds when quantizing canonically the free field scalar theory. We can then alternatively write

$$Z_0[j] = \exp \left( -\frac{1}{2} \int d^4x d^4x' j(x) G_0^{(2)}(x, x') j(x') \right) Z_0[j = 0] \quad (7.1.17)$$

from which the Wick theorem of QFT can be easily derived. This “theorem” due a work of to Gian Carlo Wick [1] (and also Freeman Dyson [2]), allows to evaluate vacuum expectation values of product of  $n$  fields (v.e.v.’s) at different points  $x_i$  ( $i = 1, 2, \dots, n$ ) in terms of the vacuum expectation values of product of just two fields.

From the partition function for the free theory defined by eq.(7.1.17) one easily finds the formula that proves the theorem. For example, for the 4-points function one gets, after differentiation

$$\begin{aligned} \langle \phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4) \rangle_0 &= G_0^{(2)}(x_1, x_2)G_0^{(2)}(x_3, x_4) + \\ &G_0^{(2)}(x_1, x_3)G_0^{(2)}(x_2, x_4) + \\ &G_0^{(2)}(x_1, x_4)G_0^{(2)}(x_2, x_3). \end{aligned} \quad (7.1.18)$$

It is evident that in the present free case, all v.e.v.’s with an odd number of points vanish. Concerning the even case with  $n > 2$ , it just gives analogous results as that for the case  $n = 4$ .

We have described how to compute in a closed form correlation functions for the free theory. When an interaction is present one cannot in general obtain exact results and one has to develop a perturbative approach to study the Green functions ( $n$ -point correlation functions in general). For simplicity just consider a cubic interaction term. The two-point function  $G^{(2)}$  will take the form

$$G^{(2)}(u, v) = \frac{1}{Z} \int \mathcal{D}\phi \exp \left( -S_0 - \frac{\lambda}{3!} \int d^4x \phi^3(x) \right) \phi(u)\phi(v) \quad (7.1.19)$$

with  $Z$  defined as

$$Z = \int \mathcal{D}\phi \exp \left( -S_0 - \frac{\lambda}{3!} \int d^4x \phi^3(x) \right) \quad (7.1.20)$$

and  $S_0$  the free (quadratic) action.

Consider first the denominator  $Z$ . Let us expand the interaction term in powers of  $\lambda$ ,

$$Z = \sum_{n=0}^{\infty} (-1)^n \frac{\lambda^n}{n!(3!)^n} \int d^4x_1 d^4x_2 \dots d^4x_n \langle \phi^3(x_1)\phi^3(x_2) \dots \phi^3(x_n) \rangle_0 \times Z_0 \quad (7.1.21)$$

The v.e.v. in the r.h.s. corresponds to a free theory. As we have seen, only if it corresponds to an even number of fields it will be different from zero. We can then write

$$Z = \sum_{k=0}^{\infty} \frac{\lambda^{2k}}{(2k)!(3!)^{2k}} \int d^4x_1 d^4x_2 \dots d^4x_{2k} \langle \phi^3(x_1)\phi^3(x_2) \dots \phi^3(x_{2k}) \rangle_0 \times Z_0 \quad (7.1.22)$$



Concerning the numerator, one has a similar expression except that in the vacuum expectation values there are insertions of  $\phi(u)$  and  $\phi(v)$ . One then proceeds to compute these free vev's developing a diagrammatic procedure and ends with an expression for  $G^{(2)}$  as a power expansion in  $\lambda$  which I present just for artistic reasons (you will have to wait your path-integral course to understand this result).

Hence, one finally has for  $G^{(2)}(u, v)$

$$G^{(2)}(u, v) = u \text{ --- } v + \frac{\lambda^2}{2} u \text{ --- } \bigcirc \text{ --- } v + \dots$$

In these

diagrams

- Each line represents a propagator which is nothing but a free two-point free Green function in coordinate space,  $G_0^{(2)}(x_1 - x_2)$  or its Fourier transform in momentum space,
- Each independent loop represents an integral.

More precisely, in this  $\lambda\phi^3$  theory case

$$k_1 \text{ --- } k_2 = \delta^{(4)}(k_1 + k_2) \frac{1}{k_1^2 + m^2}$$

$$\begin{array}{c}
 & k_2 \\
 & \diagup \\
 k_1 \text{ --- } & \\
 & \diagdown \\
 & k_3
 \end{array} = \lambda \delta^{(4)}(k_1 + k_2 + k_3)$$

$$\text{An (independent) loop} \Rightarrow \int \frac{d^4 k}{(2\pi)^4}$$

We see that only connected diagrams contribute to  $G^{(2)}(u, v)$ . Examples of disconnected diagrams are the following possible ones that in fact appear in the numerator:



Such disconnected diagrams are canceled out by contributions from the denominator leaving us with solely the connected one. This is true in general and this is the reason why one calls  $Z[j]$  the generating functional for connected diagrams.

## 7.2 Effective field theories: the 1PI effective action

We shall present here an approach that allows to study properties of quantum field theories using concepts and the intuition developed in classical mechanics and classical field theories like electrodynamics.

In quantum field theory the objects with the greatest physical interest are the Green functions from which the S matrix can be constructed. Perturbative calculations are usually expressed in terms of these functions, the Feynman diagram technique is specifically devised to compute them; their analytic properties give much information about the quantum physics.

On the other hand, certain quantum phenomenon, as for example the spontaneous symmetry breaking of a classical symmetry are difficult to understand in terms of Green functions. Of course one can follow, through the analysis of Green functions, step by step, the appearance of Goldston bosons, the acquisition of mass by certain fields without loss of symmetry, etc. But the starting classical scenario in which a symmetry breaking potential provides non trivial classical minima has a difficult translation to the quantum level in terms of Green functions.

Functional methods can be used to construct objects which will be called effective quantum actions and effective potentials. As we shall see, although closely related to Green functions, they provide the necessary bridge between classical and quantum theory. It is not hard to understand why the path-integral framework, in which the classical action plays a central role, provides the privileged tool to define and compute these objects.

It will be convenient to work with Euclidean time. The passage from Minkowski to Euclidean space can be done as follows.

Let us call for the moment Minkowski time  $t$  and Euclidean time  $x_0$ . The so-called Wick rotation (an analytic continuation) just consist in making

$$t \rightarrow x^0 \equiv it \quad (7.2.23)$$

Take for example the Minkowski action for a particle of mass  $m$  in a potential  $V$ ,

$$\frac{i}{\hbar} S_M = \frac{i}{\hbar} \int dt \left( \frac{m\dot{q}^2}{2} - V(q) \right) \quad (7.2.24)$$

(Here  $\dot{f} = df/dt$ ).

By carefully proceeding to continuation (7.2.23) to Euclidean time  $x^0$  one gets

$$\frac{i}{\hbar} S_M \rightarrow -\frac{1}{\hbar} S_E \quad (7.2.25)$$

with the Euclidean Action taking the form

$$S_E = \int dx^0 L_E = \int dx^0 \left( \frac{m\dot{q}^2}{2} + V(q) \right) \quad (7.2.26)$$

where now  $\dot{f} = df/dx^0$ . We see that the Euclidean Lagrangian looks as the Minkowskian energy, a sum of kinetic and potential terms.

Up to here we can summarize the result of passing from Minkowski to Euclidean time by stating that one has to change

- $i/\hbar \rightarrow -1/\hbar$
- $S_M \rightarrow S_E$  or  $L_M \rightarrow L_E$
- $d/dt \rightarrow d/dx^0$
- $V(q) \rightarrow -V(q) \equiv V_E$
- $T_M \rightarrow T_E$
- $L_E = T_E - V_E$
- $\varepsilon = T_E + V_E$

An intuitive way of understanding what is happening is the following. The Newton second law reads, in Minkowski space

$$m \frac{d^2 \vec{x}}{dt^2} = -\vec{\nabla} V_M \quad (7.2.27)$$

When passing to Euclidean space one changes  $t = -ix_0$  while  $\vec{x}$  remains unchanged

$$-m \frac{d^2 \vec{x}}{dx_0^2} = -\vec{\nabla} V_M \quad (7.2.28)$$

With the definition  $V_E = -V_M$  Newton's law looks unchanged

$$m \frac{d^2 \vec{x}}{dx_0^2} = -\vec{\nabla} V_E \quad (7.2.29)$$

but then the force  $\vec{F}_E = -\vec{\nabla} V_E = \vec{\nabla} V_M = -\vec{F}_M$  points in the oposite direction to that of the Minkowskian one.

The relevance of the Euclidean action goes beyond the fact that it allows to handle more properly gaussian actions. As the chemists discovered in the seventies [3] it allows to describe tunneling effect using the path-integral and to develop semiclassical approximations. This, as we shall see, has a lot to do with the fact that one can think that, in passing to Euclidean space, the potential is inverted.

We have in Euclidean space

$$Z_E = \int \mathcal{D}\phi \exp \left( -\frac{1}{\hbar} S_E[\phi] \right) \quad (7.2.30)$$

From here on, we shall not insist in the subscript  $E$  and call time  $t$  indistinctly in Euclidean or Minkowski space. The factor multiplying the action in the exponential defining the path-integral will clearly remind us the nature of space-time in which we are working.

Let us consider again the theory of a real scalar field  $\phi$  with mass  $m$  but this time with a quartic self-interaction. At the classical level, dynamics is dictated by the Lagrangian

$$\begin{aligned} L &= \frac{1}{2} (\partial_\mu \phi) (\partial^\mu \phi) + \frac{1}{2} m^2 \phi^2 + \frac{1}{4!} \phi^4 \\ S &= \int d^4 x L \end{aligned} \quad (7.2.31)$$

We have already explained that the generating functional of disconnected Green functions  $Z_{disc}[j]$  is defined in terms of path-integrals as

$$Z_{disc}[j] = \frac{1}{Z_{disc}[0]} \int \mathcal{D}\phi \exp \left( -\frac{1}{\hbar} S[j] \right) \equiv \exp \left( -\frac{1}{\hbar} W[j] \right) \quad (7.2.32)$$

Here we have defined the action  $S[j]$  in the presence of a source  $j$  as

$$S[j] = S + \int d^4x \phi(x) j(x) \quad (7.2.33)$$

The expansion of  $Z_{disc}[j]$  in powers of  $j$  gives disconnected Green functions  $G_{disc}^{2n}$ ,

$$Z_{disc}[j] = \sum_n \frac{1}{(2n)!} \int d^4x_1 \dots d^4x_{2n} G_{disc}^{2n}(x_1, x_2, \dots, x_{2n}) j(x_1) \dots j(x_{2n}) \quad (7.2.34)$$

with

$$G_{disc}^{2n}(x_1, x_2, \dots, x_{2n}) = \hbar^{2n} \frac{\delta^{2n} Z_{disc}[j]}{\delta j(x_1) \delta j(x_2) \dots \delta j(x_{2n})} \Big|_{j=0} \quad (7.2.35)$$

We have introduced  $W$  in (7.2.32) in order to pass from disconnected to connected Green functions. Indeed, being  $W$  related to  $Z_{disc}$  through

$$W[j] = -\hbar \log Z_{disc}[j] \quad (7.2.36)$$

the logarithm relating them ensures that all disconnected contributions are eliminated from  $W$  derivatives because of the  $1/Z_{disc}$  arising in the logarithm derivative,

$$G_{conn}^{2n}(x_1, x_2, \dots, x_{2n}) = -\frac{1}{\hbar} \frac{\delta^{2n} W[j]}{\delta j(x_1) \delta j(x_2) \dots \delta j(x_{2n})} \Big|_{j=0} \quad (7.2.37)$$

A possible approximate way to evaluate a path-integral like that defining  $Z_{disc}$  is to use the steepest descent method which we shall explain below. To this end, one should look for a solution  $\phi_0 = \phi_0[j]$  of the classical equations of motion deriving from the action in the path-integrand,

$$\delta S[j] = 0 \longrightarrow (-\nabla^2 + m^2) \phi_0 + \frac{\lambda}{3!} \phi_0^3 + j = 0 \quad (7.2.38)$$

Once the classical solution is found, one proceeds to a change of variables to separate the classical contribution from quantum fluctuations and usually appeals to a perturbative calculation. That is, one writes

$$\phi(x) = \phi_0[j] + \sqrt{\hbar} \varphi(x) \quad (7.2.39)$$

and then writes the action  $S[j]$  in terms of the new variable,

$$\begin{aligned} S[j] &= S[\phi_0] + \frac{\hbar}{2} \int d^4x \varphi \Delta[j; m, \lambda] \varphi + \frac{\hbar^{3/2} \lambda}{3!} \int d^4x \phi_0[j] \varphi^3(x) \\ &+ \frac{\hbar^2 \lambda}{4!} \int d^4x \varphi^4(x) + \dots \end{aligned} \quad (7.2.40)$$

where

$$\Delta[j; m, \lambda] = -\nabla^2 + \left( m^2 + \frac{\lambda \phi_0^2[j]}{2} \right) \quad (7.2.41)$$

Note that  $\Delta$  replaces the kinetic energy terms in the original action and contains an effective (in general  $x$ -dependent) mass since  $\phi_0[j] = \phi_0[j(x)]$ .

Being eq.(7.2.38) elliptic, a boundary condition of the form

$$\phi_0[j(\infty) = 0] = 0 \quad (7.2.42)$$

will lead to a unique solution. (Demanding the external source to satisfy  $j(\infty) = 0$  is a natural and allowed condition).

At the path integral level the shift (7.2.39) has no associated Jacobian so that one has

$$\begin{aligned} \exp \left( -\frac{1}{\hbar} W[j] \right) &= \frac{1}{Z[0]} \exp \left( -\frac{1}{\hbar} S[\phi_0] \right) \int \mathcal{D}\varphi \exp \left( - \int d^4x \left( \varphi \Delta \varphi + \right. \right. \\ &\quad \left. \left. \frac{\hbar^{1/2} \lambda}{3!} \phi_0 \varphi^3(x) + \frac{\hbar \lambda}{4!} \varphi^4(x) \right) \right) \end{aligned} \quad (7.2.43)$$

Up to now, no approximation has been assumed. We face however a non-quadratic path-integral over  $\varphi$  which cannot be computed in a closed form and hence, we have to develop some perturbative treatment in order to compute  $Z_{disc}$ . A possibility is to expand in powers of  $\hbar^\alpha \lambda^\beta$  which corresponds to the factors which appear in the non-quadratic terms. This will be a good approximation both for weak coupling and in the semiclassical limit. The answer is

$$\exp \left( -\frac{1}{\hbar} W[j] \right) = \frac{\exp \left( -\frac{1}{\hbar} S[\phi_0] \right) \left( \det^{-1/2} \Delta \right) (1 + O(\hbar))}{\exp \left( -\frac{1}{\hbar} S[\phi_0[j=0]] \right) \left( \det^{-1/2} \Delta_0 \right) (1 + O(\hbar))} \quad (7.2.44)$$

where

$$\Delta_0 \equiv \Delta[\phi_0[j=0]] = -\nabla^2 + m^2 \quad (7.2.45)$$

Since  $S[\phi_0[j=0]] = 0$ , we can write

$$W[j] = S[\phi_0] + \frac{\hbar}{2} \log \det \left( \frac{\Delta}{\Delta_0} \right) + O(\hbar^2) \quad (7.2.46)$$

We see from (7.2.46) that in the  $\hbar \rightarrow 0$  limit, the generating functional of connected quantum Green functions coincides with the classical action evaluated at a solution of the classical equations of motion in the presence of a source.

Now, objects appearing in (7.2.46) are all functionals of the source, an artifact introduced to get Green functions by functional differentiation. One can rework the results so as to

define objects which are not functionals of the source but of “classical” fields. This will make possible a contact with classical field dynamics. To this end, we use the so called Legendre transformation that allows to express functionals  $F[f]$  depending on functions  $f(x)$  in terms of functionals of  $g(x) = df(x)/dx$ .

Define

$$F[g] = f(x) - xg(x) \quad (7.2.47)$$

then

$$\frac{dF}{dg} = \frac{df}{dx} \frac{dx}{dg} - \frac{dx}{dg} g - x \quad (7.2.48)$$

Now, the first and second term in the l.h.s. of (7.2.48) cancel so that we can write

$$x = -\frac{dF}{dg} \quad (7.2.49)$$

and then (7.2.47) takes the form

$$F[g] = f(x) + g(x) \frac{dF}{dg} \quad (7.2.50)$$

or

$$f(x) = F[g] - g(x) \frac{dF}{dg} \quad (7.2.51)$$

which should be compared with (7.2.47)

$$F[g] = f(x) - xg(x) \Rightarrow f(x) = F[g] - g(x) \frac{dF}{dg} \quad (7.2.52)$$

We can directly use this connection identifying

$$f \mapsto W[j], \quad x \mapsto j \quad (7.2.53)$$

and introducing  $\Gamma$  and  $\phi_{cl}$  such that

$$F \mapsto \Gamma, \quad g \mapsto \phi_{cl} \quad (7.2.54)$$

so that

$$g = \frac{df}{dx} \mapsto \phi_{cl} = \frac{\delta W[j]}{\delta j} \quad (7.2.55)$$

Then we have for (7.2.47),

$$\Gamma[\phi_{cl}] = W[j] - \int \phi_{cl}(x) j(x) dx^4 \Rightarrow W[j] = \Gamma[\phi_{cl}] - \int d^4x \phi_{cl} \frac{\delta \Gamma}{\delta \phi_{cl}} \quad (7.2.56)$$

Now, if we solve

$$\phi_{cl} = \frac{\delta W[j]}{\delta j} \quad (7.2.57)$$

we have

$$j = j[\phi_{cl}] \quad (7.2.58)$$

and then we can write

$$\Gamma[\phi_{cl}] = W[j[\phi_{cl}]] - \int \phi_{cl}(x) j[\phi_{cl}] d^4x \quad (7.2.59)$$

That, is, the Legendre transformed of  $W$  has been entirely written in terms of a field  $\phi_{cl}$ .

What can we say about  $\phi_{cl}$ ? The notation “cl” signal that this object, as we shall see, plays at the quantum level the role that classical solutions play at the “classical” level. To investigate this, let us start from the expansion (7.2.46) for  $W$ ,

$$W[j] = S[\phi_0[j]] + \hbar S^{(1)}[\phi_0[j]] + \hbar^2 S^{(2)}[\phi_0[j]] + \dots \quad (7.2.60)$$

so that

$$\frac{\delta W}{\delta j} = \frac{\delta S[\phi_0[j]]}{\delta j} + \hbar \frac{\delta S^{(1)}[\phi_0[j]]}{\delta j} + \dots \quad (7.2.61)$$

But this means, after (7.2.57) that

$$\phi_{cl} = \frac{\delta S[\phi_0[j]]}{\delta j} + \hbar \frac{\delta S^{(1)}[\phi_0[j]]}{\delta j} + \dots \quad (7.2.62)$$

or, using

$$\frac{\delta S}{\delta j} = \phi_0 \quad (7.2.63)$$

$$\phi_{cl} = \phi_0 + O(\hbar) \quad (7.2.64)$$

We see that  $\phi_{cl}$  contains the classical solution to the equations of motion in the presence of a source and also quantum corrections. In order to go on in the interpretation of quantum calculations in terms of classical concepts, let us expand  $\Gamma[\phi_{cl}]$ , the Legendre transformed of the generating functional of connected Green functions, in powers of  $\hbar$ .

$$\Gamma[\phi_{cl}] = \Gamma^{(0)}[\phi_{cl}] + \hbar \Gamma^{(1)}[\phi_{cl}] + \hbar^2 \Gamma^{(2)}[\phi_{cl}] + \dots \quad (7.2.65)$$

Using the expansion (7.2.60) for  $W$  and the relation (7.2.59) between  $W$  and  $\Gamma$ , one gets

$$\Gamma[\phi_{cl}] = S[\phi_0] + \hbar S^{(1)}[\phi_0] - \int d^4x (\phi_0(x) + O(\hbar)) j(x) \quad (7.2.66)$$



To order zero in  $\hbar$  we can then write,

$$\Gamma^{(0)}[\phi_{cl}] = S[\phi_0] - \int d^4x \phi_0(x) j(x) \quad (7.2.67)$$

Now, the second term in the right hand side of (7.2.67) cancels the source term in  $S[\phi_0]$  so that we in fact have

$$\Gamma^{(0)}[\phi_{cl}] = \int d^4x L = \int d^4x \left( \frac{1}{2} (\partial_\mu \phi_0) (\partial^\mu \phi_0) + \frac{1}{2} m^2 \phi_0^2 + \frac{1}{4!} \phi_0^4 \right) \quad (7.2.68)$$

We have seen (eq.(7.2.64)) that there is no distinction between  $\phi_0$  and  $\phi_{cl}$  to order  $\hbar$ . Then, we can rewrite (7.2.68) in the form

$$\Gamma^{(0)}[\phi_{cl}] = \int d^4x \left( \frac{1}{2} (\partial_\mu \phi_{cl}) (\partial^\mu \phi_{cl}) + \frac{1}{2} m^2 \phi_{cl}^2 + \frac{1}{4!} \phi_{cl}^4 \right) \quad (7.2.69)$$

Functional  $\Gamma^{(0)}$  represents the classical action written as a functional of a classical object which we called  $\phi_{cl}$ . But which is the precise meaning of  $\phi_{cl}$  in terms of quantum concepts? To see this, let us rewrite formula (7.2.32) defining  $W$ ,

$$\exp \left( -\frac{1}{\hbar} W[j] \right) = \frac{1}{Z_{disc}[0]} \int \mathcal{D}\phi \exp \left( -\frac{1}{\hbar} \int d^4x (L + j\phi) \right) \quad (7.2.70)$$

Differentiation with respect to  $j$  shows that

$$\frac{\delta W}{\delta j(x)} = \langle \phi(x) \rangle_j \quad (7.2.71)$$

We know from eq.(7.2.57) the the l.h.s. of (7.2.71) is nothing but  $\phi_{cl}$  so that

$$\phi_{cl} = \phi_{cl}[j] = \langle \phi(x) \rangle_j = \phi_0[j] + \hbar \phi^{(1)} + \dots \quad (7.2.72)$$

or, using  $\phi_0[0] = 0$ ,

$$\phi_{cl}[0] = \langle \phi \rangle = \hbar \phi^{(1)} + O(\hbar^2) \quad (7.2.73)$$

That is, the argument  $\phi_{cl}$  in  $\Gamma[\phi_{cl}]$  can be associated, for zero source, with the vacuum expectation value of the field. But we can say more. From the definition of  $\Gamma$  as a Legendre transformation, eq.(7.2.59), we can write

$$\frac{\delta \Gamma[\phi_{cl}]}{\delta \phi_{cl}} = \frac{\delta W[j[\phi_{cl}]]}{\delta \phi_{cl}} - j[\phi_{cl}] - \int d^4x \phi_{cl} \frac{\delta j}{\delta \phi_{cl}} \quad (7.2.74)$$

but since

$$\frac{\delta W[j[\phi_{cl}]]}{\delta \phi_{cl}} = \int d^4x \frac{\delta W[j[\phi_{cl}]]}{\delta j} \frac{\delta j}{\delta \phi_{cl}} \quad (7.2.75)$$

the first and third terms in the r.h.s. of (7.2.74) cancel and one is left with

$$\frac{\delta \Gamma[\phi_{cl}]}{\delta \phi_{cl}} = -j \quad (7.2.76)$$

or

$$\left. \frac{\delta \Gamma[\phi_{cl}]}{\delta \phi_{cl}} \right|_{j=0} = 0 \quad (7.2.77)$$

So we see that the extrema of  $\Gamma[\phi_{cl}]$  as a functional of  $\phi_{cl}$  will give something that is equal to (to order  $\hbar$ ) the v.e.v. of the quantum field  $\phi$ . It is then clear why one calls  $\Gamma$  the effective action: it encodes quantum information exactly as the classical action does for classical dynamics: its minimum correspond to the v.e.v. that the field takes.

We have seen in eq.(35) that there is an expansion in  $\Gamma$  in powers of  $\hbar$  that identifies its zeroth order with the classical action. This agrees with the interpretation above and shows that, to zeroth order, we can identify the v.e.v. of the quantum field with the extrema of the classical action. This identification legalizes, at least to the lowest order, the description of quantum spontaneous breaking of gauge symmetry as described in terms of minima of a classical potential. What about the first order, second order, etc? We have seen that, to order  $\hbar$ , we have for  $W$  (see eq.(7.2.46))

$$W^{(1)} = \hbar S^{(1)} = \hbar \frac{1}{2} \log \det \left( \frac{\Delta}{\Delta_0} \right) \quad (7.2.78)$$

Then, from the relation (7.2.59) between  $W$  and  $\Gamma$  we can see that

$$\hbar \Gamma^{(1)} = \hbar S^{(1)} = \hbar \frac{1}{2} \log \det \left( \frac{\Delta}{\Delta_0} \right) \quad (7.2.79)$$

So, in the computation of v.e.v.'s using the effective action method, to order zero we identify the v.e.v. with the minimum of the classical action. To order one, we have to compute the fermion determinant in the r.h.s. of (7.2.79) and the addition of  $\Gamma^{(1)}$  to the classical action will incorporate a contribution of order  $\hbar$  to the information of the minimum of the effective action. The procedure can be systematically ameliorated.

## The effective potential

Up to now we have been working with functions  $\phi_{cl}(x)$  which were related to extrema of the effective action. Now, if one particularizes to constant functions, by analogy with the classical case in which the action becomes just (minus) the potential, here we shall say we are computing the effective potential. Minima of this quantum object will correspond to the quantum version of the minima of the classical potential.

So, let us put

$$\phi_{cl}(x) = \phi_{cl} = \text{constant} \quad (7.2.80)$$

and then define the effective potential through the identity

$$\Gamma[\phi_{cl}] = \int d^4x V_{eff}[\phi_{cl}] \quad (7.2.81)$$

Of course, being  $\phi_{cl}$  constant,  $V_{eff}$  in (7.2.81) does not depend on  $x$  and the integration gives a divergent factor which will be identified with the volume of space-time and should be consequently extracted.

Using our previous results, we can write

$$V_{eff}[\phi_{cl}] = V_0[\phi_{cl}] + \hbar V_1[\phi_{cl}] + \hbar^2 V_2[\phi_{cl}] + \dots \quad (7.2.82)$$

with

$$V_0[\phi_{cl}] = \frac{m^2}{2} \phi_{cl}^2 + \frac{\lambda}{4!} \phi_{cl}^4 \quad (7.2.83)$$

That is, the first term in the expansion of the effective potential just gives the classical potential, the Higgs potential in the present case. Concerning  $V_1$ , from our previous calculations we have (eq.(7.2.79))

$$\Gamma^{(1)} = \frac{1}{2} \log \det \left( \frac{\Delta}{\Delta_0} \right) \quad (7.2.84)$$

with

$$\Delta = -\nabla^2 + \left( m^2 + \frac{\lambda \phi_{cl}^2}{2} \right) = \Delta_0 + \frac{\lambda \phi_{cl}^2}{2} \quad (7.2.85)$$

Then, using the identity (valid both for matrices and operators)

$$\log \det A = \text{tr} \log A \quad (7.2.86)$$

we can write

$$\Gamma^{(1)} = \frac{1}{2} \text{tr} \log \left( \frac{\Delta}{\Delta_0} \right) = \frac{1}{2} \text{tr} \log \left( 1 + \frac{\lambda}{2} \Delta_0^{-1} \phi_{cl}^2 \right) \quad (7.2.87)$$

here, we have called  $\Delta_0^{-1} \equiv G_0$  the Green function for the free problem,

$$\Delta_0 G_0(x-y) = \delta^{(4)}(x-y) \quad (7.2.88)$$

The logarithm in (7.2.87) can be more precisely written as

$$\begin{aligned} \log \left( 1 + \frac{\lambda}{2} \Delta_0^{-1} \phi_{cl}^2 \right) &= \log \left( \delta^{(4)}(x-y) + \frac{\lambda}{2} \phi_{cl}^2 G_0(x-y) \right) \\ &\equiv \log \left( \delta_{xy}^{(4)} + K_{xy} \right) \end{aligned} \quad (7.2.89)$$

We can use the expansion

$$\log(1+y) = \sum_{n=1}^{\infty} (-1)^n \frac{y^n}{n} \quad (7.2.90)$$

and write

$$\log \left( \delta_{xy}^{(4)} + K_{xy} \right) = \sum_{n=0}^{\infty} \frac{1}{n+1} (-K_{xy})^{n+1} \quad (7.2.91)$$

or, explicitly

$$\begin{aligned} \mathcal{I} &\equiv \log \left( \delta^{(4)}(x-y) + K(x-y) \right) \\ &= \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{n+1} \int d^4 x_1 d^4 x_2 \dots d^4 x_n K(x-x_1) K(x_1-x_2) \dots K(x_n-x) \end{aligned} \quad (7.2.92)$$

Then, Fourier transforming in the r.h.s.,

$$\begin{aligned} \mathcal{I} &= - \sum_{n=0}^{\infty} \frac{1}{n+1} \int d^4 x_1 d^4 x_2 \dots d^4 x_n \frac{d^4 k_1}{(2\pi)^4} \frac{d^4 k_2}{(2\pi)^4} \dots \frac{d^4 k_n}{(2\pi)^4} \times \\ &\quad \exp \left( i k_1(x-x_1) + i k_2(x_1-x_2) + \dots + i k_n(x_n-x) \right) \times \\ &\quad \left( -\frac{\lambda}{2} \phi_{cl}^2 \tilde{G}_0(k_1) \right) \left( -\frac{\lambda}{2} \phi_{cl}^2 \tilde{G}_0(k_2) \right) \dots \left( -\frac{\lambda}{2} \phi_{cl}^2 \tilde{G}_0(k_n) \right) \end{aligned} \quad (7.2.93)$$

Each integral over  $x_i$  gives  $(2\pi)^4 \delta(k_i - k_{i+1})$  which in turn eliminates one integral over  $k$ . One then ends with

$$\begin{aligned} \text{tr} \mathcal{I} &= \sum_n \frac{1}{n+1} \int d^4 x \frac{d^4 k_1}{(2\pi)^4} \left( -\frac{\lambda}{2} \phi_{cl}^2 \tilde{G}_0(k_1) \right)^{n+1} \\ &= \int d^4 x \int \frac{d^4 k}{(2\pi)^4} \log \left( 1 + \frac{\lambda}{2} \phi_{cl}^2 \tilde{G}_0(k) \right) \end{aligned} \quad (7.2.94)$$

where “tr” has been explicitly included in the r.h.s. as an integral over  $x$ . Then, extracting from  $\text{tr}\mathcal{I}$  the volume factor  $\int d^4x$  we get for the first quantum correction to the classical potential,

$$V_1[\phi_{cl}] = \frac{1}{2} \int \frac{d^4k}{(2\pi)^4} \log \left( 1 + \frac{\lambda}{2} \phi_{cl}^2 \tilde{G}_0(k) \right) \quad (7.2.95)$$

Now, since

$$\tilde{G}_0(k) = \frac{1}{k^2 + m^2} \quad (7.2.96)$$

we finally have

$$V_1[\phi_{cl}] = \frac{1}{2} \int \frac{d^4k}{(2\pi)^4} \log \left( 1 + \frac{\lambda \phi_{cl}^2}{2(k^2 + m^2)} \right) \quad (7.2.97)$$

Of course, the integral over  $k$  is divergent. This is not unexpected since we are in fact computing quantum effects in a field theory and this, in general, needs renormalization. The divergence is an ultraviolet one and manifests in the first two terms in the expansion of the logarithm in powers of  $k^2$ . Since each power of  $k^2$  is accompanied by  $\phi_{cl}^2$ , only two counterterms proportional to  $\phi_{cl}^2$  and  $\phi_{cl}^4$  will suffice to cure infinities. We then write

$$V_1[\phi_{cl}]|_{ren} = V_1[\phi_{cl}] + A\phi_{cl}^2 + B\phi_{cl}^4 \quad (7.2.98)$$

We need two normalization conditions in order to fix  $A, B$ . Inspired in the form of the classical potential we chose as conditions for the complete effective potential

$$\left. \frac{d^2 V_{eff}}{d\phi_{cl}^2} \right|_{\phi_{cl}=0} = m^2, \quad \left. \frac{d^4 V_{eff}}{d\phi_{cl}^4} \right|_{\phi_{cl}=0} = \lambda \quad (7.2.99)$$

These conditions are effectively defining mass and coupling constant for the renormalized theory. This imply, for  $V_1$  the conditions

$$\left. \frac{d^2 V_1}{d\phi_{cl}^2} + 2A \right|_{\phi_{cl}=0} = 0, \quad \left. \frac{d^4 V_1}{d\phi_{cl}^4} \right|_{\phi_{cl}=0} + 4!B = 0 \quad (7.2.100)$$

Using these conditions in (7.2.98) we get,

$$\begin{aligned} A &= -\frac{\lambda}{2} \int \frac{d^4k}{(2\pi)^4} \frac{1}{k^2 + m^2} \\ B &= \frac{\lambda^2}{2} \int \frac{d^4k}{(2\pi)^4} \frac{1}{(k^2 + m^2)^2} \end{aligned} \quad (7.2.101)$$

The final renormalized formula for  $V_1$  is then

$$V_1[\phi_{cl}] = \frac{1}{2} \int \frac{d^4 k}{(2\pi)^4} \left( \log \left( 1 + \frac{\lambda \phi_{cl}^2}{2(k^2 + m^2)} \right) - \frac{\lambda \phi_{cl}^2}{2} \frac{1}{k^2 + m^2} \right) + \frac{1}{2} \left( \frac{\lambda \phi_{cl}^2}{2} \right)^2 \frac{1}{(k^2 + m^2)^2} \quad (7.2.102)$$

Up to now we can then write

$$V_{eff} = \frac{m^2}{2} \phi_{cl}^2 + \frac{\lambda}{4!} \phi_{cl}^4 + \hbar V_1[\phi_{cl}] + O(\hbar^2) \quad (7.2.103)$$

It will be convenient to write  $V_1[\phi_{cl}]$  as a function of

$$x = \frac{\lambda \phi_{cl}}{2m^2} \quad (7.2.104)$$

in the form

$$V_1[\phi_{cl}] = m^4 H(x) \quad (7.2.105)$$

with

$$H(x) = \frac{1}{2} \int \frac{d^4 k}{(2\pi)^4} \left( \log \left( 1 + \frac{x}{k^2 + 1} \right) - \frac{x}{k^2 + 1} + \frac{1}{2} \frac{x^2}{(k^2 + 1)^2} \right) \quad (7.2.106)$$

Integrals in (7.2.106) can be easily performed leading to

$$H(x) = \frac{1}{64\pi^2} \left( (x+1)^2 \log(1+x) - \frac{3}{2}x - x \right) \quad (7.2.107)$$

It is clear that the effective potential incorporates quantum corrections to the classical potential so that any analysis, as for example that starting from the minima of the potential, will then take into account quantum aspects. In particular, we can analyze whether quantum corrections to the potential can change dramatically its behavior. To start with, we shall be interested in the massless case,  $m = 0$ . The classical potential is given by just the  $\lambda \phi^4$  term and then there is not be symmetry breaking at the classical level since the minimum of the potential is located at  $\phi = 0$ . Can this behavior change because of quantum corrections?

To begin with, let us observe that QFT of massless particles is affected by infrared divergencies. These divergencies have to manifest in some way in the effective potential approach and indeed they affect the renormalization conditions we have used. Indeed, in the massive case we fixed the two required counterterms imposing the conditions

$$\left. \frac{d^2 V_{eff}}{d\phi_{cl}^2} \right|_{\phi_{cl}=0} = m^2 \quad (7.2.108)$$

$$\left. \frac{d^4 V_{eff}}{d\phi_{cl}^4} \right|_{\phi_{cl}=0} = \lambda \quad (7.2.109)$$

leading to the following formulae for the counterterms,

$$A = -\frac{1}{2} \left. \frac{d^2 V_1}{d\phi_{cl}^2} \right|_{\phi_{cl}=0}, \quad B = -\frac{1}{4!} \left. \frac{d^4 V_1}{d\phi_{cl}^4} \right|_{\phi_{cl}=0} \quad (7.2.110)$$

If we try to repeat this procedure for the massless case, conditions (7.2.108)- (7.2.109) change to

$$\left. \frac{d^2 V_{eff}}{d\phi_{cl}^2} \right|_{\phi_{cl}=0} = 0 \quad (7.2.111)$$

$$\left. \frac{d^4 V_{eff}}{d\phi_{cl}^4} \right|_{\phi_{cl}=0} = \lambda \quad (7.2.112)$$

but this produces in  $B$  an infrared divergence leading to an incorrect result. Indeed, from (??) and (7.2.112) one has

$$B = \frac{3}{2} \lambda^2 \int \frac{d^4 k}{(2\pi)^4} \frac{1}{k^4} \quad (7.2.113)$$

Although this result does its work in the ultraviolet, it introduces an infrared divergence which cannot be cured within this choice of normalization conditions. Now, instead of (7.2.112), consider the condition

$$\left. \frac{d^4 V_{eff}}{d\phi_{cl}^4} \right|_{\phi_{cl}=\mu \neq 0} = \lambda \quad (7.2.114)$$

with  $\mu$  an arbitrary mass scale. Moreover, we shall maintain  $m \neq 0$  till the end of calculations and then take the  $m \rightarrow 0$  limit, then no infrared divergences appear and one ends with the result

$$V_1[\phi_{cl}] = \frac{1}{64\pi^2} \left( \frac{\lambda \phi_{cl}^2}{2} \right)^2 \left( \log \left( \frac{\phi_{cl}^2}{\mu^2} \right) - \frac{25}{6} \right) \quad (7.2.115)$$

The following points should be remarked concerning this result

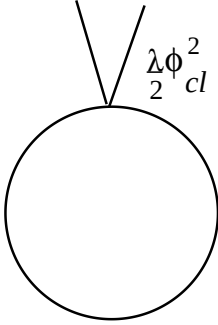
- Infrared divergencies appear here as singularities of the effective potential at  $\phi_{cl} = 0$ . They are cured introducing a mass scale.
- The first correction to the classical potential is order  $\lambda^2$ . Then, if one expected to pass from a trivial minimum classical potential to an effective potential with non-trivial minima, this does not happen for the pure scalar case. There is no choice of scale and coupling constant changing the minimum.

## A loop expansion

Till now we have not clearly analyzed what kind of approximation were we performing. To tackle this problem, let us remember that the calculation of the contribution  $V_1$  to the effective potential reduces to the evaluation of a functional determinant,

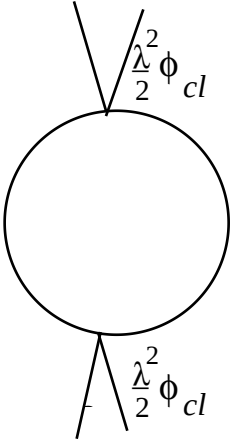
$$V_1 = \frac{1}{2} \text{tr} \log \left( 1 + \frac{\lambda}{2} \Delta_0^{-1} \phi_{cl}^2 \right) \quad (7.2.116)$$

After expansion of the logarithm we ended with a series for which the first non trivial term was a loop consisting of a closed line representing the propagator  $\Delta_0^{-1}$  with a  $\phi_{cl}^2$  insertion,

$$\frac{\lambda \phi_{cl}^2}{2} \int \frac{d^4 k}{(2\pi)^4} \Delta_0^{-1}(k) = \text{Diagram}$$


The diagram shows a circle representing a loop. A single vertex is attached to the top of the circle, with two lines extending upwards and outwards. The label  $\frac{\lambda \phi_{cl}^2}{2}$  is placed next to this vertex.

It is not difficult to guess the form of the following term

$$\left( \frac{\lambda \phi_{cl}^2}{2} \right)^2 \int \frac{d^4 k}{(2\pi)^4} \Delta_0^{-1}(k) \Delta_0^{-1}(k) = \text{Diagram}$$


The diagram shows a circle representing a loop. Two vertices are attached to the circle, one at the top and one at the bottom. Each vertex has two lines extending upwards and outwards. The label  $\frac{\lambda^2 \phi_{cl}^2}{2}$  is placed next to each vertex.

The same happens with the terms that follow: they correspond to one-loop Feynman diagrams with 3 insertions, 4 insertions, etc. Each diagram has a power of  $\lambda$  which corresponds



to the number of insertions. We can then represent what we are doing in this computation of the effective potential by just writing the following self-explanatory formula

$$V_{eff} = \frac{\lambda}{4!} \phi_{cl}^4 + \hbar \underbrace{(a[\phi_{cl}]\lambda + b[\phi_{cl}]\lambda^2 + \dots)}_{V_1} + \hbar^2 \underbrace{(\quad ? \quad)}_{V_2} + \dots \quad (7.2.117)$$

This formula shows the effective potential expressed as an expansion in powers of  $\hbar$ . Each term in the expansion contains all powers of  $\lambda$  so that the contribution say of first order in  $\hbar$  is not just the result one should obtain using the habitual perturbative in  $\lambda$  approach but contains all orders in  $\lambda$ .

In order to better understand the statement above, let us observe that, given a connected diagram with  $L$  loops,  $I$  internal lines,  $E$  external lines and  $V$  vertices, one necessarily has, on the one hand

$$2I + E = 4V \quad (7.2.118)$$

since to each vertex of a  $\lambda^4$  theory arrive 4 lines which should be counted as a 1 if they are external or as a 1/2 if they are internal (an internal line is shared by 2 vertices).

On the other hand, the number of loops obeys

$$L = I - (V - 1) \iff L - 1 = I - V \quad (7.2.119)$$

Indeed, by definition, the number of loops is the number of integrations over independent 4-momenta in a Feynman diagram. In principle, there are as many integrations as internal lines a diagram has. However, at each vertex one has a conservation delta function. This reduces the number of integrations in  $V - 1$ , the  $-1$  arising because the “last” delta function concerns external momenta and then does not reduce the number of integrations. Combining (7.2.118) and (7.2.119) one can write,

$$L = V - \frac{E}{2} + 1 \quad (7.2.120)$$

According to Feynman diagrams recipe one should include a  $\hbar$  factor for each propagator (there is a  $1/\hbar$  in the kinetic energy term of the action and the propagator is obtained from the inverse of this term). The vertices arise from the expansion of the interaction Lagrangian so that each vertex contributes with a  $1/\hbar$  power to the diagram. We are considering diagrams without external legs so that we can just write

$$\text{powers of } \hbar = I - V \quad (7.2.121)$$

or, using (7.2.119),

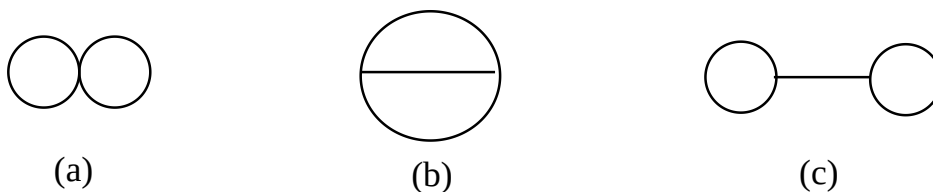
$$\text{powers of } \hbar = L - 1 \quad (7.2.122)$$

This counts the powers in  $\hbar$  in the expansion of  $\exp(-\Gamma/\hbar)$ , so that the  $-1$  appearing in (7.2.122) corresponds to the  $1/\hbar$  appearing in the argument of the exponential. We then have for diagrams in  $\Gamma$  or in  $V_{eff}$ ,

$$\text{powers of } \hbar \text{ in a diagram} = \text{number } L \text{ of loops in the diagram}$$

Then, the approximation systematically performed within the effective potential (or effective action) approach corresponds to a loopwise expansion which gives corrections to the classical potential in powers of  $\hbar$ . For  $\hbar = 0$  one just has the tree diagrams which represent the interaction terms in the classical potential.

Without making any new calculation, we can now infer the form of the contributions to  $V_2$  in expression (7.2.117). Being an  $\hbar^2$  term, it will correspond to two-loop diagrams. We can easily draw the independent two-loop diagrams arising in a  $\lambda\phi^4$  theory. They are



Now, diagram (c) is not one particle irreducible (1PI) since with just one cut in the horizontal line it becomes disconnected. Does this kind of diagrams contribute to the effective potential or the effective action? In the same way we have seen that  $Z[j]$  generates disconnected Green functions and  $W[j]$  generates the connected ones, one can see that the effective action  $\Gamma$  generates 1PI diagrams so that only diagrams (a) and (b) will contribute to the effective potential.

# Bibliography

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- [2] F.J. Dyson, Physical Review **75** (1949) 486.
- [3] K. Freed, Journal of Chemical Physics **56** (1972) 692.



# Chapter 8

## Path-integral quantization and effective field theories II

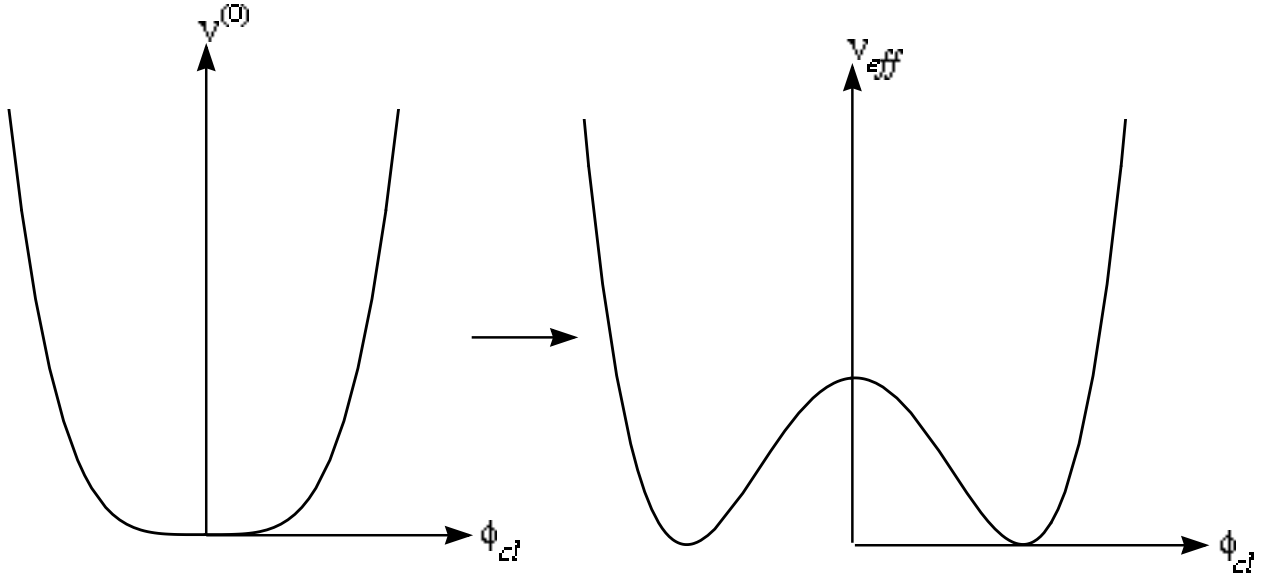
### 8.1 The Coleman-Weinberg mechanism [1]

The outcome of the phenomenon we shall study here is graphically represented in the next page. The idea is to find a model for which, at the classical level, the symmetry of the ground state is the same as that of the Lagrangian (for example gauge symmetry) but quantum corrections break the symmetry spontaneously. I.e., the vacuum state is no more symmetric under the symmetry transformation. One usually calls this phenomenon a dynamical symmetry breaking or a Coleman-Weinberg mechanism.

Such a mechanism should be of interest, for example, when studying the physics of Grand Unified Theories for which, in different circumstances, one has different symmetries and then it should exist a dynamical mechanism producing the symmetry breaking when certain physical parameters change. As we shall see, it also plays a central role in models of supersymmetry breaking. We have seen that this does not happen for the theory of a real scalar field. Coleman and Weinberg shown, in contrast, have seen that the coupling of a gauge field to a (complex) scalar does produce a dramatic effect and dynamical symmetry breaking can take place. Consider, as they did, the Abelian model

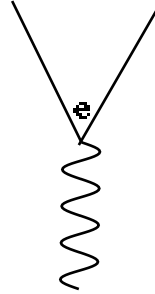
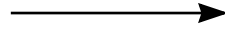
$$L = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2} |(\partial_\mu + ieA_\mu)\phi|^2 - \frac{\lambda}{4!}(\phi\phi^*)^2 \quad (8.1.1)$$

here  $\phi$  is a charged scalar field,  $\phi = \phi_1 + i\phi_2$ , and  $A_\mu$  and Abelian gauge field which couples to  $\phi$  with coupling constant (charge)  $e$ .

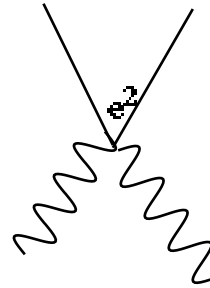
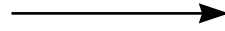


Of course one can consider *ab initio* a symmetry breaking potential but the idea is precisely to see whether quantum effects can change the classical potential with a trivial minimum to a Mexican hat potential producing symmetry breaking. Note that there are two coupling constants in the problem,  $e$  and  $\lambda$ . Apart from the  $\lambda$  vertex, one has also cubic and quartic interaction terms mixing  $\phi$  and  $A_\mu$  as represented in the Feynman rules below,

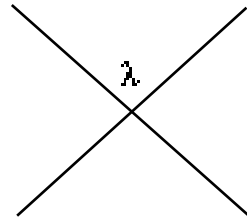
$$ieA_\mu\phi\partial_\mu\phi^* + \text{cc}$$



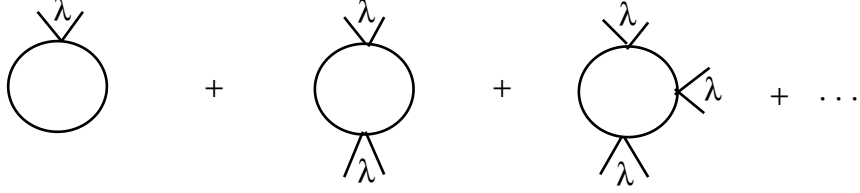
$$e^2 A_\mu\phi\phi^*$$



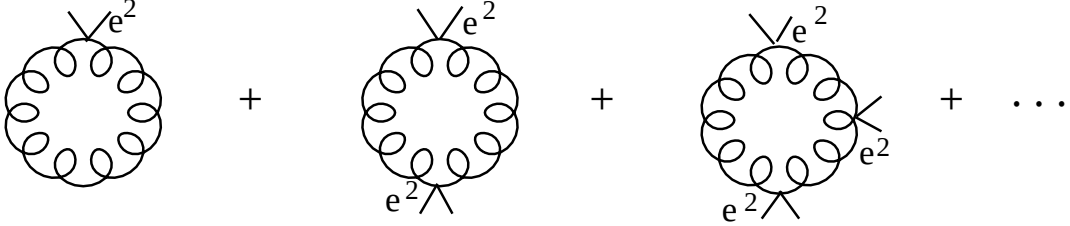
$$\frac{\lambda}{4!}(\phi\phi^*)^2$$



If we are going to compute the effective potential up to first order in  $\hbar$ , we have the same diagrams already studied for the pure scalar field (except that now the field is complex),



The new diagrams correspond to  $A_\mu$  loops with  $\phi_{cl}\phi_{cl}^*$  insertions,



It is important to note that the diagrams are identical except that, while for scalars the propagator is given by (including a  $+i\epsilon$  to handle infrared divergences)

$$\Delta_\phi^{-1}(k) = \frac{1}{k^2 + i\epsilon} \quad (8.1.2)$$

for the gauge field one has

$$\Delta_{A_\mu}^{-1}(k) = \frac{g_{\mu\nu} + (\alpha - 1)k_\mu k_\nu / (k^2 + i\epsilon)}{k^2 + i\epsilon} \quad (8.1.3)$$

In particular, in the Feynman gauge,  $\alpha = 1$  and the only difference is the  $g_{\mu\nu}$  factor. One can then copy the result for one scalar field taking care of replacing the  $\lambda$  coupling constant by an  $e^2$ . Also, one has to take into account that one is dealing in fact with two scalars. The final answer is

$$V_{eff} = \frac{\lambda}{4!} (\phi_{cl}\phi_{cl}^*)^2 + \left( \frac{5\lambda^2}{1152\pi^2} + \frac{3e^4}{64\pi^2} \right) (\phi_{cl}\phi_{cl}^*)^2 \left( \log \left( \frac{\phi_{cl}\phi_{cl}^*}{\mu^2} \right) - \frac{25}{6} \right) \quad (8.1.4)$$



To see how the Coleman-Weinberg phenomenon occurs, suppose that  $\lambda \sim e^4$ . In that case, the  $\lambda^2$  term in (8.1.4) can be disregarded and one then has

$$V_{eff} = \frac{\lambda}{4!} (\phi_{cl} \phi_{cl}^*)^2 + \frac{3e^4}{64\pi^2} (\phi_{cl} \phi_{cl}^*)^2 \left( \log \left( \frac{\phi_{cl} \phi_{cl}^*}{\mu^2} \right) - \frac{25}{6} \right) \quad (8.1.5)$$

Let us search for the value  $\langle \phi \rangle$  that minimizes this effective potential. We have

$$\begin{aligned} \left. \frac{dV_{eff}}{d\phi_{cl}^*} \right|_{\phi_{cl}^* = \langle \phi \rangle^*} &= \frac{\lambda}{3!} \langle \phi \rangle^* \langle \phi \rangle^2 + \frac{12e^4}{64\pi^2} \langle \phi \rangle^* \langle \phi \rangle^2 \left( \log \left( \frac{|\langle \phi \rangle|^2}{\mu^2} \right) - \frac{25}{6} \right) \\ &\quad + \frac{6e^4}{64\pi^2} \langle \phi \rangle^* \langle \phi \rangle^2 \end{aligned} \quad (8.1.6)$$

Now, since the normalization point  $\mu^2$  is arbitrary, we can chose it so as to cancel the logarithm term. This amounts to put  $\mu^2 = \langle \phi \rangle^* \langle \phi \rangle$ . One then has for the minima

$$\left. \frac{dV_{eff}}{d\phi_{cl}^*} \right|_{\phi_{cl}^* = \langle \phi \rangle^*} = \left( \frac{\lambda}{6} - \frac{11e^4}{16\pi^2} \right) \langle \phi \rangle^* \langle \phi \rangle^2 = 0 \quad (8.1.7)$$

Apart from the trivial solution  $\langle \phi \rangle = 0$  there is a second possibility: a non-trivial minimum for  $\langle \phi \rangle \neq 0$  whenever the following relation between coupling constants hold

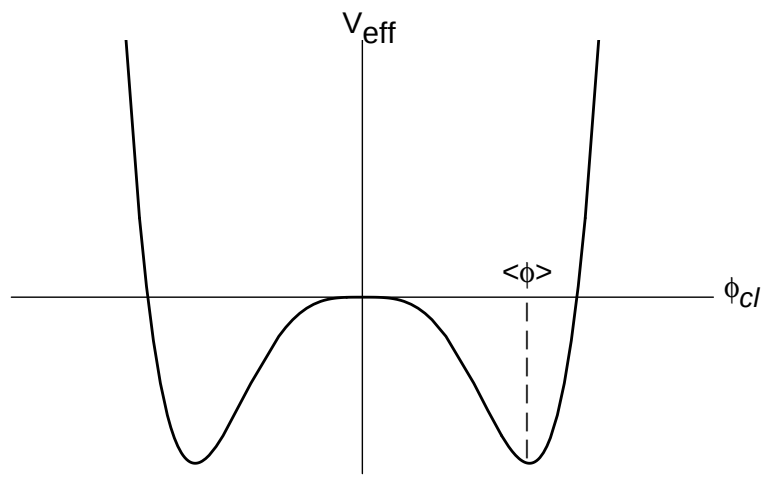
$$\lambda = \frac{33}{8\pi^2} e^4 \quad (8.1.8)$$

Hence, the effective potential including the first order quantum corrections supports a non-trivial minimum  $\langle \phi \rangle \neq 0$  provided the coupling constant satisfy (8.1.8). It is important to note that we started with two dimensionless coupling constants and we ended with just one! Instead, we have gained a dimensionfull quantity,  $\langle \phi \rangle$ .

This phenomenon is known as *dimensional transmutation*: we traded one of the two dimensionless coupling constant ( $e, \lambda$ ) for a dimensionfull constants, say ( $e, \langle \phi \rangle$ ). And we have an effective potential which has a non-trivial minimum

$$V_{eff} = \frac{3e^4}{64\pi^2} |\phi_{cl}|^2 \left( \log \left( \frac{|\phi_{cl}|^2}{|\langle \phi \rangle|^2} \right) - \frac{1}{2} \right) \quad (8.1.9)$$

The shape of this effective potential is



## 8.2 Effective field theories: the Wilsonian effective action

We shall describe here Wilson path-integral approach to Renormalization theory [2]-[3]. We shall follow Kogut-Wilson famous review article [4] as well as the chapter 12 of [5] and lecture 1 in an article by Polchinski [6].

There is a crucial subtlety to be mentioned before starting the presentation of the Wilsonian effective action [7]. When there are no interacting massless particles it coincides with the 1PI effective action previously discussed. However, when interacting massless particles are present the 1PI effective action suffers from infrared ambiguities and may suffer from holomorphic anomalies which are absent in the Wilsonian action. The difference between the Wilsonian and 1PI effective actions is due to the fact that in the vacuum loops for the former one keeps only the contribution of high virtual momenta so that infrared problems are avoided (see below, formula ()).

The concept of Wilsonian effective action is simple. Any physical process has a typical scale. The idea of Wilson was to give the Lagrangian describing the dynamics of some physical process at the scale  $\Lambda$ ,  $[\Lambda] = m^1$  at which it takes place. Consider the case of scalar fields  $\phi(x)$ . The microscopic theory will be described by a *bare* Lagrangian,  $L^{\Lambda_0}$  where  $\Lambda_0$  is an ultraviolet cutoff. The microscopic degrees of freedom will be  $\phi(x, \Lambda_0)$ .

At scale  $\Lambda \ll \Lambda_0$  the effective Lagrangian will take the form

$$L^\Lambda = \sum_i \lambda^i(\Lambda) O_i(x, \Lambda) \quad (8.2.10)$$

where  $O_i(x, \Lambda)$  are relevant composite local operators of the effective fields which in momentum space are  $\phi = \phi(p, \Lambda)$ . These are the effective degrees of freedom at scale  $\Lambda$ , with momentum modes running between 0 and  $\Lambda$ . The constants  $\lambda^i(\Lambda)$  measure the strength of interactions  $O_i$  at scale  $\Lambda$ .

Since the bare Lagrangian encodes the dynamics of scales below  $\Lambda_0$ , the effective Lagrangian is completely determined by  $L^{\Lambda_0}$ . It is obtained from it by integrating between  $\Lambda$  and  $\Lambda_0$ . This integration gives the values of the effective couplings in terms of the bare couplings  $\lambda^i(\Lambda_0)$ ,

$$\lambda^i(\Lambda) = \lambda^i(\lambda_0^i(\Lambda_0); \Lambda, \Lambda_0) \quad (8.2.11)$$

Since physics at scale  $\Lambda$  should be independent of the ultraviolet cutoff  $\Lambda_0$  one has

$$\frac{\partial \lambda^i}{\partial \Lambda_0} = 0 \quad (8.2.12)$$

This is so because the  $\Lambda_0$  dependence of  $g_0^i$  cancels the explicit  $\Lambda_0$  dependence. This is the action of the renormalization group. It allows to take the  $\Lambda_0 \rightarrow \infty$  limit without changing the low energy physics.

Let us show how this effectively works in the simplest case of a  $\phi^4$  scalar theory with Lagrangian

$$L = \frac{1}{2}(\partial_\mu \phi(x))^2 + \frac{1}{2}m_0^2 \phi^2(x) + \frac{\lambda_0}{4!} \phi^4(x) \quad (8.2.13)$$

where  $m_0$  and  $\lambda_0$  are bare parameters (no renormalization counterterms are included).

Consider the Euclidean path integral-version of the generating functional

$$Z[j] = \int D\phi \exp \left( - \int (L + j\phi) d^4k \right) = \left( \prod_k \int d\phi(k) \right) \exp \left( - \int (L + j\phi) d^4k \right) \quad (8.2.14)$$

here we have written the Lagrangian in its Fourier transformed version so as to work with momentum variables.

In the extreme right of (8.2.14) we have explicitly indicated that the path-integral measure can be defined as an infinite product of integrals over  $\phi(k)$  with  $k$  running over all possible  $\mathbb{R}^4$  values. Once this is accepted, we shall introduce an ultraviolet momentum cutoff which in practice restricts the number of integration variables in (8.2.14). That is, instead of integrating  $k$  over  $\mathbb{R}^4$ , we integrate over  $|k| \leq \Lambda$  and set  $\phi(k) = 0$  for  $|k| > \Lambda$ . We then write, instead of (8.2.14)

$$Z_\Lambda = \int [D\phi]_\Lambda \exp \left( - \int d^d k L \right) \quad (8.2.15)$$

where

$$[D\phi]_\Lambda = \prod_{|k| \leq \Lambda} d\phi(k) \quad (8.2.16)$$

where we have taken  $j = 0$  and written  $d^d k$  instead of  $d^4 k$  in order to regularize integrals (so an ultraviolet cutoff  $\Lambda_0$  is not necessary here)

Let us call  $b$  a fraction such that  $b < 1$ . Then, we divide integration variables  $\phi(k)$  into two groups. Those in the interval  $b\Lambda \leq |k| \leq \Lambda$  correspond to high-momentum degrees of freedom that will be integrated over. We label the variables in this region using a hat

$$\hat{\phi}(k) = \begin{cases} \phi(k) & \text{for } b\Lambda \leq |k| \leq \Lambda \\ 0 & \text{otherwise} \end{cases} \quad (8.2.17)$$

while we keep the symbol  $\phi(k)$  for the variables corresponding to  $|k| < b\Lambda$ . Outside this domain the new  $\phi$  vanish.

The prescription above can be interpreted as a change of variables  $\phi \rightarrow \phi + \hat{\phi}$  after which (8.2.15) takes the form

$$\begin{aligned}
Z_\Lambda &= \int D\phi \int D\hat{\phi} \exp \left( - \int d^d k \left( \frac{1}{2} (\partial_\mu \phi + \partial_\mu \hat{\phi})^2 + \frac{m_0^2}{2} (\phi + \hat{\phi})^2 + \frac{\lambda_0}{4!} (\phi + \hat{\phi})^4 \right) \right) \\
&= \int D\phi \exp \left( - \int d^d k L[\phi] \right) \int D\hat{\phi} \exp \left( - \int d^d k \left( \frac{1}{2} (\partial_\mu \hat{\phi})^2 + \frac{m_0^2}{2} \hat{\phi}^2 \right. \right. \\
&\quad \left. \left. \lambda_0 \left( \frac{1}{6} \phi^3 \hat{\phi} + \frac{1}{4} \phi^2 \hat{\phi}^2 + \frac{1}{4!} \hat{\phi}^4 \right) \right) \right)
\end{aligned} \tag{8.2.18}$$

(Note that terms  $\phi \hat{\phi}$  vanish since Fourier components with different wave-lengths are orthogonal).

Summarizing, we have arrived to the following recipe for computing  $Z$

$$Z = \int D\phi \big|_{b\Lambda} \exp \left( - \int d^d x L_{eff} \right) \tag{8.2.19}$$

where  $L_{eff}$  involves only the Fourier components  $\phi(k)$  with  $|k| < b\Lambda$ . In fact  $L_{eff}$  is nothing but  $L$  plus corrections proportional to powers of  $\lambda$  which compensate the removal of the large- $k$  Fourier components  $\hat{\phi}$ .

We shall now proceed to effectively integrate out over  $\hat{\phi}$ . To this end we treat the quartic terms as perturbations in  $\lambda_0$ . Also the  $m_0^2 \hat{\phi}^2$  term will be treated as a perturbation since  $m_0^2 \ll \Lambda^2$ .

The only quadratic (“free Lagrangian term”) that remains will be just the kinetic energy term which leads to the propagator

$$\hat{G}(k, p) = \frac{1}{k^2} (2\pi)^d \delta^{(d)}(k + p) \Theta(k) \tag{8.2.20}$$

where

$$\Theta(k) = \begin{cases} 1 & \text{if } b\Lambda \leq |k| \leq \Lambda \\ 0 & \text{otherwise} \end{cases} \tag{8.2.21}$$

Expanding the  $\phi^2 \hat{\phi}^2$  term, we get, to first order

$$I_{22}^{(1)} = -\frac{\lambda_0}{4} \int d^d x \phi^2(x) \hat{G}(x) = -\frac{1}{2(2\pi)^d \mu} \int d^d k \phi(k) \phi(-k) \tag{8.2.22}$$

where

$$\mu = \frac{\lambda}{2(2\pi)^d} \int_{b\Lambda \leq |k| \leq \Lambda} d^d k \frac{1}{k^2} = \frac{\lambda_0}{(4\pi)^d \Gamma(\frac{d}{2})} \frac{1 - b^{d-2}}{d-2} \Lambda^{d-2} > 0 \quad (8.2.23)$$

Such a term could have arrived from an exponential of the form

$$\exp \left( - \int d^d x \frac{\mu}{2} \phi^2 \right) \quad (8.2.24)$$

so that in fact this term in the expansion (and all the higher ones) will just give a positive correction to the  $m_0^2$  mass term.

To order  $\lambda_0^2$  there are two possible contributions coming from  $\lambda_0 \phi^2 \hat{\phi}^2$ , one coming from the square of the tadpole which supplies the order  $\lambda_0^2$  in the exponential (8.2.24), the other one a new contribution correcting the  $\lambda_0 \phi^4$  interaction. This last diagram gives, in the approximation in which the external momenta carried by  $\phi$  are very small compared to  $b\Lambda$ ,

$$I_{22}^{(4)} = -\frac{1}{4!} \int d^d x \zeta \phi^4 \quad (8.2.25)$$

where

$$\begin{aligned} \zeta &= -4! \frac{2}{2!} \left( \frac{\lambda_0^2}{4} \right) \int_{b\Lambda \leq |k| \leq \Lambda} \frac{d^d k}{(2\pi)^d} \left( \frac{1}{k^2} \right)^2 = \frac{-3\lambda_0^2}{(4\pi)^{d/2} \Gamma(d/2)} \frac{1 - b^{4-d}}{d-4} \Lambda^{d-4} \Big|_{d \rightarrow 4} \\ &\longrightarrow -\frac{3\lambda_0^2}{16\pi^2} \log \frac{1}{b} \end{aligned} \quad (8.2.26)$$

The diagrammatic perturbation theory we have described, apart from terms with quadratic and quartic powers of  $\phi$  generates higher (even) powers and also derivative interaction terms.

We can then finally write the effective Lagrangian in the form

$$\begin{aligned} L_{eff} &= \frac{1}{2} (\partial_\mu \phi)^2 + \frac{1}{2} m_0^2 \phi^2 + \frac{\lambda_0}{4!} \phi^4 + (\text{sum of connected diagrams}) \\ &= \frac{1}{2} (1 + \Delta Z) (\partial_\mu \phi)^2 + \frac{1}{2} (m_0^2 + \Delta m_0^2) \phi^2 + \frac{1}{4!} (\lambda_0 + \Delta \lambda) \phi^4 + \Delta C (\partial_\mu \phi)^4 + \Delta D \phi^6 \end{aligned} \quad (8.2.27)$$

We now change variables according to

$$x \rightarrow x' = bx, \quad k' = k/b \quad (8.2.28)$$

so that the new momentum variable  $k'$  is integrated over  $|k'| < \Lambda$ . With this, the effective action

$$S_{eff} = \int d^4 x L_{eff} \quad (8.2.29)$$

becomes

$$S_{eff} = \int d^d x' b^{-d} \left( \frac{1}{2} (1 + \Delta Z) b^2 (\partial'_\mu \phi)^2 + \frac{1}{2} (m_0^2 + \Delta m^2) \phi^2 + \frac{1}{4!} (\lambda_0 + \Delta \lambda) \phi^4 + \Delta C b^4 (\partial'_\mu \phi)^4 + \Delta D \phi^6 + \dots \right) \quad (8.2.30)$$

$$(8.2.31)$$

In order to get from the effective action the same propagator as in the original model we have to rescale fields according to

$$\phi' = (b^{2-d} (1 + \Delta Z))^{1/2} \phi \quad (8.2.32)$$

With this we return to the original form

$$S_{eff} = \int d^d x' \left( \frac{1}{2} (\partial'_\mu \phi')^2 + \frac{1}{2} m^2 \phi'^2 + \frac{\lambda}{4!} \phi'^4 + C' (\partial'_\mu \phi')^4 + D' \phi'^6 + \dots \right) \quad (8.2.33)$$

where the new parameters are

$$\begin{aligned} m^2 &= (m_0^2 + \Delta m^2) (1 + \Delta Z)^{-1} b^{-2} \\ \lambda &= (\lambda_0 + \Delta \lambda_0) (1 + \Delta Z)^{-1} b^{d-4} \\ C' &= (C + \Delta C) (1 + \Delta Z)^{-1} b^d \\ D' &= (D + \Delta D) (1 + \Delta Z)^{-1} b^{2d-6} \end{aligned} \quad (8.2.34)$$

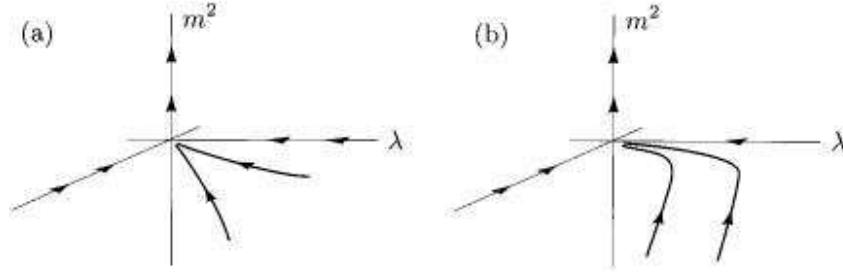
All the corrections  $\Delta m^2$ ,  $\Delta \lambda$ , etc come from diagrams and then are small compared to the leading terms if perturbation theory is justified.

Given the action (8.2.33), we can restart the process of integrating out high-momentum degrees of freedom and then rescale according to (8.2.28) thus iterating the procedure as many times as necessary. If we take  $b$  to be close to 1, the shells in momentum space are made infinitesimally thin and the transformation becomes continuous. **The result of integrating over high-momentum degrees of freedom can then be interpreted as a trajectory or a flow in the space of all possible Lagrangians.** This set of transformations are referred to as the renormalization group (even if the transformations are not invertible and hence they do not form a group).

Let us start from the simplest case and consider a Lagrangian in the vicinity of the point  $m^2 = \lambda = C = D = \dots = 0$ . In view of the transformations (8.2.34) and the fact that all corrections imply a  $\lambda$  factor, the point is unchanged and in this sense the free Lagrangian

$$L_0 = \frac{1}{2} (\partial_\mu \phi)^2 \quad (8.2.35)$$

has a fixed point.



**Figure 4:** Renormalization group flows near the free-field fixed point in scalar field theory: (a)  $d > 4$ ; (b)  $d = 4$ .

Let us study the vicinity of  $L_0$ , where we can ignore terms  $\Delta m^2$ ,  $\Delta \lambda$ , etc, so that (8.2.34) simplify to

$$\begin{aligned} m^2 &= m_0^2 b^{-2} \\ \lambda &= \lambda_0 b^{d-4} \\ C' &= C b^d \\ D' &= D b^{2d-6} \end{aligned} \tag{8.2.36}$$

Now,  $b < 1$  and then parameters multiplied by negative powers of  $b$  grow while those multiplied by positive powers of  $b$  decay. When the Lagrangian contains growing coefficients these will eventually carry it away from  $L_0$ .

Let us call local operators the different terms appearing in a Lagrangian. Then

- **relevant operator:** An operator whose coefficient grows during the recursion operations.
- **irrelevant operator:** An operator whose coefficient dies away during the recursion operations.
- **marginal operator:** An operator multiplied by  $b^0$

Examples:



- $\phi^2$  (mass operator) is always relevant
- $\phi^4$  is relevant if  $d < 4$
- $\phi^4$  is marginal if  $d = 4$



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# Chapter 9

## Quantum aspects of supersymmetric models

27/6/2010

### 9.1 Brief description of the supergraph approach to non-renormalization theorems

The original proofs of non-renormalization theorems [1] in supersymmetric models were based in studies of counterterms analogous to those one performs in ordinary (non-supersymmetric) quantum field theories. As other symmetries do, supersymmetry is able to cancel many divergences usually present in QFT, leading to the so-called “non-renormalization theorems”.

In the process of renormalization of supersymmetric QFT, it turns to be easy to obtain independent scale parameters  $m_i$  corresponding to masses of scalar particles or to scales of supersymmetry breaking, with possibly very large ratios  $m_i/m_j$ . In non-supersymmetric theories loop corrections bring back these ratios to  $m_i/m_j \sim 1$  but this is no more true for supersymmetric theories.

Supersymmetric theories can then give rise to hierarchies of scales that are necessary to unify all 4 fundamental interactions with rather different scales (electroweak interaction scale is  $\sim 10^2$  GeV while gravity scale is  $M_P \sim 10^{19}$  GeV).

As it is well established, the renormalization idea consists in adding to the classical Lagrangian a number of counterterms that cancel the divergencies and leads introduction of physical parameters (masses and coupling constants). Also a rescale of fields (wave function

renormalization) is needed.

$$L_{qantum} = L_{classical} + L_{counterterms} \quad (9.1.1)$$

This scheme can be also applied within the superfield formalism: all parameters in the superpotential and gauge coupling constants get renormalized and also a rescale of the superfield is needed.

Counterterms fall into two classes: those who have the form of a  $\theta\theta$  component of a chiral superfields, which are called  $F$  counterterms,

$$L_C = C(\Phi_1, \dots, \Phi_n)\theta\theta + \text{c.c.} \quad (9.1.2)$$

and those who are obtained as  $\theta\theta\bar{\theta}\bar{\theta}$  of a vector superfield and are consequently called  $D$  counterterms.  $F$  terms will give rise to superpotential and kinetic term renormalization.

In the loop expansion of a supersymmetric theory a given loop Feynman diagram say with scalar fields belonging to a chiral multiplet running in the loop will have its partner diagram with a chiral fermion in the loop. All these different diagrams can be encoded in a superdiagram by defining appropriate superfield Feynman rules and a loop integration that includes Grassman  $\theta, \bar{\theta}$  variables, ie, with a measure  $d^4k d^2\theta d^2\bar{\theta}$ . Studying the form of the counterterms in terms of superfields one can prove the following theorem:

- Theorem: Any perturbative quantum contribution to the effective action must be expressible as one integral over the whole superspace.

This implies that the loop corrections correspond always to  $D$  counterterms and as a consequence, the parameters in the superpotential are not renormalized. That is, they do not receive any quantum correction (even finite) as long as supersymmetry is preserved.

The only necessary renormalization constants are then wave function renormalization of the chiral and vector superfields:

$$\begin{aligned} \Phi_i &\rightarrow \Phi_i^{(r)} = \left(Z_\Phi^{-1/2}\right)_i^j \Phi_j \\ V_a &\rightarrow V_a^{(r)} = \left(Z_V^{-1/2}\right)_a^b V_b \end{aligned} \quad (9.1.3)$$

As in ordinary QED, the renormalization of the gauge coupling constant is related to  $Z_V$  through Ward identities.

## 9.2 Another way to derive non renormalization theorems for supersymmetric theories

The results described above, apart from being rather complicated to prove, are limited to perturbation theory. We shall present here the proof given by Seiberg and collaborators [2]-[4] more recently, based on holomorphic properties of the superpotential.

In supersymmetric theories there are some operators  $O_i$  which only depend on the chiral combination of  $(x, \theta)$  and not on  $\bar{\theta}$ . Clearly their field content can only be made of chiral superfields. Those of more relevant physical importance are the superpotential  $W[\Phi_i, \tau_0, m_f]$  and the gauge kinetic operator  $\tau(\Lambda/\Lambda_0)$  where  $\tau_0$  is the bare complex gauge coupling constants and  $m_f$  complex parameters appearing in  $W$ . The Kähler potential is a real function of  $\Phi_i$  but in what the vacuum states concerns, it is determined by the superpotential (in the  $\Lambda \rightarrow 0$  limit).

We know that complex analysis is substantially more powerful than real analysis. For example, there are a lot of real functions that go at  $x \rightarrow 0$  and  $x \rightarrow \infty$  like  $f(x) = x$  but just one holomorphic function (i.e.  $\partial_{\bar{z}}f(z) = 0$  with those properties:  $f(z) = z$  and no other one. The holomorphic constraint is so strong that in many cases just symmetries and some consistency conditions are enough to determine the unique superpotential.

Let us start by assigning dimensions (in mass units) to superspace as follows

| object                             | dimension |         |
|------------------------------------|-----------|---------|
| $x^\mu, dx^\mu$                    | -1        |         |
| $\partial/\partial x^\mu$          | +1        |         |
| $\theta$                           | -1/2      |         |
| $d\theta, \partial/\partial\theta$ | +1/2      | (9.2.4) |

Then the classical scaling dimension of a chiral superfield  $\Phi$  is  $[\Phi] + 1$ , the same as that of its scalar component  $\phi$ ; the fermion  $\psi$  as dimension  $[\psi] = 3/2$ .

If the action is to be invariant, from the Kähler part of the action one infers

$$[S_K] = [d^4x] [d^2\theta] [d^2\bar{\theta}] [K] = -4 + 2 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} + [K] = 0 \Rightarrow [K] = 2 \quad (9.2.5)$$

while for the superpotential one has

$$\left[ \int d^4x d^2\theta W \right] = -4 + 2 \cdot \frac{1}{2} + [W] = 0 \Rightarrow [W] = 3 \quad (9.2.6)$$

The basic strategy in Seiberg approach to non renormalization theorems starts by promoting the coupling constants in the classic superpotential  $W_{class}$  to the status of background

fields. This includes coupling constants and masses that we shall generically call  $\lambda_I$  appearing in monomials constructed from chiral fields  $\Phi_i$  that appear in the Lagrangian and also and also gauge coupling constants  $4\pi/g^2 = \log \Lambda$ .

Note that even the mass term appearing in the superfield action is a complex parameter that can be viewed as a vacuum expectation value of the  $F$ -term of an auxiliary chiral superfield and it is in this sense that they are taken as background fields.

This idea is inspired in string theory, where the coupling constant is a dynamical field, the dilaton. We can think that the background chiral fields representing coupling constants are dynamical but very massive, with their own superpotential terms which fix their vacuum expectation values of their scalar component to the values  $\lambda_i$  (or, alternatively, one can do this with the  $F$  term). They should then be taken as classical fields having the same dimensions that the original coupling constants have, so that the theory remains renormalizable.

Since only chiral fields appear in the superpotential, coupling constants, now taken as chiral fields should also appear holomorphically: only  $\lambda_i$  and not  $\lambda_i^*$  can appear in any quantum correction to the superpotential since the superpotential is a function only of chiral fields, not of anti-chiral fields.

The following conditions on the quantum renormalized superpotential  $W_{eff}(\Phi_i, \lambda_I, \Lambda)$  should be taken into account

1. Symmetries of the classical theory (with certain transformations laws for fields and coupling constants) should also hold in the effective Lagrangian defining the quantum theory.

For  $\lambda_i = 0$  the theory has a large global symmetry group  $G$ . A non-zero value for these coupling constants can be interpreted as producing a spontaneous breaking of  $G$ . The fact that is spontaneous implies that the Lagrangian should remain invariant under  $G$ .

2. Holomorphy<sup>1</sup>: The superpotential  $W_{class}[\Phi_i]$  should be also and holomorphic function in the coupling constants (the background fields) (We already knew that it is an holomorphic function function of  $\Phi_i$  (it does not depend on  $\Phi_i^+$ ).
3.  $W_{eff}$  can be studied approximately at weak coupling and singularities that arise can be controlled.

---

<sup>1</sup>If  $f$  is complex differentiable at every point  $z_0$  in  $Z$ , we say that  $f$  is holomorphic on  $Z$ . We say that  $f$  is holomorphic at the point  $z_0$  if it is holomorphic on some neighborhood of  $z_0$ . We say that  $f$  is holomorphic on some non-open set  $A$  if it is holomorphic in an open set containing  $A$ . Given  $f = u + iv$ , if  $u$  and  $v$  have continuous first partial derivatives and satisfy the CauchyRiemann equations, then  $f$  is holomorphic. If a function is holomorphic on an open subset  $D$  of the complex plane except a set of isolated points (poles for the function) we say it is meromorphic.



Often, the three constraints above enable to determine  $W_{eff}$  completely. The point is that a holomorphic function is determined by its asymptotic behavior and its singularities.

It is convenient to recall here a subtle point we have already discussed. It concerns the difference between the 1PI effective action and the Wilsonian one. As we have explained, when there are no interacting massless particles, these two effective actions are identical in the sense that they are related by the formula

$$\exp(-S_{eff}^{1PI}) = \langle \exp(-S_{eff}^{Wilson}) \rangle \quad (9.2.7)$$

in the sense that to arrive to the 1PI effective action starting from the Wilsonian one it remains to integrate over the modes with small momenta flowing in the loops. Now, when interacting massless particles are present, this integration is problematic, the 1PI effective action suffers from IR ambiguities and might suffer from holomorphic anomalies. All these problems are absent in the Wilsonian effective action.

Let us take again as a simple example the Wess-Zumino model with a classical superpotential of the form

$$W^{classical} = \frac{1}{2}m\Phi^2 + \frac{1}{3}\lambda\Phi^3 \quad (9.2.8)$$

We can quickly understand Seiberg's strategy by recalling that if the  $\Phi^3$  were absent, since the R-charge of  $\Phi$  is  $R[\Phi] = 1$ ,  $R[W_{\lambda=0}] = 2$  and the theory would have an  $U_R(1)$  R global symmetry. For  $\lambda \neq 0$  the cubic term has charge +1 and R symmetry is explicitly broken. Now, if we think that  $\lambda \neq 0$  correspond to the non-zero vacuum expectation value of some chiral field  $\Lambda(x)$ ,

$$\langle \Lambda(x) \rangle = \lambda \quad (9.2.9)$$

and the charge of such field is  $R[\Lambda] = -1$ , the cubic term will also lead to an R symmetric Lagrangian since  $R[\langle \Lambda \rangle \Phi^3] = 2$ .

Polynomial terms  $\Delta W$  allowed in the quantum effective potential should take the form

$$\Delta W = \mu \Phi^2 f(\langle \Lambda \rangle \Phi / \mu) \quad (9.2.10)$$

where  $\mu$  is the scale one introduces in the renormalization process. Since  $\lambda$  is a dimensionless coupling constant so is the associated superfield  $\Lambda$ . Then  $\Phi$  is the only field with dimensions. One can give the correct R-charge 2 to  $\Delta W$  with a factor  $\Phi^2$  and then accommodate the correct mass dimensions  $m^3$  with an extra factor  $m$ . Then, function  $f$  can only depend on dimensionless variables  $(\langle \Lambda \rangle, \Phi/\mu)$  but in order to do have additional R-charge contributions one should have as argument the product  $\langle \Lambda \rangle \Phi / \mu$ .

Let us consider the limit of small  $\lambda$  (a free theory in this limit) where perturbation theory holds so that no negative powers can arise

$$\Delta W = \sum_{n=0}^{\infty} a_n \mu^{1-n} \lambda^n \Phi^{n+2} \quad (9.2.11)$$

A  $n^{th}$  term in this expansion can arise solely from a tree diagram (no loops) with  $n + 2$  external legs,  $n$  vertices and  $n - 1$  propagators.

### 9.3 The supersymmetric effective potential

We have seen in chapter 7 that the effective potential for a scalar theory with classical potential

$$V_0 = \frac{1}{2}m^2\phi^2 + \frac{1}{4!}\lambda\phi^4 \quad (9.3.12)$$

takes (before renormalization) the form

$$V_1[\phi_{cl}] = \int \frac{d^4k}{(2\pi)^4} \log \left( 1 + \frac{\lambda\phi_{cl}^2/2}{k^2 + m^2} \right) \quad (9.3.13)$$

If instead of making the subtractions necessary to renormalize the theory that led us to the finite expression (7.2.115) we just introduce an ultraviolet cutoff  $\Lambda$ , the integral in (9.3.13) can be performed and the answer is

$$V_1 = \frac{1}{64\pi^2} (m^2 + \lambda\phi_{cl}^2/2)^2 \log \left( \frac{m^2 + \lambda\phi_{cl}^2/2}{\Lambda^2} \right) \quad (9.3.14)$$

Recall that this expression was obtained by expanding the potential around the “classical” configuration  $\phi_{cl}$

$$\phi = \phi_{cl} + \hbar\varphi \quad (9.3.15)$$

The quadratic term in  $\varphi$  which will be identified with the mass term of the “quantum fluctuation field”  $\varphi$

$$V^{quad} = \frac{1}{2} \left. \frac{\partial^2 V_0}{\partial \phi_{cl}^2} \right|_{\varphi=0} = \frac{1}{2} m_{scalar}^2 [\phi_{cl}] \varphi^2 \quad (9.3.16)$$

with

$$m_{scalar}^2 \equiv m^2 + \lambda\phi_{cl}^2/2 \quad (9.3.17)$$

We then see that the 1-loop effective potential (9.3.14) can be compactly written in the form

$$V_1 = \frac{1}{64\pi^2} m_{scalar}^4 \log \left( \frac{m_{scalar}^2}{\Lambda^2} \right) \quad (9.3.18)$$

If instead of just one scalar field we have  $N$  of them, formula (9.3.18) is simply extended to

$$V_1 = \frac{1}{64\pi^2} \text{Tr} M_S^4 \log \left( \frac{M_S^2}{\Lambda^2} \right) \quad (9.3.19)$$

where now  $M_S^2$  is the square mass matrix, defined as

$$M_S^{ab} = \left. \frac{\partial^2 V_0}{\partial \phi_{cl}^a \partial \phi_{cl}^b} \right|_{\varphi=0}, \quad a, b = 1, 2, \dots, N \quad (9.3.20)$$

The trace arises because what one should do is to diagonalize the mass matrix and then sum the contributions of the diagonal which is nothing but the (invariant under diagonalization) trace.

If apart from scalars, other boson fields like gauge fields were present, formula (9.3.19) should include them,

$$V_1 = \frac{1}{64\pi^2} \text{Tr} \sum_i M_{Bi}^4 \log \left( \frac{M_{Bi}^2}{\Lambda^2} \right) \quad (9.3.21)$$

where the sum runs over all helicity states of the boson fields  $B$  (and will be omitted in what follows). Now, if fermions are included, their contribution will be similar to that of bosons, except for the minus term always arising when one considers fermion loops

$$V_1 = \frac{1}{64\pi^2} \text{Tr} M_B^4 \log \left( \frac{M_B^2}{\Lambda^2} \right) - \frac{1}{64\pi^2} \text{Tr} M_F^4 \log \left( \frac{M_F^2}{\Lambda^2} \right) \quad (9.3.22)$$

This formula can be of course used in the case of supersymmetric theories. Let us rewrite eqs.(6.5.96) and (6.5.97) in the form

$$M_B^2 = \begin{pmatrix} \overline{W}^{ac} W_{cb} & \overline{W}^{abc} W_c \\ W^{abc} \overline{W}^c & W_{ac} \overline{W}^{cb} \end{pmatrix}, \quad M_F^2 = \begin{pmatrix} \overline{W}^{ac} W_{cb} & 0 \\ 0 & W_{ac} \overline{W}^{cb} \end{pmatrix} \quad (9.3.23)$$

In terms of a supermatrix  $\mathcal{M}$ , constructed from  $M_B^2$  and  $M_F^2$ , and a supertrace  $\mathcal{STr}$  that includes a factor  $(-1)^F$  ( $F = 0$  for bosons and  $F = 1$  for fermions), eq.(9.3.22) can be compactly written in the form

$$V_1 = \frac{1}{64\pi^2} \mathcal{STr} \mathcal{M}^4 \log \left( \frac{\mathcal{M}^2}{\Lambda^2} \right) \quad (9.3.24)$$

Note that according to (9.3.23) mass degeneracy (both matrices coincide only if ) takes place whenever  $W_a = \partial W / \partial \phi^a = 0$  which is precisely the requirement of unbroken supersymmetry. Nevertheless, even when supersymmetry is broken the supertrace of the mass supermatrix  $\mathcal{M}^2$  vanishes

$$\mathcal{STr} \mathcal{M}^2 \equiv \text{Tr} (M_B^2 - M_F^2) = 0 \quad (9.3.25)$$

## 9.4 Lifting the pseudo-moduli

We have discussed in section 5.5 the case of a generalized O’Raifeartaigh model with a metastable vacuum or better, metastable vacua since there was a flat direction. The superpotential was taken as

$$W = \lambda X + m\Phi_1\Phi_2 + gX\Phi_1^2 + \frac{1}{2}\epsilon m\phi_2^2 \quad (9.4.26)$$

with  $|\epsilon| \ll 1$ .

Such model has two supersymmetric vacua at

$$\phi_{1S} = \pm\sqrt{-\lambda/g}, \quad \phi_{2S} = \mp\frac{1}{\epsilon}\sqrt{-\lambda/g}, \quad X_S = \frac{m}{2g\epsilon} \quad (9.4.27)$$

Concerning non-supersymmetric vacua, the one existing when  $\epsilon = 0$  which corresponded to

$$\phi_1 = \phi_2 = 0, \quad X \text{ arbitrary} \quad (9.4.28)$$

will continue to exist with a small modification provided  $|\epsilon| \ll 1$  and  $m^2 > 2|\lambda g|$  (a no-tachyon condition, see (5.2.30)).

The masses of the fermionic and scalar components of  $\Phi_1$  and  $\Phi_2$  around (9.4.28) are easily calculated:

$$\begin{aligned} m_b^2 &= \frac{1}{2} \left( |2gX|^2 + m^2(2 + |\epsilon|^2\eta|2g\lambda| \right. \\ &\quad \left. + \sqrt{(|2gX|^2 + m^2(2 + |\epsilon|^2) + \eta|2g\lambda|)^2 - 4m^2(|2gX\epsilon - m|^2 + \eta|2g\lambda|(1 + |\epsilon|^2))} \right) \\ m_f^2 &= m_b^2|_{\lambda=0} \end{aligned} \quad (9.4.29)$$

with  $\eta = \pm 1$ .

One can see that no negative eigenvalues (tachyons) arise if

$$\left| 1 - \frac{2\epsilon X}{m} \right| > (1 + |\epsilon|)y \quad (9.4.30)$$

with

$$y = \left| \frac{2g\lambda}{m^2} \right| \quad (9.4.31)$$

a region that includes the origin  $X = 0$  for all

$$y < \frac{1}{1 + |\epsilon|^2} \quad (9.4.32)$$

We can now compute the one loop correction to the classical potential by using formula (9.3.24) and the masses (9.4.29). We restrict the calculation to the range (9.4.30) and (9.4.32) where the pseudomoduli is locally stable. The effective potential is found to still have a minimum near the origin,

$$V_{eff} = V_0 + m_X^2 |X - X_{min}|^2 + \mathcal{O}(\epsilon^2, |X - X_{min}|^4) \quad (9.4.33)$$

with

$$X_{min} = -\frac{\epsilon m}{2g} \frac{(1+y) \log(1+y) - (1-y) \log(1-y) - 2y}{(1+y)^2 \log(1+y) - (1-y) \log(1-y) - 2y} + \mathcal{O}(\epsilon^3) \quad (9.4.34)$$

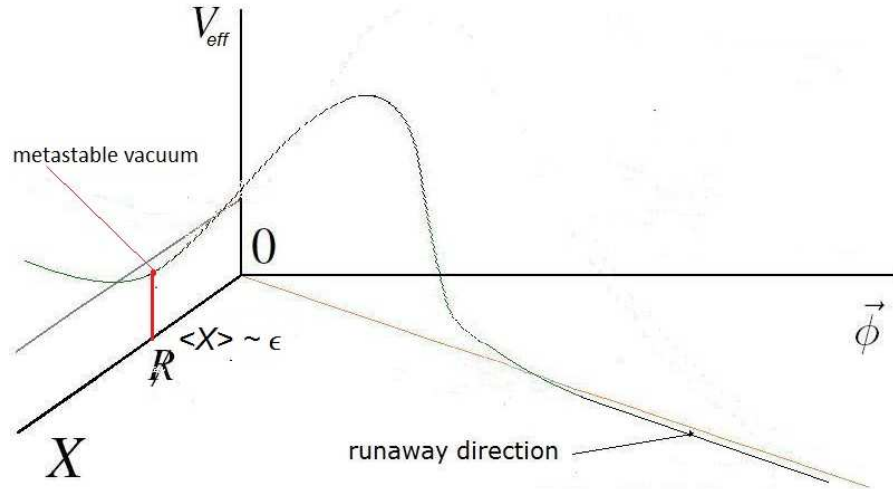
This is a metastable minimum with broken supersymmetry. There is a massless Goldstone boson fermion  $\psi_X$  and all other modes have  $\mathcal{O}(\epsilon^0)$  masses. In particular  $X$  becomes massive with a mass

$$m_X^2 = \frac{1}{8\pi^2} |g^2 m^2| f_2(y) \quad (9.4.35)$$

with

$$f_1(y) \equiv y^{-1} ((1+y)^2 \log(1+y) - (1-y)^2 \log(1-y) - 2y) \quad (9.4.36)$$

We illustrate in the next figure how the (horizontal) flat  $X$  direction has been lifted in such a way that a local minimum exists for a certain value  $X_{min} \sim \epsilon$  determined by a particular choice of parameters.





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# Chapter 10

## Soft supersymmetry breaking and gauge mediation

June 27, 2010

### 10.1 The need of soft supersymmetry breaking

The extension of the Standard Model (SM) to its minimal supersymmetric version (MSSM) obviously requires supersymmetry breaking: a fully supersymmetric model would contain a massless fermion partner of the photon and a charged scalar particle with the mass of the electron. These particles manifestly do not exist.

It seems natural to resort to a spontaneous symmetry breaking mechanism as those we have discussed in previous lectures after introducing the O’Raifeartaigh model (Lecture 5). Let us review this.

We shall consider the superpotential for a very simple supersymmetric O’Raifeartaigh model with 3 supermultiplets is (see (5.2.19))

$$W = \lambda X + m\Phi_1\Phi_2 + gX\Phi_1^2 \quad (10.1.1)$$

which, as we have seen, leads to a scalar potential of the form

$$V = |F_X|^2 + |F_{\Phi_1}|^2 + |F_{\Phi_2}|^2 \quad (10.1.2)$$

The supersymmetric vacuum condition  $V = 0$  implies

$$F_i = 0 \quad (10.1.3)$$

but we have seen that this homogeneous system is incompatible and hence  $V > 0$  and supersymmetry is spontaneously broken.

The absolute minimum was found to be

$$\phi_{1m} = \phi_{2m} = 0, \quad X_m : \text{arbitrary} \quad (10.1.4)$$

$$V[\phi_{1m}, \phi_{2m}, X_m] = |\lambda|^2 \quad (10.1.5)$$

There is then at the classical level a continuous manifold of vacua, a moduli space of non-supersymmetric vacua. We have seen that this degeneracy can be lifted once quantum corrections are taken into account so we used the more precise term of “pseudomoduli space”.

From the second order derivatives at the minimum we derived the masses of the 6 boson fields

$$m_{x^R} = m_{x^I} = 0, \quad m_{\varphi_1^R} = \sqrt{m^2 + 2\lambda g}, \quad m_{\varphi_1^I} = \sqrt{m^2 - 2\lambda g}, \quad m_{\varphi_2^R} = m_{\varphi_2^I} = m \quad (10.1.6)$$

The fact that  $m_{\varphi_1^R} \neq m_{\varphi_1^I}$  is an indication of supersymmetry breaking (Recall that  $m^2 > 2\lambda g$  in order to avoid tachyons).

Concerning fermions,  $\psi_1$  and  $\psi_2$  were combined forming a Dirac fermion  $\Psi_D$  with mass  $m$ . As for  $\psi_X$ , it remained massless and was identified with which is the Goldstone particle associated with spontaneously supersymmetry breaking (the goldstino).

Although supersymmetry is spontaneously broken, the sum rule for fermion and boson masses remains valid

$$\text{Str}[m^2] = \sum_i m_{bi}^2 - \sum_i m_{fi}^2 = 0 \quad (10.1.7)$$

In fact if  $Q$  is a conserved charge associated to a given symmetry of the model, the sum rule is valid in each charge sector  $Q = q$

$$\text{Str}_q[m^2] = 0 \quad (10.1.8)$$

Although this guarantees that divergence cancelation in the vacuum energy (in which the supertrace appears as a factor) will continue to take place, the sum rule is a disaster for the prospect of finding a simple model of spontaneous symmetry breaking leading to an acceptable supersymmetric extension of the SM. For the charged sector of the bosonic supersymmetric partner of the  $d$ -quarks (the  $d$  squarks) one would need to have masses below 5 GeV. For the charge sector of the charged leptons, we need all sleptons to have masses less than 2 GeV.

The solution to this problem is to introduce a field sector in a much higher mass scale than that of the SM. It is in this “hidden sector” that supersymmetry is broken. If this be

sector is weakly coupled to the SM particles it will communicate supersymmetry breaking to the SM without the byproduct of light experimentally undetected sparticles. This kind of scenario is known as “soft supersymmetry breaking”.

It can be shown that since gravity and supergravity couple to all fields, it could do the job of transmitting supersymmetry breaking. Since we have not yet discussed supergravity models, we cannot describe in detail this issue but just give an idea of how it works.

## 10.2 Supersymmetry breaking in supergravity models

We have seen that in (globally) supersymmetric models with chiral fields, the scalar potential can be written as

$$V = \sum_i g_{ij} \frac{\partial W}{\partial \phi_j} \frac{\partial \bar{W}}{\partial \phi_j^*} \quad (10.2.9)$$

where  $W$  is the superpotential and  $g_{ij}$  the metric associated with the Kähler potential  $K$ ,

$$g_{ij} = \frac{\partial^2 K}{\partial \phi_i \partial \phi_j^*} \quad (10.2.10)$$

If one imposes supersymmetry to be a local symmetry then gravitation necessarily enters in the game and for this reason the resulting theory is known as Supergravity. In that case, formula (10.2.9) is modified taking the form

$$V = \exp(K) \sum_i \left( g_{ij} \left( \frac{\partial W}{\partial \phi_i} + \frac{\partial K}{\partial \phi_i} W \right) \left( \frac{\partial \bar{W}}{\partial \phi_j^*} + \frac{\partial K}{\partial \phi_j^*} \bar{W} \right) - 3 \bar{W} W \right) \quad (10.2.11)$$

The potential (10.2.11) has an obvious stationary point when the “Kähler derivative” vanishes.

$$D_i W \equiv \frac{\partial W}{\partial \phi_i} + \frac{\partial K}{\partial \phi_i} W = 0 \quad (10.2.12)$$

However, when this happens, the vacuum energy takes a negative value,

$$V_0 = -\exp(K) |W|^2 \quad (10.2.13)$$

This is a characteristic difference between theories of supergravity and ordinary supersymmetry. In fact, there is an algebraic reason why vacuum states with unbroken supersymmetry in supergravity theories cannot have positive energy density. The solution of Einstein equations for a uniform non-zero vacuum energy density take the form of a de Sitter or anti de Sitter space with a spacetime symmetry which is no more the Poincaré group but  $O(4, 1)$  or

$O(3, 2)$  respectively. It has been proven by Nahm that in the  $N = 1$  case we are considering (only one supersymmetry) the only possible supersymmetry is  $O(3, 2)$  and then one has to deal with an anti-de Sitter space leading to a negative vacuum energy density.

To find a stationary point with  $D_i W = 0$  giving an asymptotically flat space would require  $W = \partial W / \partial \phi_i = 0$ .

The strategy to build supergravity models with broken supersymmetry is similar in spirit to what we have already discussed when we considered a hidden sector with a field  $X$  which produces symmetry breaking and a “visible sector” with ordinary matter fields + superpartners included in chiral fields  $\Phi_i$ . Up to order  $1/M_P$  one considers a superpotential of the form

$$W = W(X) + W_\Phi[\Phi] \quad (10.2.14)$$

and a canonical Kähler potential

$$K = X^\dagger X + \sum_i \Phi_i^\dagger \Phi_i \quad (10.2.15)$$

The idea is to tune the  $F_X$  expectation value to

$$\langle F \rangle = M_W M_P \quad (10.2.16)$$

and, in order to have a vanishing cosmological constant to demand that

$$V = 0 \quad \Rightarrow \quad 3|W|^2 = \sum_i g_{ij} D_i W D_{\bar{j}} \bar{W} \quad (10.2.17)$$

Because of this,

$$\langle W \rangle = M_W M_P^2 \quad (10.2.18)$$

The simplest model where one does this is the Polonyi model [1]

$$W = m^2(X + \beta) \quad (10.2.19)$$

with

$$\beta = 2 + \sqrt{3} M_P \quad (10.2.20)$$

This would be to a rather trivial superpotential in global supersymmetry but not in supergravity. The minimum of the superpotential is in this case at

$$X = (\sqrt{3} - 1) M_P \quad (10.2.21)$$

and one can see that this gives a gravitino mass of the form

$$m_{3/2} = \frac{m^2}{M_P} \exp\left((\sqrt{3} - 1)^2/2\right) \quad (10.2.22)$$

masses for other fields ( $\Phi_i$  fields and gauginos, if they are coupled to the  $\Phi_i$ 's are also generated and all have the correct TeV scale. Indeed, if the supersymmetry breaking scale is of the order  $\mu \approx 10^{11}$  GeV then the resulting gravitino mass is  $m_g \approx 10^3$  GeV which is of the same order of the weak interaction symmetry breaking scale  $m_W/g$ . Even if the supersymmetry breaking scale is large, the mass splitting between the components of a multiplet is of order  $m_g$  [2].

### 10.3 The building blocks in gauge mediation

We have seen that in the gravity-mediated approach, the soft terms are generated at the Planck scale, and therefore necessarily at a scale larger than or equal to the scale  $\Lambda_F$  at which one expects that new Physics originated by supersymmetry arises.

In contrast, in gauge-mediated theories, soft terms are generated at the messenger scale  $M$ , which is a priori unrelated to  $\Lambda_F$ . In particular,  $M \ll \Lambda_F$ . Apart from the observable sector, we have explained that in gauge mediated supersymmetry breaking there is a hidden sector in which (simplifying) there is a chiral superfield  $X$ , which acquires a vacuum expectation value along the scalar and auxiliary components

$$\langle X \rangle = M + \theta^2 F \quad (10.3.23)$$

Here  $M$  and  $\sqrt{F}$  are the fundamental mass scales in the theory and can vary from several tens of TeV to almost the GUT scale. For simplicity we shall consider the case in which  $X$  coincides with the goldstino superfield.

The third ingredient of the gauge mediation theory is a messenger sector, formed by some new superfields that transform under the gauge group as a real non-trivial representation and couple at tree level with the goldstino superfield  $X$ . This coupling generates a supersymmetric mass of order  $M$  for the messenger fields, through couplings of the form

$$\text{messenger--messenger--}X \quad (10.3.24)$$

and mass-squared splittings inside the messenger supermultiplets of order  $F$ .

This messenger sector is unknown and it is the main source of model dependence. It is fairly reasonable to expect that the hidden and messenger sectors have a common origin. The simplest messenger sector is described by  $N_f$  flavours of chiral superfields  $\Phi_i$  and  $\bar{\Phi}_i$  ( $i = 1, 2, \dots, N_f$ ) transforming as the representation  $r + \bar{r}$  under the gauge group. In order to preserve gauge coupling-constant unification, one usually requires that the messengers form complete GUT multiplets. If this is the case, the presence of messenger fields at an intermediate scale does not modify the value of  $M_{GUT}$ , but the inverse gauge coupling strength at the unification scale  $\alpha_{GUT}^{-1}$  does receive an extra contribution.

The interaction between the chiral messenger superfields  $\Phi$  and  $\bar{\Phi}$  and the goldstino superfield  $X$  is given by the superpotential term

$$W_{int} = \lambda_{ij} \bar{\Phi}_i X \bar{\Phi}_j \quad (10.3.25)$$

After replacing in this equation the  $X$  vev, see eq.(10.3.23), we find that the spinor components of  $\Phi$  and  $\bar{\Phi}$  form Dirac fermions with masses  $\lambda M$ , while the scalar components have a squared-mass matrix

$$(\Phi^\dagger \bar{\Phi}) \begin{pmatrix} (\lambda M)^\dagger (\lambda M) & -(\lambda F)^\dagger \\ -(\lambda F) & (\lambda M)(\lambda M)^\dagger \end{pmatrix} \begin{pmatrix} \Phi \\ \bar{\Phi}^\dagger \end{pmatrix} \quad (10.3.26)$$

## Soft terms in the observable sector

The mass scale  $\sqrt{F}$  is the measure of supersymmetry breaking in the messenger sector. However, we are of course mainly interested in the amount of supersymmetry breaking in the observable sector. Ordinary particle supermultiplets are degenerate at the tree level, since they do not directly couple to  $X$ , responsible of supersymmetry breaking. However, splittings in the observable sector arise at the quantum level because of gauge interactions between observable and messenger fields.

We know that vector bosons and matter fermion masses are protected by gauge invariance. In contrast, once supersymmetry is broken, gauginos, squarks, and sleptons can acquire masses consistently with the gauge symmetry. Gaugino masses are generated at one loop, but squark and slepton masses can only arise at two loops, since the exchange of both gauge and messenger particles is necessary. The corresponding Feynman diagrams are drawn in Fig. 1. They were first evaluated in [3], where it was shown that they lead to an acceptable particle spectrum with positive scalar squared masses.

Higher-loop calculations in theories with many particles are quite involved. Therefore it is often useful to employ a simple and systematic method [4] to extract the soft terms for observable. We shall very briefly comment on it.

The positivity of the messenger squared masses requires  $F < M^2$ . Moreover, in most realistic cases one can see that it is appropriate to assume  $F \ll M^2$ . With this assumption, in the effective field theory below the messenger scale  $M$ , supersymmetry breaking can be treated as a small effect. This allows us to use a manifestly supersymmetric formalism to keep track of supersymmetry-breaking effects as well.

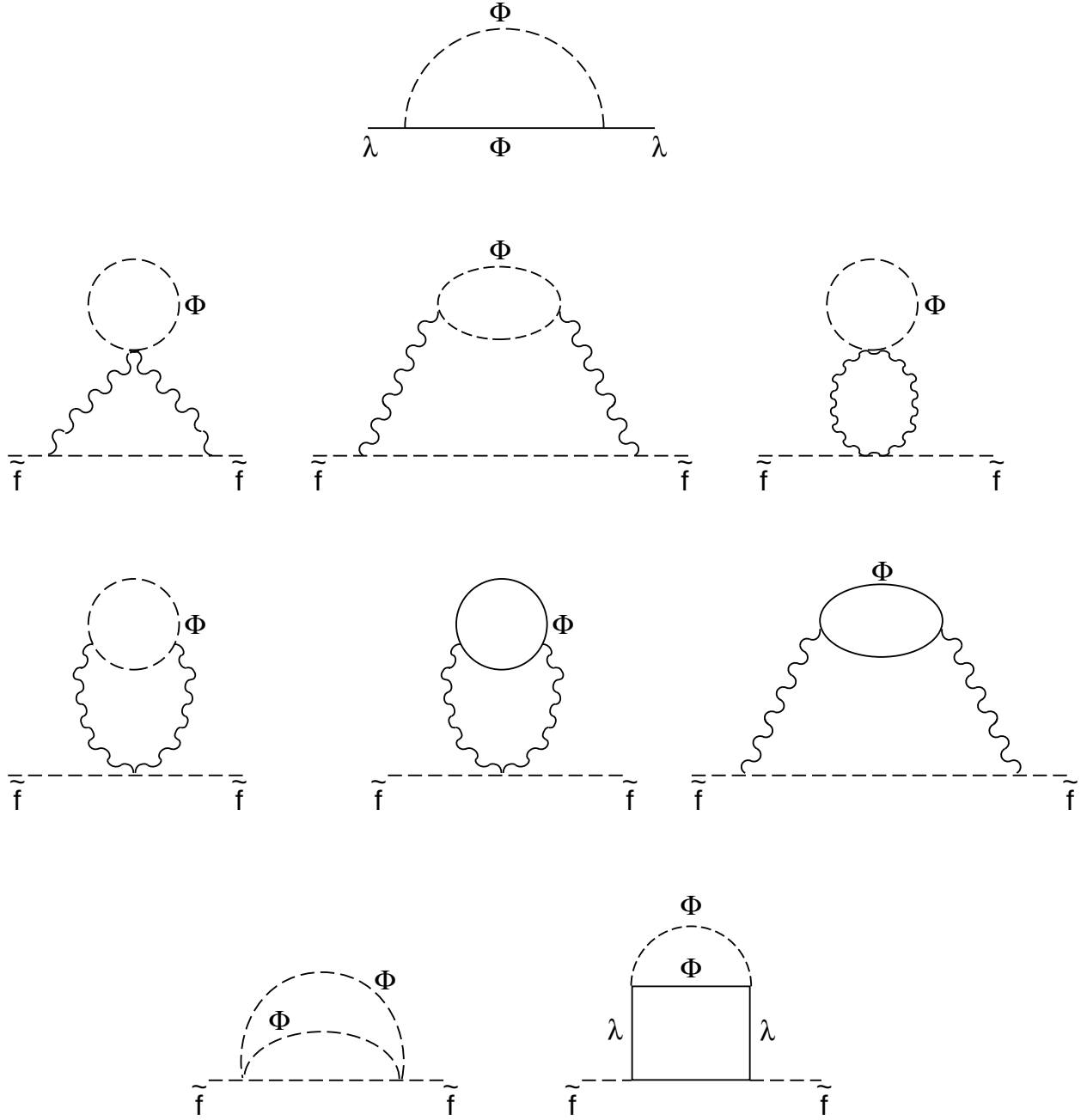


Fig. 1: Feynman diagrams contributing to supersymmetry-breaking gaugino ( $\lambda$ ) and sfermion ( $\tilde{f}$ ) masses. The scalar and fermionic components of the messenger fields  $\Phi$  are denoted by dashed and solid lines, respectively; ordinary gauge bosons are denoted by wavy lines.

First one defines the effective theory, valid below the mass scale  $M$ , by integrating out the heavy messenger fields. One is interested only in renormalizable terms, which are not suppressed by powers of  $M$ . Due to the non-renormalization of the superpotential, all the relevant  $M$  dependence of the low-energy effective theory is contained in the gauge and matter wave-function renormalizations. In the presence of a single mass scale  $M$ , this dependence is logarithmic, and it can be calculated by solving the renormalization-group (RG) equations in the exact supersymmetric theory. At the end, the mass parameter  $M$  can be analytically continued in superspace into a chiral superfield  $X$ . As we have already explained, holomorphy dictates the correct analytic continuation  $M \rightarrow X$  to be performed in the gauge wave function renormalization. On the other hand, the only substitution in the matter wave function renormalization that is consistent with the chiral reparametrization  $X \rightarrow \exp(i\varphi)X$  is given by  $M \rightarrow \sqrt{XX^\dagger}$ . By replacing  $X$  with its background value, given in eq. 10.3.23), one can derive all the relevant supersymmetry-breaking effects. In this way one obtains that the supersymmetry-breaking gaugino masses, squark and slepton masses, and coefficients of the trilinear terms.

We shall just quote the result, in the leading log approximation, for the soft terms generated for chiral messengers communicating via gauge interaction. The supersymmetry-breaking gaugino masses are

$$\begin{aligned}\tilde{M}_{\tilde{\lambda}_r}(t) &= k_r \frac{\alpha_r(t)}{4\pi}, \quad r = 1, 2, 3\Lambda_G \\ \Lambda_G &= \sum_{i=1}^{N_F} n_i \frac{F_i}{M_i} (1 + \mathcal{O}(F_i^2/M_i^4)) \\ k_1 &= \frac{5}{3}, \quad k_2 = k_3 = 1\end{aligned}\tag{10.3.27}$$

Here  $t = \log M^2/Q^2$  with  $Q$  the scale at which one is computing the running coupling constant. The gauge coupling constants  $\alpha_r(t)$  are normalized so that  $k_r \alpha_r$  are all equal at the GUT scale. When there is only one  $X$  chiral superfield and the ratio  $F_i/M_i$  is independent of the flavor index  $i$  (10.3.27) simplifies to

$$\Lambda_G = N \frac{F}{M} (1 + \mathcal{O}(F^2/M^4))\tag{10.3.28}$$

where  $N$  is a calculable factor called the messenger index and has to do with the properties of the gauge group representation.

Concerning the supersymmetry breaking scalar masses and neglecting Yukawa coupling



effects are

$$\begin{aligned}
m_{\tilde{f}}^2(t) &= 2 \sum_{r=1}^3 C_r^{\tilde{f}} \frac{r^2(0)}{4\pi^2} (\Lambda_S^2 + h_r \Lambda_G^2) \\
h_r &= \frac{k_r}{b_r} \left( 1 - \frac{\alpha_r(0)}{4\pi} \right)^{-1}
\end{aligned} \tag{10.3.29}$$

where

$$\Lambda_S^2 = N \frac{F^2}{M^2} (1 + \mathcal{O}(F^2/M^4)) \tag{10.3.30}$$

An attractive feature of the gauge-mediated mass formulae is that gaugino mass terms are generated at one loop, and (positive) squark mass terms are generated at two loops. Since fermion and boson bilinears have different canonical dimensions, all supersymmetry-breaking mass parameters have the same scaling property  $\tilde{m} = (\alpha/\pi)F/M$ .

Another important property of the mass formulae derived here is that they allow a high degree of predictivity. The whole supersymmetric spectrum is determined by the effective supersymmetry breaking scale  $\Lambda = F/M$ , the messenger index  $N$ , the messenger mass  $M$ , and  $\tan \beta$ . The parameter  $k$  can be fixed from the electroweak-breaking condition (up to a phase ambiguity), if we assume the absence of any non-minimal contribution to  $m_{H_{1,2}}^2$ .



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# Chapter 11

## Extended ( $N > 1$ ) supersymmetry

6/7/10

One can consider supersymmetry generators carrying, apart from the spinor index  $\alpha$ , an additional index  $I$ ,  $I = 1, 2, \dots, N$ . In this case one refers to extended ( $N > 1$ ) supersymmetry.

Since in four dimensions a spinor has 4 degrees of freedom, the minimal number of supersymmetric generators is 4 and this corresponds to the  $N = 1$  supersymmetry we have been discussing up to now.

The supersymmetry generators of extended supersymmetry will be written in the form

$$Q_\alpha^I, \quad \bar{Q}_{\dot{\alpha}}^I \quad (11.0.1)$$

with  $I = 1, 2, \dots, N$ . Note that the total number of generators is in this case  $4N$ . Instead of anticommutation relation (5.3.50) for  $Q$  and  $\bar{Q}$ , one now has

$$\{Q_\alpha^I, \bar{Q}_{\dot{\beta}}^J\} = 2\delta^{IJ}\sigma_{\alpha\dot{\beta}}^\mu P_\mu \quad (11.0.2)$$

Concerning the  $Q$ - $Q$  anticommutator, the relation given in (2.7.77) is now extended to

$$\{Q_\alpha^I, Q_\beta^J\} = 2Z^{IJ}\varepsilon_{\alpha\beta} \quad (11.0.3)$$

where  $Z^{IJ}$  is an  $N \times N$  antisymmetric Lorentz scalar matrix.

This is so because a possible symmetric part of  $Z$  would have to appear on the r.h.s. of (11.0.3) in a term which necessarily has to have spin 1. The only available factor to multiply this symmetric  $Z$  to give spin 1 is the Lorentz generator  $S_{\alpha\beta}$ . But then, from the fact that  $Q_\alpha$  commutes with  $P_\mu$  it follows that the symmetric part of  $Z^{IJ}$  should vanish.

Concerning the  $\bar{Q}_{\dot{\alpha}}\bar{Q}_{\dot{\alpha}}$  anticommutator, we have learnt that, carrying Weyl fermions indices one has (see (2.4.58))

$$(Q_{\alpha}^I)^{\dagger} = \bar{Q}_{\dot{\alpha}I} \quad (11.0.4)$$

so that daggering (11.0.3) we have (after raising and lowering indices using the corresponding  $\epsilon$  "tensor")

$$\{\bar{Q}_{\dot{\alpha}}^I, \bar{Q}_{\dot{\beta}}^J\} = -2Z^{IJ*}\epsilon_{\dot{\alpha}\dot{\beta}} \quad (11.0.5)$$

The  $Z^{IJ}$  Lorentz scalars are called central charges. Using Jacobi identity one can prove that they commute with the  $Q_{\alpha}^I$  and  $\bar{Q}_{I\dot{\alpha}}$ . The existence of central extensions in  $N > 1$  supersymmetry algebra was shown by Haag, Lapuszanski and Sohnius [2].

## The $N = 2$ case in more detail

In the  $N = 2$  case there is only one central charge,  $Z^{12} \equiv \sqrt{2}Z$  so that one has

$$\begin{aligned} \{Q_{\alpha}^I, Q_{\beta}^J\} &= 2\sqrt{2}\epsilon_{\alpha\beta}\epsilon^{IJ}Z \\ \{\bar{Q}_{I\dot{\alpha}}, \bar{Q}_{J\dot{\beta}}\} &= 2\sqrt{2}\epsilon_{\dot{\alpha}\dot{\beta}}\epsilon_{IJ}Z^* \end{aligned} \quad (11.0.6)$$

Let us start from the case in which the central charges vanish. Then the supersymmetry generator algebra becomes

$$\begin{aligned} \{Q_{\alpha}^I, \bar{Q}_{J\dot{\beta}}\} &= 2\sigma_{\alpha\dot{\beta}}^{\mu}P_{\mu}\delta_I^J \\ \{Q_{\alpha}^I, Q_{\beta}^J\} &= 0 \end{aligned} \quad (11.0.7)$$

This algebra has a new symmetry: we can perform a unitary rotation of the two supercharges  $Q_{\alpha}^I$

$$Q_{\alpha}^I \rightarrow R^{IJ}Q_{\alpha}^J \quad \text{with } R \in U(2) \quad (11.0.8)$$

that leaves the anticommutator relations invariant. Using that  $U(2) = U(1) \times SU(2)$ , we identify the Abelian factor with the familiar  $U(1)_R$  R-symmetry that rotates the global phase of the 2 supercharges  $Q_{\alpha}^1$  and  $Q_{\alpha}^2$

$$Q \rightarrow Q' = \exp(i\alpha)Q \quad (11.0.9)$$

With respect of the remaining  $SU(2)$  group, the supercharges  $Q_{\alpha}^I$  are in a doublet representation.

Recall that in the massless case, being  $P_\mu = (E, 0, 0, -E)$  the  $N = 1$  algebra takes the form

$$\begin{aligned}\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} &= 2 \begin{pmatrix} 2E & 0 \\ 0 & 0 \end{pmatrix} \\ \{Q_\alpha, Q_\beta\} &= \{\bar{Q}_{\dot{\alpha}}, \bar{Q}_{\dot{\beta}}\} = 0\end{aligned}\tag{11.0.10}$$

So that we identified  $Q_1$  and  $\bar{Q}_1$  (after rescaling with a  $1/2\sqrt{E}$  factor) with creation-annihilation operators  $a, a^\dagger$

$$\begin{aligned}a &= \frac{1}{2\sqrt{E}}Q_1 \\ a^\dagger &= \frac{1}{2\sqrt{E}}\bar{Q}_1\end{aligned}\tag{11.0.11}$$

satisfying

$$\{a, a^\dagger\} = 1, \quad \{a, a\} = \{a^\dagger, a^\dagger\} = 0\tag{11.0.12}$$

while the  $Q_2$  were just vanishing operators. In the  $N = 2$  case we do the same, by defining

$$a^I = \frac{1}{2\sqrt{E}}Q_1^I\tag{11.0.13}$$

Proceeding as in the  $N = 1$  case we then define a Clifford ground state, where states were classified by the energy and helicity

$$\lambda = \frac{1}{E}\vec{J} \cdot \vec{P}\tag{11.0.14}$$

we consider a vacuum state  $|\lambda_0\rangle$ , such that

$$a^I|\lambda_0\rangle = 0, \quad \lambda|\lambda_0\rangle = J_3|\lambda_0\rangle = \lambda_0|\lambda_0\rangle\tag{11.0.15}$$

and then generate a state of helicity  $\lambda_0 - \frac{1}{2}$  by applying  $a^{\dagger I}$  which, from the point of view of helicity is a lowering operator

$$a^{\dagger I}|\lambda_0\rangle = |\lambda_0 - \frac{1}{2}\rangle\tag{11.0.16}$$

For the massless vector multiplet we chose  $\lambda_0 = 1$  and acting with  $a^{\dagger I}$  we generate two fermionic states

$$|1/2\rangle >^I = a^{\dagger I}|1\rangle\tag{11.0.17}$$

Acting with the two  $a^\dagger$ 's we get one scalar boson

$$|0\rangle = a^{\dagger 1} a^{\dagger 2} |1\rangle \quad (11.0.18)$$

We learnt that in n CPT invariant theories, the number of states must in general be doubled since CPT reverses the sign of the helicity. Then each one of the states described above should be doubled.

$$\left( \begin{array}{c} \{ |1\rangle, | -1\rangle_{CPT} \} \\ \{ |\frac{1}{2}\rangle^{(1)}, | -\frac{1}{2}\rangle_{CPT}^{(1)} \} \quad \{ |\frac{1}{2}\rangle^{(2)}, | -\frac{1}{2}\rangle_{CPT}^{(2)} \} \\ \{ |0\rangle, |0\rangle_{CPT} \} \end{array} \right) \quad (11.0.19)$$

In terms of local fields there are:

- a vector  $A_\mu$  (the gauge bosons of some gauge group  $G$ , since we consider massless representations), which is  $SU(2)_R$  singlet
- two Weyl spinors  $\lambda^{(I)}$ , the gauginos, arranged in an  $SU(2)_R$  doublet
- a complex scalar  $\phi$ , playing the role of the Higgs, a singlet of  $SU(2)_R$  but in the adjoint of the gauge group  $G$ .

These fields arrange as

$$\left( \begin{array}{ccc} & A_\mu & \\ & \swarrow & \\ \lambda^{(1)} & & \lambda^{(2)} \\ & \searrow & \\ & \phi & \end{array} \right) \quad (11.0.20)$$

where the arrows indicate the action of the supercharge  $\overline{Q}_\alpha^{(1)}$ .

Concerning the  $SU(2)_R$  symmetry we discovered above,  $A_\mu$  is a singlet while the two gauginos can be arranged as an  $SU(2)_R$  doublet and the scalar is a singlet.

We can interpret this as an  $N = 1$  supersymmetry representation by looking to the  $N = 2$  vector multiplet as composed of an  $N = 1$  vector multiplet  $W_\alpha = (A_\mu, \lambda^{(1)})$  and an  $N = 1$  chiral multiplet  $\Phi = (\phi, \lambda^{(2)})$ .

Coming now to the case  $Z \neq 0$ , let us define

$$a_\alpha = \frac{1}{2} \{ Q_\alpha^1 + \epsilon_{\alpha\beta} (Q_\beta^2)^\dagger \}, \quad (11.0.21)$$

$$b_\alpha = \frac{1}{2} \{ Q_\alpha^1 - \epsilon_{\alpha\beta} (Q_\beta^2)^\dagger \}. \quad (11.0.22)$$



Then, in the rest frame (and using eq.(11.0.4) in order to write formulæ in terms of  $Q^\dagger$  instead of  $\bar{Q}$ ) the  $N = 2$  supersymmetry algebra reduces to

$$\{a_\alpha, a_\beta^\dagger\} = \delta_{\alpha\beta}(m + \sqrt{2}Z), \quad (11.0.23)$$

$$\{b_\alpha, b_\beta^\dagger\} = \delta_{\alpha\beta}(m - \sqrt{2}Z), \quad (11.0.24)$$

with all other anti-commutators vanishing.

Since all physical states have positive definite norm, it follows that for massless states, the central charge is trivially realized (*i.e.*,  $Z = 0$ ), and this case was already discussed.

For massive states, eq.(11.0.24) leads to the bound on the mass  $m \geq \sqrt{2}|Z|$ . Note that when  $m = \sqrt{2}|Z|$ , the operators in (11.0.24) are trivially realized and the algebra resembles the massless case. The dimension of the representation is greatly reduced. For example, a reduced massive  $N = 2$  multiplet has the same number of states as a massless  $N = 2$  multiplet. Thus the representations of the  $N = 2$  algebra with a central charge can be classified as either long multiplets (when  $m > \sqrt{2}|Z|$ ) or short multiplets (when  $m = \sqrt{2}|Z|$ ).

States that saturate the bound are called BPS states (from the names of Bogomol'nyi, Prasad and Sommerfield [3]-[4]):

$$\{b_\alpha, b_\beta^\dagger\}|BPS\rangle = 0 \quad (11.0.25)$$

From this equations it is clear that the BPS states are annihilated by half of the supersymmetry generators and thus belong to reduced representations of the supersymmetry algebra. An important consequence of this is that, for BPS states, the relationship between their charges and masses is dictated by supersymmetry and does not receive perturbative or nonperturbative corrections in the quantum theory. This is so because a modification of this relation implies that the states no longer belong to a short multiplet. On the other hand, quantum corrections are not expected to generate the extra degrees of freedom needed to convert a short multiplet into a long multiplet. Since there is no other possibility, we conclude that for short multiplets the relation  $m = \sqrt{2}|Z|$  is not modified either perturbatively or nonperturbatively.

The first example of theories with  $N = 2$  extended supersymmetry was given by Pierre Fayet [1]. Following a french tradition, he called it "hypersymmetry" but this name had not success and was replaced by "extended supersymmetry".

Any theory with  $N = 2$  supersymmetry also has  $N = 1$  supersymmetry so its Lagrangian has to be a special case of the  $N = 1$  Lagrangians we were discussing.

Let us end this lecture discussing supersymmetry in arbitrary spacetime dimensions In four dimensions  $N = 1$  supersymmetry means that there are four independent SUSY generators:  $Q_1, Q_2, \bar{Q}_1, \bar{Q}_2$ . This is of course the minimum number of supersymmetries in four dimensions since the minimum spinor dimension is four.

Let us refer to this as  $N = (1)_4$  supersymmetry. More generally,  $N = (p)_{p*n_{min}}$  denotes  $p*n_{min}$  supersymmetries, where  $n_{min}$  is the minimum spinor dimension in a given space-time dimensions. In this new notation the possible global supersymmetries in four dimensions are:

$$\begin{aligned}
N &= (1)_4 \\
N &= (2)_8 \\
N &= (4)_{16} \\
N &= (8)_{32} \\
&\dots
\end{aligned}
\tag{11.0.26}$$

In  $4k + 2$  dimensions we can have independent chiral and antichiral SUSY generators, since CPT conjugates have the same chirality. Thus we need a notation which distinguishes chiral from antichiral SUSY generators:

$$N = (p, q)_{(p+q)*n_{min}} \tag{11.0.27}$$

where  $p, q$  are the number of chiral/antichiral SUSY generators, respectively. Analyzing the properties of spinors in different spacetime dimensions one can infer that in any spacetime dimension there is a minimum number of supersymmetries (other than zero). Thus:

As few as 4 supersymmetries can only occur for: . . . . .  $d \leq 4$

As few as 8 supersymmetries can only occur for: . . . . .  $d \leq 6$

As few as 16 supersymmetries can only occur for: . . . . .  $d \leq 10$

As few as 32 supersymmetries can only occur for: . . . . .  $d \leq 11$

Furthermore, in any spacetime dimension, the maximum number of supersymmetries of physical interest is always 32 or less. This is because for more than 32 supersymmetries all massless multiplets contain unphysical higher spin particles, i.e. particles with spin greater than that of the d-dimensional graviton.

## 11.1 Problems

- 1- From the fact that  $[Q_\alpha, P_\mu] = 0$ . show that the symmetric part of  $Z^{IJ}$  should vanish
- 2- Using the Jacobi identity prove that the  $Z$ s are central charges (they commute with all  $Q$ s, with  $P_\mu$  and among them).



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## Chapter 12

### Solitons in condensed matter





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